



Aegis AutoAssure®

AUDIT DEFENSE PACK

IATF 16949:2016 / ISO 9001:2015

Ridgeway Lightweight Structures

Findlay, Ohio

Scope: Manufacture to customer design of friction stir welded, brazed, riveted and bonded aluminium battery enclosures and structures for Ford, BMW and Mercedes-Benz

Audit Type: Registrar

Generated from data as of: 2026-10-08 21:46 UTC

Prepared by: Aegis AutoAssure (Recovery consultant)

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1. Executive Summary

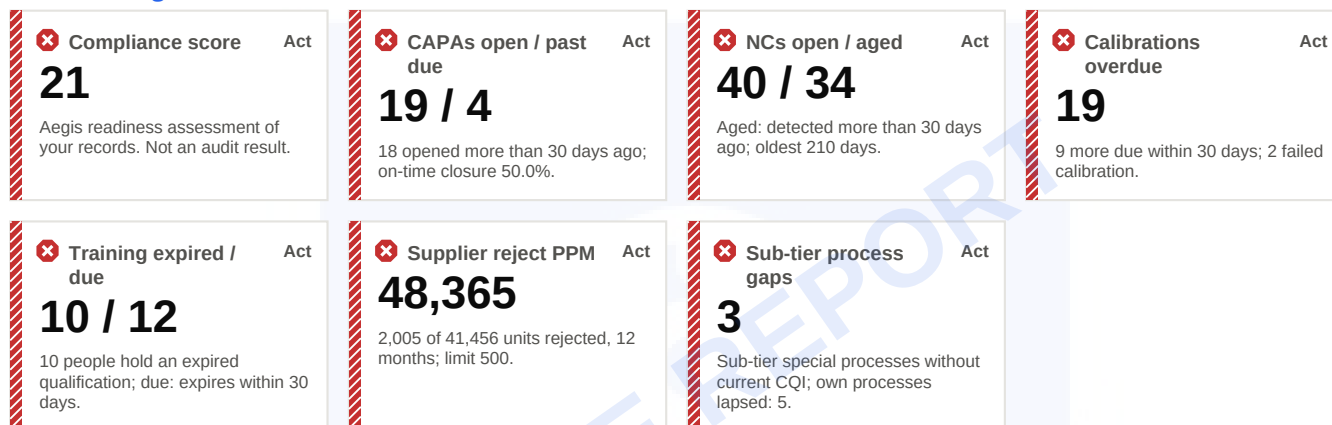
80 issues ranked for action from the records; 17 groups of them share one cause.

The IATF 16949 compliance score is 21. 118 findings open across audits, nonconformances and self-assessment; 4 CAPAs past their due date; 19 measuring instruments past calibration; 10 people with an expired qualification.

QMS snapshot

Every number below is a key figure counted from the registers on 2026-10-08; the key figures table that follows says what each measures.

Headline figures



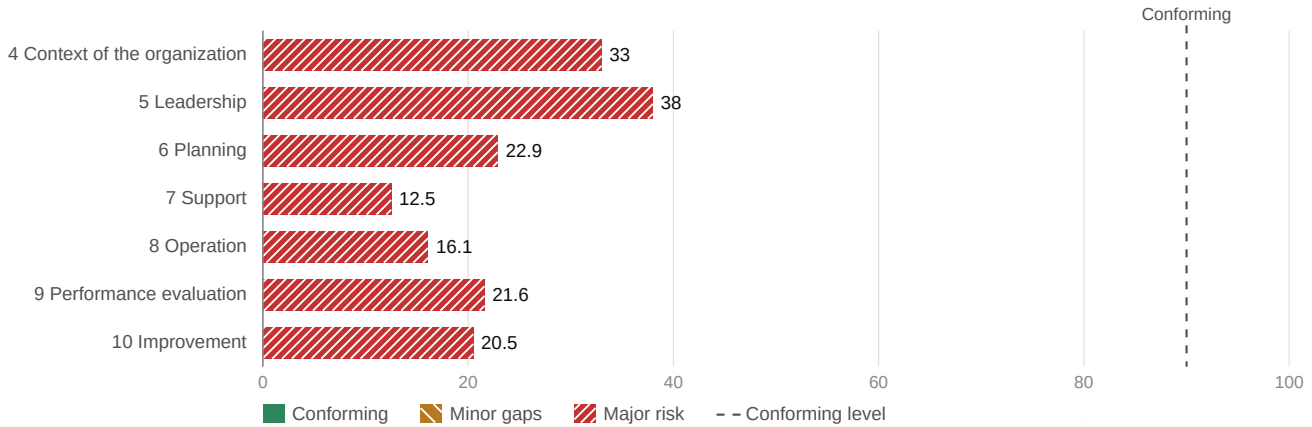
Compliance score: 21 (act, Aegis readiness assessment of your records. Not an audit result.); CAPAs open / past due: 19 / 4 (act, 18 opened more than 30 days ago; on-time closure 50.0%.); NCs open / aged: 40 / 34 (act, Aged: detected more than 30 days ago; oldest 210 days.); Calibrations overdue: 19 (act, 9 more due within 30 days; 2 failed calibration.); Training expired / due: 10 / 12 (act, 10 people hold an expired qualification; due: expires within 30 days.); Supplier reject PPM: 48,365 (act, 2,005 of 41,456 units rejected, 12 months; limit 500.); Sub-tier process gaps: 3 (act, Sub-tier special processes without current CQI; own processes lapsed: 5.).

How the customer sees the plant



Customer PPM: 212 PPM (act, Helium Leak Escapes; target 0 PPM.); Delivery to the customer: 71% (act, On-Time Delivery; target 95%.); Controlled shipping: CS1 (watch, In force with Ford. From RISK-RDG-02 (risk register). 1 of 2 suspect product cases shipped to a customer; 1 told.); Customer complaints: 16 (act, Raised in the last 12 months, 8 still open. From nonconformances raised from a customer complaint.).

Clause scores



Top-level IATF 16949:2016 clause scores: Aegis's proprietary assessment of the documents, evidence and open findings recorded against each clause. Not an audit result, and not a certification decision.

Clause scores: 4 33, 5 38, 6 22.9, 7 12.5, 8 16.1, 9 21.6, 10 20.5. Major risk: 4 (33), 5 (38), 6 (22.9), 7 (12.5), 8 (16.1), 9 (21.6), 10 (20.5).

Every QMS element at a glance

	Current	⚠ Due in 30 days	✖ Overdue	✖ Missing
Documents	19	2	2	11
Records	108	n/a	n/a	10
Training	35	12	10	7
Calibration	67	9	19	0
CAPA	5	10	4	0
Nonconformances	5	13	22	0
Suppliers	0	0	10	13
Special processes (CQI)	7	2	7	1
PPAP and layout inspection	7	4	6	4
Internal audits	6	1	0	0
Risk and PFMEA	7	3	0	18
Management review	1	0	0	0

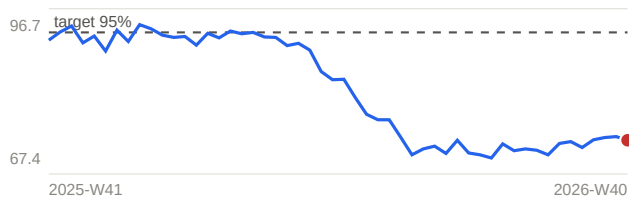
Count: 0 1 to 108, darker is more

Each QMS element's records by state on the report date. Current: in date, or not yet due. n/a: the state does not apply to that element, or its records were not read.

Documents: 19 current, 2 due in 30 days, 2 overdue, 11 missing; Records: 108 current, 10 missing; Training: 35 current, 12 due in 30 days, 10 overdue, 7 missing; Calibration: 67 current, 9 due in 30 days, 19 overdue, 0 missing; CAPA: 5 current, 10 due in 30 days, 4 overdue, 0 missing; Nonconformances: 5 current, 13 due in 30 days, 22 overdue, 0 missing; Suppliers: 0 current, 0 due in 30 days, 10 overdue, 13 missing; Special processes (CQI): 7 current, 2 due in 30 days, 7 overdue, 1 missing; PPAP and layout inspection: 7 current, 4 due in 30 days, 6 overdue, 4 missing; Internal audits: 6 current, 1 due in 30 days, 0 overdue, 0 missing; Risk and PFMEA: 7 current, 3 due in 30 days, 0 overdue, 18 missing; Management review: 1 current, 0 due in 30 days, 0 overdue, 0 missing.

Documents: In force and approved, review date more than 30 days off; review due within 30 days; review date passed; not approved or no review date. Records: Closed records naming where their evidence is kept; those naming none are missing. Training: Qualifications current; expiring within 30 days; expired; assigned and not started. Calibration: Gauges in service due after 30 days; due within 30; past due; no due date. CAPA: Open CAPAs by due date: more than 30 days off; within 30; past due; no due date. Nonconformances: Open nonconformances by due date, as CAPAs. Suppliers: Re-audited within a year; re-audit due within 30 days; past due or under watch (probation, conditional, suspended); never audited. Special processes (CQI): Own processes and sub-tier scope: CQI assessment current; due within 30 days; overdue, or a Needs Immediate Action, late or unrated finding open; no CQI assessment on record. PPAP and layout inspection: PPAP submissions and layout inspection reports: approved and signed, or not yet due; required within 30 days; past the required date, or rejected; approved with no signed warrant or no signatories. Internal audits: Internal audits completed or scheduled more than 30 days off; scheduled within 30 days; scheduled date passed, not completed; no date. Risk and PFMEA: Open risks by due date; missing counts high-priority failure modes no control plan names. Management review: Reviews held in the last year; scheduled within 30 days; one missing when none was held in the year.

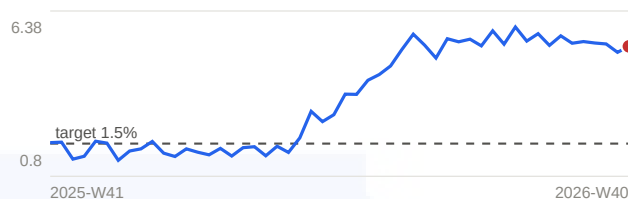
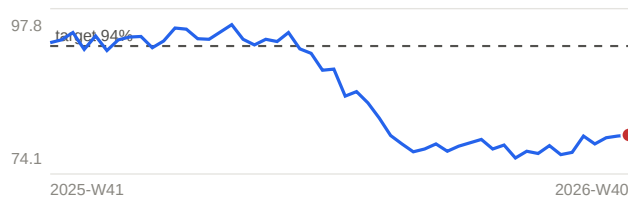
KPI trend



■ Weekly value - - Target

Weekly KPI trend, each on its own scale. Dashed: the scorecard target. The latest week is printed and called on or off target.

On-time delivery 71.3% in 2026-W40 against a target of 95%, off target; First-pass yield 78.2% in 2026-W40 against a target of 94%, off target; CAPA closure 40.1% in 2026-W40 against a target of 90%, off target; Scrap 5.57% in 2026-W40 against a target of 1.5%, off target.



in scale. Dashed: the scorecard target. The latest week is printed and called on or off target.

26-W40 against a target of 95%, off target; First-pass yield 78.2% in 26-W40 against a target of 90%, off target; Scrap 5.57% in 26-W40 against a target of 1.5%, off target

Key figures

Measure	Value	What it measures
IATF 16949 compliance score	21	Aegis readiness score: a proprietary assessment of the documents, evidence and open findings recorded against each IATF 16949:2016 clause. Not an audit result, and not a certification decision.
Open findings, all sources	118 (12 audit findings, 40 nonconformances, 66 assessment nonconformities)	Open audit findings, open nonconformances and nonconforming self-assessment answers, together.
Open nonconformances	40	Nonconformances not yet closed, at any stage from open to verification.
Open nonconformances detected more than 30 days ago	34	Open nonconformances whose detected date is more than 30 days before the report date.
Oldest open nonconformance (days since detected)	210	Days from the detected date of the oldest open nonconformance to the report date.
Open CAPAs	19	CAPAs not closed or cancelled.
Open CAPAs past their due date	4	Open CAPAs whose due date is before the report date.
Open CAPAs opened more than 30 days ago	18	Open CAPAs whose opened date is more than 30 days before the report date.
CAPA on-time closure (%)	50	22 of 44 CAPAs closed in the last 12 months were closed by their due date.
Measuring equipment past calibration due date	19	Equipment whose next calibration date has passed.
Measuring equipment due for calibration within 30 days	9	Equipment whose next calibration date is today or in the next 30 days.
Equipment that failed calibration	2	Calibration failures with an impact evaluation raised.
People with an expired qualification	10	Distinct people holding at least one expired qualification.
Qualifications expiring within 30 days	12	Qualification records whose expiry date is today or in the next 30 days.
Training assignment completion (%)	71.9	46 of 64 training assignments completed.
Layout inspection reports not signed	12	Layout inspection reports neither signed by a reviewer and an approver nor recorded as accepted.
Part submission warrants not signed	5	PPAP submissions whose warrant is not signed.
Special processes at risk	5	Processes whose CQI special process assessment is overdue or was never made, or that carry an open Needs Immediate Action, late or unrated finding.
Sub-tier special processes not verified	3	Special processes performed by a supplier's mapped sub-tier whose CQI assessment is overdue, was never made, or has an open Needs Immediate Action finding.
Customer requirements not flowed down	13	Accepted customer requirements not recorded as flowed to the supplier or the internal process they apply to.
Customer requirements only partly flowed down	6	Of the requirements not recorded as flowed, those the register records as partly in place. Counted within the requirements not flowed down, not beside them.
Supplier reject PPM, from receipts	48365 (2005 rejected, 41456 received)	2,005 of 41,456 units received in the last 12 months were rejected, across all suppliers, per million.

Measure	Value	What it measures
Open suspect product cases	2	Suspect product cases (8.7.1.3) not closed.
Suppliers on probation, conditional or disqualified	10	Suppliers whose status is probation, conditional, disqualified or suspended.



Priority actions

Most pressing first, in the order of Aegis's proprietary prioritisation of your findings. The same records always give the same order.

#	Action	Why now	Owner	Clause
1	Owner H. Lindgren (QD) (7 findings) • Clear the outstanding containment zones on NCR-RDG-016 and establish whether suspect product shipped • Complete the disposition record for RDG-5110 (MRB-RDG-316) • Complete the disposition record for RDG-5109 (MRB-RDG-313) • Complete the disposition record for RDG-5106 (MRB-RDG-318) • Complete the disposition record for RDG-5107 (MRB-RDG-325) • Complete the disposition record for RDG-5105 (MRB-RDG-306) • Complete the disposition record for RDG-5105 (MRB-RDG-321)	H. Lindgren (QD) owns 7 open items across containment, nonconforming product. Where one person holds every overdue item in several areas, the constraint is capacity, and reassignment will move more than escalation will.	H. Lindgren (QD)	8.7.1.1, 8.7.1.3, 8.7.1.4, 8.7.1.7
2	Owner S. Kaur (6 findings) • Clear the outstanding containment zones on NCR-RDG-066 and establish whether suspect product shipped • Re-plan or escalate CAPA-RDG-058: past due and still open • Complete the disposition record for RDG-5107 (MRB-RDG-315) • Complete the disposition record for RDG-5103 (MRB-RDG-322) • Complete the disposition record for RDG-5104 (MRB-RDG-314) • Complete the disposition record for RDG-5115 (MRB-RDG-310)	S. Kaur owns 6 open items across corrective action, containment, nonconforming product. Where one person holds every overdue item in several areas, the constraint is capacity, and reassignment will move more than escalation will.	S. Kaur	8.7.1.1, 8.7.1.3, 8.7.1.4, 8.7.1.7, 10.2.3
3	Part RDG-5102 (5 findings) • Complete the open PPAP elements and sign the part submission warrant for RDG-5102 (PPAP-RDG-5102) • Add the High-priority failure modes on RDG-5102 (PFMEA-RDG-5102) to the control plan, each with a control method and reaction plan • Qualify the measurement system (Gage R&R) before relying on the capability result: FSW penetration depth on RDG-5102 measured with MS-1407 • Complete the layout inspection report for RDG-5102 (LAY-RDG-102) and have it signed by a reviewer and an approver • Complete the disposition record for RDG-5102 (MRB-RDG-341)	Part RDG-5102 carries 5 findings across APQP, capability, layout inspection, nonconforming product. A single part accumulating issues in several controls usually means the process was released before it was ready, rather than 5 independent lapses.	Not assigned	7.1.5.1.1, 8.3.4.4, 8.5.1.1, 8.6.2, 8.7.1.7

#	Action	Why now	Owner	Clause
4	Part RDG-5105 (6 findings) • Add the High-priority failure modes on RDG-5105 (PFMEA-RDG-5105) to the control plan, each with a control method and reaction plan • Complete the disposition record for RDG-5105 (MRB-RDG-306) • Complete the disposition record for RDG-5105 (MRB-RDG-321) • Complete the layout inspection report for RDG-5105 (LAY-RDG-105) and have it signed by a reviewer and an approver • Complete the open PPAP elements and sign the part submission warrant for RDG-5105 (PPAP-RDG-5105) • Complete the disposition record for RDG-5105 (MRB-RDG-342)	Part RDG-5105 carries 6 findings across APQP, layout inspection, nonconforming product. A single part accumulating issues in several controls usually means the process was released before it was ready, rather than 6 independent lapses.	H. Lindgren (QD)	8.3.4.4, 8.5.1.1, 8.6.2, 8.7.1.4, 8.7.1.7
5	Supplier Hocking Brazed Components (3 findings) • Quarantine the material in suspect product case SPR-RDG-002, trace where it went, and notify the customers whose parts share the route • Flow down CSR-RDG-02 (Ford) to Hocking Brazed Components, and record the supplier's acknowledgement • Flow down CSR-RDG-01 (Ford) to Hocking Brazed Components, and record the supplier's acknowledgement	Supplier Hocking Brazed Components appears in 3 separate findings across customer requirements, traceability. These sit in different registers and read as unrelated; together they describe one source that is not under control, and addressing them as one supplier action closes all 3.	Not assigned	8.4.2.3, 8.4.3.1, 8.7.1.3
6	Part RDG-5104 (5 findings) • Add the High-priority failure modes on RDG-5104 (PFMEA-RDG-5104) to the control plan, each with a control method and reaction plan • Complete the disposition record for RDG-5104 (MRB-RDG-303) • Complete the disposition record for RDG-5104 (MRB-RDG-314) • Complete the layout inspection report for RDG-5104 (LAY-RDG-104) and have it signed by a reviewer and an approver • Complete the open PPAP elements and sign the part submission warrant for RDG-5104 (PPAP-RDG-5104)	Part RDG-5104 carries 5 findings across APQP, layout inspection, nonconforming product. A single part accumulating issues in several controls usually means the process was released before it was ready, rather than 5 independent lapses.	E. Vasko, S. Kaur	8.3.4.4, 8.5.1.1, 8.6.2, 8.7.1.4
7	Part RDG-5103 (4 findings) • Add the High-priority failure modes on RDG-5103 (PFMEA-RDG-5103) to the control plan, each with a control method and reaction plan • Complete the disposition record for RDG-5103 (MRB-RDG-322) • Complete the layout inspection report for RDG-5103 (LAY-RDG-103) and have it signed by a reviewer and an approver • Complete the open PPAP elements and sign the part submission warrant for RDG-5103 (PPAP-RDG-5103)	Part RDG-5103 carries 4 findings across APQP, layout inspection, nonconforming product. A single part accumulating issues in several controls usually means the process was released before it was ready, rather than 4 independent lapses.	S. Kaur	8.3.4.4, 8.5.1.1, 8.6.2, 8.7.1.1

#	Action	Why now	Owner	Clause
8	Part RDG-5101 (5 findings) • Add the High-priority failure modes on RDG-5101 (PFMEA-RDG-5101) to the control plan, each with a control method and reaction plan • Complete the layout inspection report for RDG-5101 (LAY-RDG-101) and have it signed by a reviewer and an approver • Complete the open PPAP elements and sign the part submission warrant for RDG-5101 (PPAP-RDG-5101) • Complete the disposition record for RDG-5101 (MRB-RDG-307) • Complete the disposition record for RDG-5101 (MRB-RDG-343)	Part RDG-5101 carries 5 findings across APQP, layout inspection, nonconforming product. A single part accumulating issues in several controls usually means the process was released before it was ready, rather than 5 independent lapses.	N. Oyelaran	8.3.4.4, 8.5.1.1, 8.6.2, 8.7.1.4, 8.7.1.7
9	Part RDG-5110 (4 findings) • Qualify the measurement system (Gage R&R) before relying on the capability result: Powder coat thickness on RDG-5110 measured with CTG-2710 • Complete the disposition record for RDG-5110 (MRB-RDG-316) • Complete the layout inspection report for RDG-5110 (LAY-RDG-110) and have it signed by a reviewer and an approver • Complete the disposition record for RDG-5110 (MRB-RDG-311)	Part RDG-5110 carries 4 findings across capability, layout inspection, nonconforming product. A single part accumulating issues in several controls usually means the process was released before it was ready, rather than 4 independent lapses.	H. Lindgren (QD), L. Ferreira	7.1.5.1.1, 8.6.2, 8.7.1.4
10	Part RDG-5106 (4 findings) • Qualify the measurement system (Gage R&R) before relying on the capability result: Conductivity on RDG-5106 measured with ECM-2911 • Complete the disposition record for RDG-5106 (MRB-RDG-318) • Obtain the customer's concession for RDG-5106 (MRB-RDG-308) before the product is used • Record who reviewed and approved the accepted layout inspection report for RDG-5106 (LAY-RDG-106)	Part RDG-5106 carries 4 findings across capability, layout inspection, nonconforming product. A single part accumulating issues in several controls usually means the process was released before it was ready, rather than 4 independent lapses.	H. Lindgren (QD), Eng Auth (pending)	7.1.5.1.1, 8.6.2, 8.7.1.1, 8.7.1.4
11	Part RDG-5107 (3 findings) • Complete the disposition record for RDG-5107 (MRB-RDG-325) • Complete the disposition record for RDG-5107 (MRB-RDG-315) • Record who reviewed and approved the accepted layout inspection report for RDG-5107 (LAY-RDG-107)	Part RDG-5107 carries 3 findings across layout inspection, nonconforming product. A single part accumulating issues in several controls usually means the process was released before it was ready, rather than 3 independent lapses.	H. Lindgren (QD), S. Kaur	8.6.2, 8.7.1.1
12	Part RDG-5109 (3 findings) • Complete the disposition record for RDG-5109 (MRB-RDG-313) • Record who reviewed and approved the accepted layout inspection report for RDG-5109 (LAY-RDG-109) • Complete the disposition record for RDG-5109 (MRB-RDG-312)	Part RDG-5109 carries 3 findings across layout inspection, nonconforming product. A single part accumulating issues in several controls usually means the process was released before it was ready, rather than 3 independent lapses.	H. Lindgren (QD), P. Ionescu	8.6.2, 8.7.1.4

#	Action	Why now	Owner	Clause
13	Part RDG-5108 (2 findings) - Complete the disposition record for RDG-5108 (MRB-RDG-305) - Complete the layout inspection report for RDG-5108 (LAY-RDG-108) and have it signed by a reviewer and an approver	Part RDG-5108 carries 2 findings across layout inspection, nonconforming product. A single part accumulating issues in several controls usually means the process was released before it was ready, rather than 2 independent lapses.	D. Morrow	8.6.2, 8.7.1.4
14	Owner N. Oyelaran (2 findings) - Re-plan or escalate CAPA-RDG-057: past due and still open - Complete the disposition record for RDG-5101 (MRB-RDG-307)	N. Oyelaran owns 2 open items across corrective action, nonconforming product. Where one person holds every overdue item in several areas, the constraint is capacity, and reassignment will move more than escalation will.	N. Oyelaran	8.7.1.7, 10.2.3
15	Owner P. Ionescu (2 findings) - Re-plan or escalate CAPA-RDG-060: past due and still open - Complete the disposition record for RDG-5109 (MRB-RDG-312)	P. Ionescu owns 2 open items across corrective action, nonconforming product. Where one person holds every overdue item in several areas, the constraint is capacity, and reassignment will move more than escalation will.	P. Ionescu	8.7.1.4, 10.2.3
16	Part RDG-5112 (2 findings) - Complete the layout inspection report for RDG-5112 (LAY-RDG-112) and have it signed by a reviewer and an approver - Complete the disposition record for RDG-5112 (MRB-RDG-309)	Part RDG-5112 carries 2 findings across layout inspection, nonconforming product. A single part accumulating issues in several controls usually means the process was released before it was ready, rather than 2 independent lapses.	T. Marchetti	8.6.2, 8.7.1.4
17	Part RDG-5115 (2 findings) - Complete the layout inspection report for RDG-5115 (LAY-RDG-115) and have it signed by a reviewer and an approver - Complete the disposition record for RDG-5115 (MRB-RDG-310)	Part RDG-5115 carries 2 findings across layout inspection, nonconforming product. A single part accumulating issues in several controls usually means the process was released before it was ready, rather than 2 independent lapses.	S. Kaur	8.6.2, 8.7.1.7
18	Quarantine the material in suspect product case SPR-RDG-001, trace where it went, and notify the customers whose parts share the route	Case SPR-RDG-001 is recorded against this scope. Until it is dispositioned, product from the same supply route is suspect, and every customer that route reached may need to be notified.	Not assigned	8.7.1.3
19	Contain the parts made through Artificial Aging — Aging Oven 2, Friction Stir Welding — Floor Panel Cell, Powder Coat — Busbar Covers and Shields, Vacuum Brazing — Cooling Plates (Hocking Brazed Components) and close its Needs Immediate Action findings	4 records; for example: Powder Coat — Busbar Covers and Shields is performed and the parts that need it are at risk: CQI-12 assessment was due 2026-09-14; 1 open Needs Immediate Action finding. A special process is one whose result cannot be verified by inspection afterwards, which is why the CQI special process assessment (CQI-12) is the control.	Not assigned	4.3.2, 8.5.1
20	Contain the parts Fremont Impregnation & Heat Treat (heat treating for Sandusky Die Casting) treated and close its Needs Immediate Action findings	Fremont Impregnation & Heat Treat performs heat treating for Sandusky Die Casting, and its CQI-9 assessment has 1 open Needs Immediate Action finding. The customer-specific requirements reach past the direct supplier to whoever performs the special process, so the parts it treats are at the same risk as the supplier's own.	Not assigned	8.4.2.1
21	Finish the impact evaluation for Helium Leak Detector (vacuum chamber) (CAL-RDG-01), Measuring Microscope (FSW macro-etch) (CAL-RDG-06): identify product accepted on its measurements since the last good calibration and disposition it	2 records; for example: Helium Leak Detector (vacuum chamber) (CAL-RDG-01) failed calibration and the impact evaluation is not complete. Every measurement taken with it since its last passing calibration is in doubt. Product accepted on those measurements has not been dispositioned.	Not assigned	7.1.5.2.1

#	Action	Why now	Owner	Clause
22	Have Mercer Fastener Finishing (plating for Keystone Fastening Systems), Mercer Fastener Finishing (heat treating for Keystone Fastening Systems) assessed to its CQI special process standard, or move the work to an assessed source	2 records; for example: Mercer Fastener Finishing performs heat treating for Keystone Fastening Systems, and no CQI-9 assessment is on record. The customer-specific requirements reach past the direct supplier to whoever performs the special process, so the parts it treats are at the same risk as the supplier's own.	Not assigned	8.4.2.1
23	Complete the CQI special process assessment of Outsourced Aging (Maumee Thermal Processing)	Outsourced Aging (Maumee Thermal Processing) is performed and the parts that need it are at risk: CQI-9 assessment was due 2026-07-06. A special process is one whose result cannot be verified by inspection afterwards, which is why the CQI special process assessment (CQI-9) is the control.	Not assigned	4.3.2
24	Implement CSR-RDG-03 (Ford) in Change Control, CSR-RDG-04 (BMW) in PPAP, CSR-RDG-05 (BMW) in Inspection, CSR-RDG-06 (Mercedes-Benz) in Traceability, CSR-RDG-07 (Mercedes-Benz) in Launch, CSR-RDG-09 (BMW) in Powder Coat, CSR-RDG-10 (Ford) in FSW, and record the evidence	7 records; for example: BMW requirement CSR-RDG-04 — Part approval for every engineering change — is not recorded as in place in PPAP: Lid rev C submitted without a capability study. A requirement the organisation took on and did not put in place is a requirement nobody is meeting.	Not assigned	7.1.5.1.1, 7.2.1, 8.3.4.4, 8.5.1, 8.5.2.1, 8.5.6.1
25	Flow down CSR-RDG-11 (Ford) to Cuyahoga Extrusions, CSR-RDG-12 (BMW) to Portage Precision Machining, CSR-RDG-15 (BMW) to Maumee Thermal Processing, CSR-RDG-16 (Mercedes-Benz) to Ashtabula Anodizing, and record the supplier's acknowledgement	4 records; for example: BMW requirement CSR-RDG-12 — Right of access for BMW and its supplier quality engineers — is not recorded as flowed to Portage Precision Machining: Recorded as Partial in the register: not all of it is in place. A requirement the organisation took on and did not pass on is a requirement nobody is meeting.	Not assigned	4.3.2, 8.4.2.3, 8.4.3.1
26	Re-plan or escalate CAPA-RDG-059: past due and still open	CAPA-RDG-059 (CMT weld porosity — RDG-5116) is 19 days past its due date and still open. An overdue corrective action is the finding an auditor uses to question whether the system corrects anything.	C. Ibsen	10.2.3
27	Complete the layout inspection report for RDG-5113 (LAY-RDG-113), RDG-5114 (LAY-RDG-114), RDG-5116 (LAY-RDG-116) and have it signed by a reviewer and an approver	3 records; for example: The layout inspection report LAY-RDG-114 for RDG-5114 has no completed reviewer and approver signature, and cannot be signed until: 70 of 74 characteristics are measured; every one must be measured against its specification before the report is signed.	Not assigned	8.6.2
28	Record who reviewed and approved the accepted layout inspection report for RDG-5111 (LAY-RDG-111)	The layout inspection report LAY-RDG-111 for RDG-5111 is recorded as accepted, but no reviewer or approver is recorded against it. An auditor will ask who signed it.	Not assigned	8.6.2

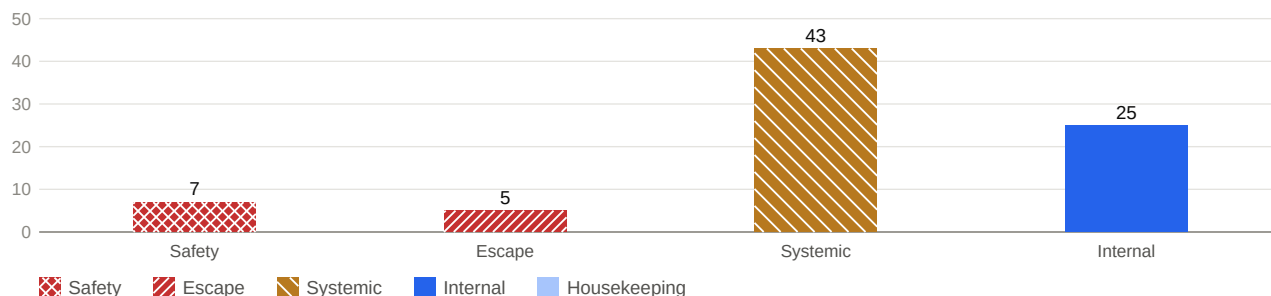
Appendix D turns these into the auditor's own questions: 69 questions across the 21 clauses where your records show a gap, with the evidence to have ready for each.

What these have in common

Shared cause	What closes it	Areas
Owner H. Lindgren (QD) (7 findings)	• Clear the outstanding containment zones on NCR-RDG-016 and establish whether suspect product shipped • Complete the disposition record for RDG-5110 (MRB-RDG-316) • Complete the disposition record for RDG-5109 (MRB-RDG-313) • Complete the disposition record for RDG-5106 (MRB-RDG-318) • Complete the disposition record for RDG-5107 (MRB-RDG-325) • Complete the disposition record for RDG-5105 (MRB-RDG-306) • Complete the disposition record for RDG-5105 (MRB-RDG-321)	Containment, nonconforming product
Owner S. Kaur (6 findings)	• Clear the outstanding containment zones on NCR-RDG-066 and establish whether suspect product shipped • Re-plan or escalate CAPA-RDG-058: past due and still open • Complete the disposition record for RDG-5107 (MRB-RDG-315) • Complete the disposition record for RDG-5103 (MRB-RDG-322) • Complete the disposition record for RDG-5104 (MRB-RDG-314) • Complete the disposition record for RDG-5115 (MRB-RDG-310)	Corrective action, containment, nonconforming product
Part RDG-5102 (5 findings)	• Complete the open PPAP elements and sign the part submission warrant for RDG-5102 (PPAP-RDG-5102) • Add the High-priority failure modes on RDG-5102 (PFMEA-RDG-5102) to the control plan, each with a control method and reaction plan • Qualify the measurement system (Gage R&R) before relying on the capability result: FSW penetration depth on RDG-5102 measured with MS-1407 • Complete the layout inspection report for RDG-5102 (LAY-RDG-102) and have it signed by a reviewer and an approver • Complete the disposition record for RDG-5102 (MRB-RDG-341)	APQP, capability, layout inspection, nonconforming product
Part RDG-5105 (6 findings)	• Add the High-priority failure modes on RDG-5105 (PFMEA-RDG-5105) to the control plan, each with a control method and reaction plan • Complete the disposition record for RDG-5105 (MRB-RDG-306) • Complete the disposition record for RDG-5105 (MRB-RDG-321) • Complete the layout inspection report for RDG-5105 (LAY-RDG-105) and have it signed by a reviewer and an approver • Complete the open PPAP elements and sign the part submission warrant for RDG-5105 (PPAP-RDG-5105) • Complete the disposition record for RDG-5105 (MRB-RDG-342)	APQP, layout inspection, nonconforming product
Supplier Hocking Brazed Components (3 findings)	• Quarantine the material in suspect product case SPR-RDG-002, trace where it went, and notify the customers whose parts share the route • Flow down CSR-RDG-02 (Ford) to Hocking Brazed Components, and record the supplier's acknowledgement • Flow down CSR-RDG-01 (Ford) to Hocking Brazed Components, and record the supplier's acknowledgement	Customer requirements, traceability
Part RDG-5104 (5 findings)	• Add the High-priority failure modes on RDG-5104 (PFMEA-RDG-5104) to the control plan, each with a control method and reaction plan • Complete the disposition record for RDG-5104 (MRB-RDG-303) • Complete the disposition record for RDG-5104 (MRB-RDG-314) • Complete the layout inspection report for RDG-5104 (LAY-RDG-104) and have it signed by a reviewer and an approver • Complete the open PPAP elements and sign the part submission warrant for RDG-5104 (PPAP-RDG-5104)	APQP, layout inspection, nonconforming product
Part RDG-5103 (4 findings)	• Add the High-priority failure modes on RDG-5103 (PFMEA-RDG-5103) to the control plan, each with a control method and reaction plan • Complete the disposition record for RDG-5103 (MRB-RDG-322) • Complete the layout inspection report for RDG-5103 (LAY-RDG-103) and have it signed by a reviewer and an approver • Complete the open PPAP elements and sign the part submission warrant for RDG-5103 (PPAP-RDG-5103)	APQP, layout inspection, nonconforming product

Shared cause	What closes it	Areas
Part RDG-5101 (5 findings)	• Add the High-priority failure modes on RDG-5101 (PFMEA-RDG-5101) to the control plan, each with a control method and reaction plan • Complete the layout inspection report for RDG-5101 (LAY-RDG-101) and have it signed by a reviewer and an approver • Complete the open PPAP elements and sign the part submission warrant for RDG-5101 (PPAP-RDG-5101) • Complete the disposition record for RDG-5101 (MRB-RDG-307) • Complete the disposition record for RDG-5101 (MRB-RDG-343)	APQP, layout inspection, nonconforming product
Part RDG-5110 (4 findings)	• Qualify the measurement system (Gage R&R) before relying on the capability result: Powder coat thickness on RDG-5110 measured with CTG-2710 • Complete the disposition record for RDG-5110 (MRB-RDG-316) • Complete the layout inspection report for RDG-5110 (LAY-RDG-110) and have it signed by a reviewer and an approver • Complete the disposition record for RDG-5110 (MRB-RDG-311)	Capability, layout inspection, nonconforming product
Part RDG-5106 (4 findings)	• Qualify the measurement system (Gage R&R) before relying on the capability result: Conductivity on RDG-5106 measured with ECM-2911 • Complete the disposition record for RDG-5106 (MRB-RDG-318) • Obtain the customer's concession for RDG-5106 (MRB-RDG-308) before the product is used • Record who reviewed and approved the accepted layout inspection report for RDG-5106 (LAY-RDG-106)	Capability, layout inspection, nonconforming product
Part RDG-5107 (3 findings)	• Complete the disposition record for RDG-5107 (MRB-RDG-325) • Complete the disposition record for RDG-5107 (MRB-RDG-315) • Record who reviewed and approved the accepted layout inspection report for RDG-5107 (LAY-RDG-107)	Layout inspection, nonconforming product
Part RDG-5109 (3 findings)	• Complete the disposition record for RDG-5109 (MRB-RDG-313) • Record who reviewed and approved the accepted layout inspection report for RDG-5109 (LAY-RDG-109) • Complete the disposition record for RDG-5109 (MRB-RDG-312)	Layout inspection, nonconforming product
Part RDG-5108 (2 findings)	• Complete the disposition record for RDG-5108 (MRB-RDG-305) • Complete the layout inspection report for RDG-5108 (LAY-RDG-108) and have it signed by a reviewer and an approver	Layout inspection, nonconforming product
Owner N. Oyelaran (2 findings)	• Re-plan or escalate CAPA-RDG-057: past due and still open • Complete the disposition record for RDG-5101 (MRB-RDG-307)	Corrective action, nonconforming product
Owner P. Ionescu (2 findings)	• Re-plan or escalate CAPA-RDG-060: past due and still open • Complete the disposition record for RDG-5109 (MRB-RDG-312)	Corrective action, nonconforming product
Part RDG-5112 (2 findings)	• Complete the layout inspection report for RDG-5112 (LAY-RDG-112) and have it signed by a reviewer and an approver • Complete the disposition record for RDG-5112 (MRB-RDG-309)	Layout inspection, nonconforming product
Part RDG-5115 (2 findings)	• Complete the layout inspection report for RDG-5115 (LAY-RDG-115) and have it signed by a reviewer and an approver • Complete the disposition record for RDG-5115 (MRB-RDG-310)	Layout inspection, nonconforming product

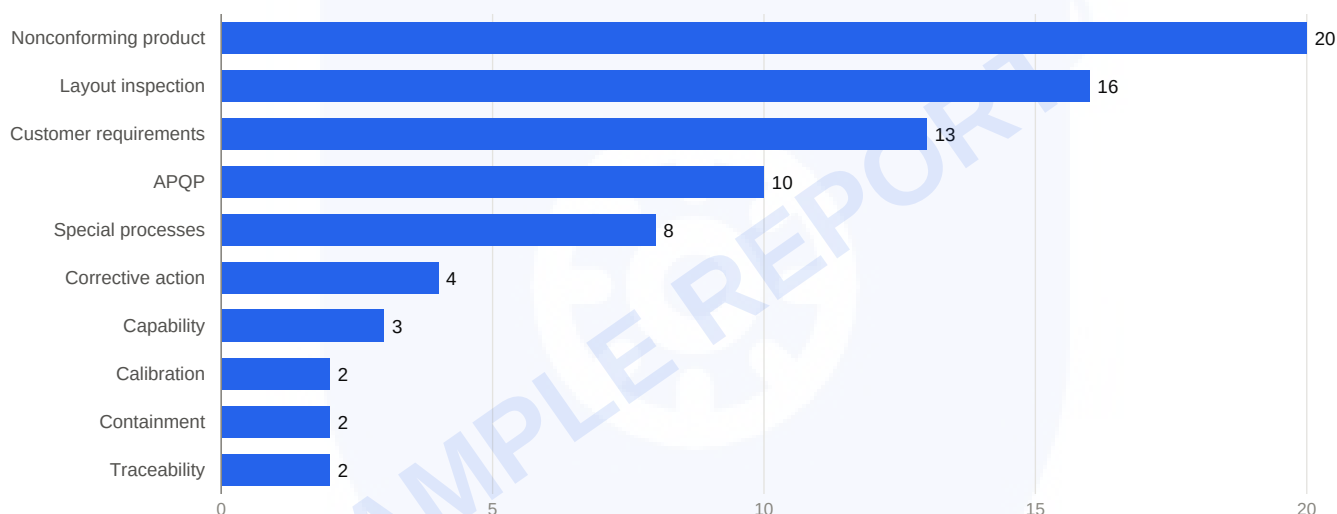
Issues ranked, by severity



Issues ranked, by severity: the ranked findings this section lists, by the worst consequence each could have.

80 ranked issues: 7 safety, 5 escape, 43 systemic, 25 internal.

Issues ranked, by area



Issues ranked, by area of the quality system, most first.

By area: Nonconforming product 20, Layout inspection 16, Customer requirements 13, APQP 10, Special processes 8, Corrective action 4, Capability 3, Calibration 2, Containment 2, Traceability 2.

Every figure and citation is drawn from your records; the priority order is Aegis's assessment.

Where the system is drifting

19 measuring equipment past its calibration due date — the worst is 143 days past due (clause 7.1.5.2).

#	Open lapse	Why it leads	Severity	Clause
1	Measuring equipment past its calibration due date	19 records against clause 7.1.5.2; the worst is 143 days past due	Critical	7.1.5.2
2	Nonconformances past their due date	22 records against clause 8.7; the worst is 180 days past due	High	8.7
3	Corrective actions past their due date	4 records against clause 10.2; the worst is 70 days past due	High	10.2
4	Assigned training past its due date	8 records against clause 7.2; the worst is 99 days past due	Medium	7.2

#	Open lapse	Why it leads	Severity	Clause
5	Nonconforming product not contained within the required period	2 records against clause 8.7.1; the worst is 0 days past due	Critical	8.7.1
6	Documents waiting on approval	11 records against clause 7.5.2; how long each has waited is not recorded	Low	7.5.2



2. Audit Scope & QMS Context

Scope Statement: Manufacture to customer design of friction stir welded, brazed, riveted and bonded aluminium battery enclosures and structures for Ford, BMW and Mercedes-Benz

Process landscape

The pack follows IATF 16949:2016: its sections, their clauses, and the automotive clauses the evidence sections of this pack are filed under.

Clause 4: Context of the organization

- 4.1 Understanding the organization and its context
- 4.2 Understanding the needs and expectations of interested parties
- 4.3 Determining the scope of the quality management system
- 4.3.2 Customer-specific requirements
- 4.4 Quality management system and its processes

Clause 6: Planning

- 6.1 Actions to address risks and opportunities
- 6.2 Quality objectives and planning to achieve them
- 6.3 Planning of changes

Clause 8: Operation

- 8.1 Operational planning and control
- 8.2 Requirements for products and services
- 8.3 Design and development of products and services
- 8.3.4.4 Product approval process
- 8.3.5.2 Manufacturing process design output
- 8.4 Control of externally provided processes, products and services
- 8.5 Production and service provision
- 8.5.1 Control of production and service provision
- 8.5.1.1 Control plan
- 8.5.2.1 Identification and traceability — supplemental
- 8.6 Release of products and services
- 8.6.2 Layout inspection and functional testing
- 8.7 Control of nonconforming outputs
- 8.7.1.3 Control of suspect product
- 8.7.1.7 Nonconforming product disposition

Clause 10: Improvement

- 10.1 Improvement: general
- 10.2 Nonconformity and corrective action
- 10.2.3 Problem solving
- 10.3 Continual improvement

Clause 5: Leadership

- 5.1 Leadership and commitment
- 5.2 Policy
- 5.3 Organizational roles, responsibilities and authorities

Clause 7: Support

- 7.1 Resources
- 7.1.5.1.1 Measurement systems analysis
- 7.1.5.2.1 Calibration/verification records
- 7.2 Competence
- 7.3 Awareness
- 7.4 Communication
- 7.5 Documented information

Clause 9: Performance evaluation

- 9.1 Monitoring, measurement, analysis and evaluation
- 9.1.1 Monitoring and measurement: general
- 9.1.1.1 Monitoring and measurement of manufacturing processes
- 9.2 Internal audit
- 9.3 Management review

3. IATF 16949 Clause Compliance Posture

Aegis readiness score: a proprietary assessment of the documents, evidence and open findings recorded against each IATF 16949:2016 clause. Not an audit result, and not a certification decision. Each clause is shown as Conforming, Minor gaps or Major risk, with what is on record against it and what would close each gap.

Clause	Title	Score	Status	Sub-clauses met	With evidence	Lowest gap
4	Context of the organization	33	Major risk	0 of 4	4 of 4	4.2 Understanding the needs and expectations of interested parties: scored 26
5	Leadership	38	Major risk	0 of 3	3 of 3	5.3 Organizational roles, responsibilities and authorities: scored 26.1
6	Planning	22.9	Major risk	0 of 3	2 of 3	6.3 Planning of changes: scored 0
7	Support	12.5	Major risk	0 of 5	5 of 5	7.5 Documented information: scored 5.5
8	Operation	16.1	Major risk	0 of 7	7 of 7	8.4 Control of externally provided processes, products and services: scored 0
9	Performance evaluation	21.6	Major risk	0 of 3	3 of 3	9.2 Internal audit: scored 17.5
10	Improvement	20.5	Major risk	0 of 3	3 of 3	10.2 Nonconformity and corrective action: scored 0

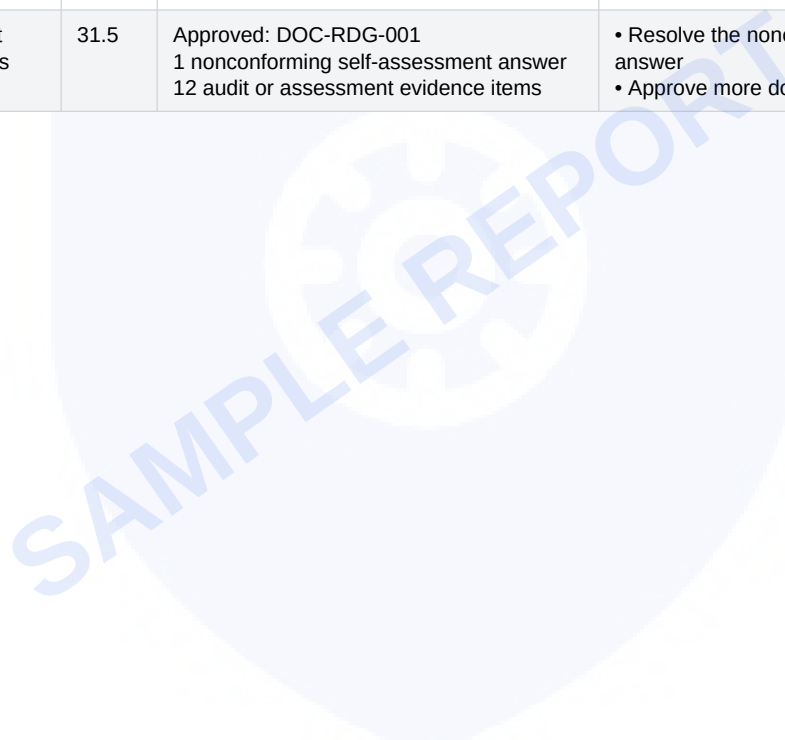
"With evidence" counts the sub-clauses that have at least one approved document or evidence record; it is not a measure of whether that evidence is sufficient.



Clause 4: Context of the organization

Major risk, scored 33. Sub-clauses met: 0 of 4.

Sub-clause	Score	On record	To close it
4.1 Understanding the organization and its context	49	Approved: DOC-RDG-013 8 audit or assessment evidence items	• Approve more documentation for 4.1 • Record more audit or assessment evidence against 4.1
4.2 Understanding the needs and expectations of interested parties	26	No approved document 2 audit or assessment evidence items	• Issue an approved document for 4.2 (none on record) • Record more audit or assessment evidence against 4.2
4.3 Determining the scope of the quality management system	26.1	Approved: DOC-RDG-027 1 nonconforming self-assessment answer 8 audit or assessment evidence items	• Resolve the nonconforming self-assessment answer • Approve more documentation for 4.3 • Record more audit or assessment evidence against 4.3
4.4 Quality management system and its processes	31.5	Approved: DOC-RDG-001 1 nonconforming self-assessment answer 12 audit or assessment evidence items	• Resolve the nonconforming self-assessment answer • Approve more documentation for 4.4



Clause 5: Leadership

Major risk, scored 38. Sub-clauses met: 0 of 3.

Sub-clause	Score	On record	To close it
5.1 Leadership and commitment	50	No approved document 18 audit or assessment evidence items	• Issue an approved document for 5.1 (none on record)
5.2 Policy	32	No approved document 4 audit or assessment evidence items	• Issue an approved document for 5.2 (none on record) • Record more audit or assessment evidence against 5.2
5.3 Organizational roles, responsibilities and authorities	26.1	Approved: DOC-RDG-014 1 nonconforming self-assessment answer 8 audit or assessment evidence items	• Resolve the nonconforming self-assessment answer • Approve more documentation for 5.3 • Record more audit or assessment evidence against 5.3



Clause 6: Planning

Major risk, scored 22.9. Sub-clauses met: 0 of 3.

Sub-clause	Score	On record	To close it
6.1 Actions to address risks and opportunities	24	No approved document Awaiting approval: DOC-RDG-015 2 nonconforming self-assessment answers 16 audit or assessment evidence items	• Resolve the 2 nonconforming self-assessment answers • Approve DOC-RDG-015 (awaiting approval) • Approve more documentation for 6.1
6.2 Quality objectives and planning to achieve them	44	No approved document 8 audit or assessment evidence items	• Issue an approved document for 6.2 (none on record) • Record more audit or assessment evidence against 6.2
6.3 Planning of changes	0	No approved document 1 nonconforming self-assessment answer No audit or assessment evidence	• Resolve the nonconforming self-assessment answer • Issue an approved document for 6.3 (none on record) • Record audit or assessment evidence against 6.3 (none on file)



Clause 7: Support

Major risk, scored 12.5. Sub-clauses met: 0 of 5.

Sub-clause	Score	On record	To close it
7.1 Resources	10.5	Approved: DOC-RDG-034 Awaiting approval: DOC-RDG-006, DOC-RDG-016 Open finding: FIND-RDG-004 6 nonconforming self-assessment answers 20 audit or assessment evidence items	• Close FIND-RDG-004 • Resolve the 6 nonconforming self-assessment answers • Approve DOC-RDG-006, DOC-RDG-016 (awaiting approval) • Approve more documentation for 7.1
7.2 Competence	7.2	No approved document Awaiting approval: DOC-RDG-017 Open finding: FIND-RDG-006 5 nonconforming self-assessment answers 6 audit or assessment evidence items	• Close FIND-RDG-006 • Resolve the 5 nonconforming self-assessment answers • Approve DOC-RDG-017 (awaiting approval) • Approve more documentation for 7.2 • Record more audit or assessment evidence against 7.2
7.3 Awareness	14.4	No approved document 2 nonconforming self-assessment answers 6 audit or assessment evidence items	• Resolve the 2 nonconforming self-assessment answers • Issue an approved document for 7.3 (none on record) • Record more audit or assessment evidence against 7.3
7.4 Communication	32	No approved document 4 audit or assessment evidence items	• Issue an approved document for 7.4 (none on record) • Record more audit or assessment evidence against 7.4
7.5 Documented information	5.5	Approved: DOC-RDG-018 5 nonconforming self-assessment answers 2 audit or assessment evidence items	• Resolve the 5 nonconforming self-assessment answers • Approve more documentation for 7.5 • Record more audit or assessment evidence against 7.5

Clause 8: Operation

Major risk, scored 16.1. Sub-clauses met: 0 of 7.

Sub-clause	Score	On record	To close it
8.1 Operational planning and control	44	No approved document 8 audit or assessment evidence items	• Issue an approved document for 8.1 (none on record) • Record more audit or assessment evidence against 8.1
8.2 Requirements for products and services	50	No approved document 16 audit or assessment evidence items	• Issue an approved document for 8.2 (none on record)
8.3 Design and development of products and services	3.5	Approved: DOC-RDG-010 Awaiting approval: DOC-RDG-028 Open findings: FIND-RDG-009 and FIND-RDG-010 7 nonconforming self-assessment answers 30 audit or assessment evidence items	• Close FIND-RDG-009 and FIND-RDG-010 • Resolve the 7 nonconforming self-assessment answers • Approve DOC-RDG-028 (awaiting approval) • Approve more documentation for 8.3
8.4 Control of externally provided processes, products and services	0	Approved: DOC-RDG-019, DOC-RDG-033 Open findings: FIND-RDG-011, NCR-RDG-020 and NCR-RDG-066 9 nonconforming self-assessment answers 28 audit or assessment evidence items	• Close FIND-RDG-011, NCR-RDG-020 and NCR-RDG-066 • Resolve the 9 nonconforming self-assessment answers • Approve more documentation for 8.4
8.5 Production and service provision	0	Approved: DOC-RDG-005, DOC-RDG-020, DOC-RDG-021, DOC-RDG-029, DOC-RDG-030, DOC-RDG-031, DOC-RDG-032 Awaiting approval: DOC-RDG-002, DOC-RDG-009 Open findings: FIND-RDG-003, FIND-RDG-001, FIND-RDG-002, FIND-RDG-007, FIND-RDG-005 and 9 nonconformances 14 nonconforming self-assessment answers 34 audit or assessment evidence items	• Close FIND-RDG-003, FIND-RDG-001, FIND-RDG-002, FIND-RDG-007, FIND-RDG-005 and the 9 open nonconformances • Resolve the 14 nonconforming self-assessment answers • Approve DOC-RDG-002, DOC-RDG-009 (awaiting approval) • Approve more documentation for 8.5
8.6 Release of products and services	24	No approved document Awaiting approval: DOC-RDG-003, DOC-RDG-035 Open finding: NCR-RDG-018 1 nonconforming self-assessment answer 24 audit or assessment evidence items	• Close NCR-RDG-018 • Resolve the nonconforming self-assessment answer • Approve DOC-RDG-003, DOC-RDG-035 (awaiting approval) • Approve more documentation for 8.6
8.7 Control of nonconforming outputs	0	Approved: DOC-RDG-007, DOC-RDG-011, DOC-RDG-022, DOC-RDG-023 Open findings: 16 nonconformances 2 nonconforming self-assessment answers 20 audit or assessment evidence items	• Close the 16 open nonconformances • Resolve the 2 nonconforming self-assessment answers • Approve more documentation for 8.7

Clause 9: Performance evaluation

Major risk, scored 21.6. Sub-clauses met: 0 of 3.

Sub-clause	Score	On record	To close it
9.1 Monitoring, measurement, analysis and evaluation	27	No approved document Awaiting approval: DOC-RDG-012 1 nonconforming self-assessment answer 22 audit or assessment evidence items	• Resolve the nonconforming self-assessment answer • Approve DOC-RDG-012 (awaiting approval) • Approve more documentation for 9.1
9.2 Internal audit	17.5	Approved: DOC-RDG-024 Open finding: FIND-RDG-008 4 nonconforming self-assessment answers 10 audit or assessment evidence items	• Close FIND-RDG-008 • Resolve the 4 nonconforming self-assessment answers • Approve more documentation for 9.2
9.3 Management review	20.4	Approved: DOC-RDG-025, DOC-RDG-036 Open finding: FIND-RDG-012 3 nonconforming self-assessment answers 8 audit or assessment evidence items	• Close FIND-RDG-012 • Resolve the 3 nonconforming self-assessment answers • Approve more documentation for 9.3 • Record more audit or assessment evidence against 9.3



Clause 10: Improvement

Major risk, scored 20.5. Sub-clauses met: 0 of 3.

Sub-clause	Score	On record	To close it
10.1 Improvement: general	32	No approved document 4 audit or assessment evidence items	• Issue an approved document for 10.1 (none on record) • Record more audit or assessment evidence against 10.1
10.2 Nonconformity and corrective action	0	No approved document Awaiting approval: DOC-RDG-026 Open findings: 12 nonconformances 1 nonconforming self-assessment answer 22 audit or assessment evidence items	• Close the 12 open nonconformances • Resolve the nonconforming self-assessment answer • Approve DOC-RDG-026 (awaiting approval) • Approve more documentation for 10.2
10.3 Continual improvement	50	No approved document 12 audit or assessment evidence items	• Issue an approved document for 10.3 (none on record)



4. Evidence Index

Each controlled document and audit evidence record, with the IATF 16949 clause it supports. Documents that are awaiting approval, in draft or withdrawn are listed because an auditor will ask about them; only approved documents are treated as evidence of a clause.

Clause	Reference	Evidence	Type	Status	Owner
4.1	DOC-RDG-013	QP-4.1 Context and Interested Parties	Procedure	Approved	C. Ibsen
4.3.2	DOC-RDG-027	QP-4.3.2 Customer-Specific Requirements Matrix	Procedure	Approved	T. Marchetti
4.4	DOC-RDG-001	QMS-001 Quality Management System Manual	Procedure	Approved	H. Lindgren (QD)
5.3	DOC-RDG-014	QP-5.3 Roles and Authorities	Procedure	Approved	J. Pemberton
6.1	DOC-RDG-015	QP-6.1 Risk and Contingency Planning	Procedure	Under review	D. Morrow
7.1.5	DOC-RDG-016	QP-7.1.5 Monitoring and Measuring Resources	Procedure	Under review	S. Kaur
7.1.5	DOC-RDG-034	FRM-310 Calibration Recall Notice	Form	Approved	F. Calloway
7.1.5.1.1	DOC-RDG-006	CAL-510 Calibration and Measurement Systems Control	Procedure	Under review	F. Calloway
7.2	DOC-RDG-008	TRN-710 Operator Qualification and Records	Procedure	Obsolete	M. Whitlock
7.2	DOC-RDG-017	QP-7.2 Competence	Procedure	Under review	C. Ibsen
7.5	DOC-RDG-018	QP-7.5 Control of Documented Information	Procedure	Approved	J. Pemberton
8.3.4.4	DOC-RDG-010	PPAP-910 Production Part Approval and Run-at-Rate	Procedure	Approved	D. Morrow
8.3.4.4	DOC-RDG-028	QP-8.3.4.4 Production Part Approval (PPAP)	Procedure	Under review	D. Morrow
8.4	DOC-RDG-019	QP-8.4 Control of External Providers	Procedure	Approved	E. Vasko
8.4	DOC-RDG-033	FRM-205 Supplier Scorecard	Form	Approved	D. Morrow
8.5.1	DOC-RDG-004	HT-310 Solution Heat Treat and Aging per CQI-9	Procedure	Obsolete	C. Ibsen
8.5.1	DOC-RDG-029	WI-FSW-115 FSW Macro-Etch Sampling	Work instruction	Approved	P. Ionescu
8.5.1	DOC-RDG-030	WI-PC-330 Powder Coat Oven Profiling	Work instruction	Approved	T. Marchetti
8.5.1	DOC-RDG-031	WI-SPR-140 Rivet Joint Cross-Section	Work instruction	Approved	T. Marchetti
8.5.1	DOC-RDG-032	WI-HT-315 Aging Oven Load Thermocouples	Work instruction	Approved	J. Pemberton
8.5.1.6	DOC-RDG-002	FSW-110 Friction Stir Welding Set-Up and Tool Life	Work instruction	Under review	T. Marchetti
8.5.2	DOC-RDG-009	TRC-810 Lot Genealogy and Traceability	Procedure	Under review	J. Pemberton
8.5.2	DOC-RDG-020	QP-8.5.2 Identification and Traceability	Procedure	Approved	F. Calloway
8.5.4.1	DOC-RDG-005	ADH-410 Structural Adhesive Handling and Shelf Life	Work instruction	Approved	N. Oyelaran
8.5.6	DOC-RDG-021	QP-8.5.6 Control of Changes	Procedure	Approved	T. Marchetti

Clause	Reference	Evidence	Type	Status	Owner
8.6	DOC-RDG-003	LT-210 Helium Leak Test Operation and Daily Verification	Work instruction	Under review	N. Oyelaran
8.6.2	DOC-RDG-035	FRM-415 Layout Inspection Report	Form	Under review	F. Calloway
8.7	DOC-RDG-011	FRM-120 Nonconformance Report Form	Form	Approved	H. Lindgren (QD)
8.7	DOC-RDG-022	QP-8.7 Control of Nonconforming Output	Procedure	Approved	N. Oyelaran
8.7.1.3	DOC-RDG-023	QP-8.7.1.3 Control of Suspect Product	Procedure	Approved	D. Morrow
8.7.1.7	DOC-RDG-007	NCP-610 Nonconforming Product and MRB	Procedure	Approved	H. Lindgren (QD)
9.1	DOC-RDG-012	RPT-220 Weekly Quality and Launch Report	Record	Under review	Recovery Lead
9.2	DOC-RDG-024	QP-9.2 Internal Audit and Layered Process Audits	Procedure	Approved	N. Oyelaran
9.3	DOC-RDG-025	QP-9.3 Management Review	Procedure	Approved	T. Marchetti
9.3	DOC-RDG-036	RPT-225 Management Review Minutes	Record	Approved	J. Pemberton
10.2	DOC-RDG-026	QP-10.2 Nonconformity and Corrective Action	Procedure	Under review	J. Pemberton

Proof on File for Closed Records

The documents behind your records stay in your systems; the registers say where each is kept. Of 118 closed or held records, 108 name where the proof is kept (CAPA 42 of 44; Layout inspection 3 of 4; MRB 21 of 24; Training 42 of 46). An auditor samples closed records and asks to see that proof, so before the audit pull a sample of the referenced documents and confirm each exists and says what the register records.

10 records closed or held with nowhere named for the proof. Find each before the audit and add its reference; the same list is 12_Evidence_Requests in the pack.

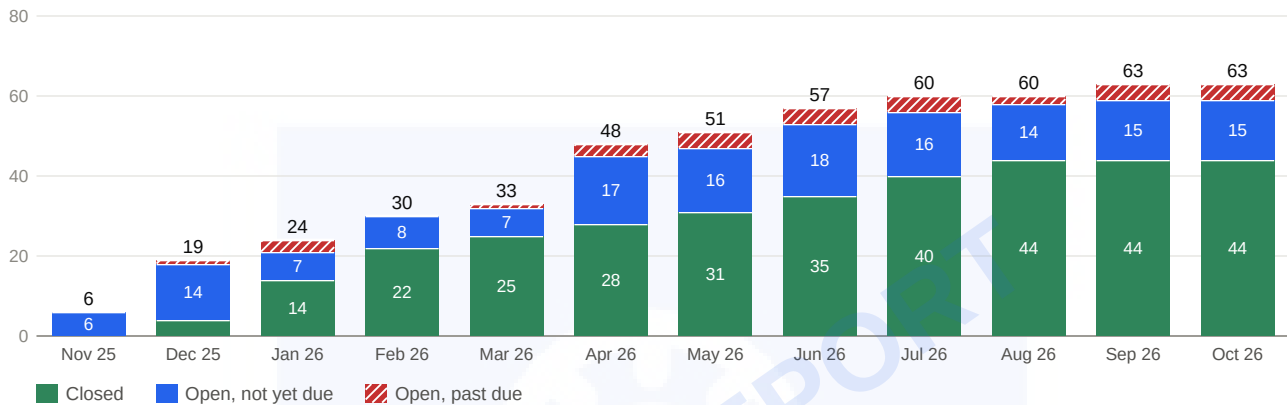
Register	Clause	Record	What	State	Owner
CAPA	10.2.3	CAPA-RDG-016	Conductivity below temper spec — RDG-5103	Closed	E. Vasko
CAPA	10.2.3	CAPA-RDG-029	Helium leak above spec — RDG-5111	Closed	P. Ionescu
Layout inspection	8.6.2	LAY-RDG-109	RDG-5109 Module Mounting Bracket	Approved	Not recorded
MRB	8.7.1.7	MRB-RDG-304	Side sill extrusion — twist 0.9 mm/m against 0.5 mm/m	Closed	S. Kaur
MRB	8.7.1.7	MRB-RDG-317	Adhesive bead break — RDG-5107	Closed	H. Lindgren (QD)
MRB	8.7.1.7	MRB-RDG-323	Rivet head height out — RDG-5108	Closed	H. Lindgren (QD)
Training	7.2	RDG-E008	Sofia Marino — Material Cert and Temper Verification	Held	Sofia Marino
Training	7.2	RDG-E024	Omar Pereira — Laser Scan and CMM Inspection	Held	Omar Pereira
Training	7.2	RDG-E042	Lena Nwosu — GD&T per ASME Y14.5	Held	Lena Nwosu
Training	7.2	RDG-E058	Isaiah Ekwueme — FSW Operator Qualification per ISO 25239-3	Held	Isaiah Ekwueme

5. Open Issues Package

Every open corrective action, audit finding and risk, with who holds it and when it is due. An item with no owner is shown as not assigned rather than left blank.

5.1 Open CAPAs

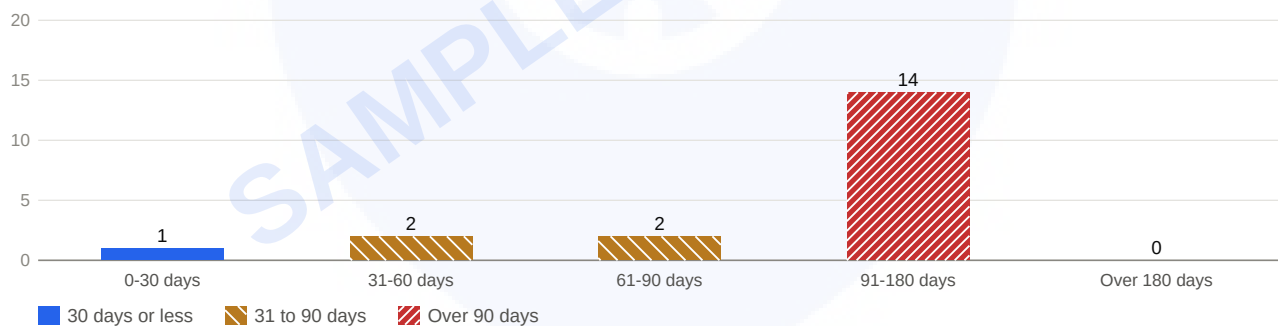
CAPAs month by month



The CAPA register at each month end, from its opened, closed and due dates; the last month ends on the report date.

CAPAs at Oct 26 (report date): 44 closed, 15 open and not yet due, 4 past due; at Nov 25: 0 closed, 6 open, 0 past due.

Open CAPAs by age



Open CAPAs by days since opened.

19 open CAPAs by age: 1 0-30 days, 2 31-60 days, 2 61-90 days, 14 91-180 days, 0 Over 180 days.

19 open, 4 past their due date.

CAPA	NCR	Title	Root cause	Effectiveness	Status	Owner	Due	Days past due
CAPA-RDG-057	NCR-RDG-067	Tunnel void in FSW seam — RDG-5105	Tool shoulder worn past its life with no tool life count on the cell	Pending — recurrence check due 2026-10-08	Effectiveness review	N. Oyelaran	2026-07-30	70

CAPA	NCR	Title	Root cause	Effectiveness	Status	Owner	Due	Days past due
CAPA-RDG-058	NCR-RDG-039	Hole position out after welding — RDG-5113	Weld distortion not compensated in the CNC datum	Pending — action in implementation, past its due date	Implementation	S. Kaur	2026-08-15	54
CAPA-RDG-059	NCR-RDG-004	CMT weld porosity — RDG-5116	Shielding gas flow low after a regulator change	Pending — action in implementation, past its due date	Implementation	C. Ibsen	2026-09-19	19
CAPA-RDG-060	NCR-RDG-040	Rivet head height out — RDG-5114	Die worn past its life; die changes not counted	Pending — action in implementation, past its due date	Implementation	P. Ionescu	2026-09-29	9
CAPA-RDG-006	NCR-RDG-206	Helium reference leak certificate expired — leak detectors unverified since November	Not recorded	Pending — replacement reference leak on order	Implementation	F. Calloway	2026-10-14	-
CAPA-RDG-061	NCR-RDG-005	Rivet head height out — RDG-5104	Die worn past its life; die changes not counted	Pending — root cause found, actions being planned	Action planning	E. Vasko	2026-10-16	-
CAPA-RDG-005	NCR-RDG-205	Aging oven TUS failure at location 7 and solution furnace SAT overdue	Not recorded	Pending — aging oven derated to 8 of 9 zones	Implementation	C. Ibsen	2026-10-21	-
CAPA-RDG-011	NCR-RDG-211	FSW tool source changed without customer notification	Not recorded	Pending — change request to Ford not submitted	Action planning	P. Ionescu	2026-10-25	-
CAPA-RDG-003	NCR-RDG-203	Leak test results and lot genealogy for three weeks of July 2025 lost in the ERP cutover	Not recorded	Pending — record reconstruction started	Implementation	J. Pemberton	2026-10-28	-
CAPA-RDG-010	NCR-RDG-210	Structural adhesive used past shelf life on the tray line — two drums	Not recorded	Pending — FIFO labels added, not verified	Verification	N. Oyelaran	2026-11-01	-
CAPA-RDG-062	NCR-RDG-057	Flatness out of tolerance — RDG-5109	Not recorded	Pending — root cause not validated	Root cause	J. Pemberton	2026-11-02	-

CAPA	NCR	Title	Root cause	Effectiveness	Status	Owner	Due	Days past due
CAPA-RDG-001	NCR-RDG-201	Coolant leak escapes on RDG-5105 cooling plates — 212 PPM at the Ford pack line	Not recorded	Pending — containment (100% helium re-test) only	Initiated	S. Kaur	2026-11-04	-
CAPA-RDG-007	NCR-RDG-207	FSW operators producing without qualification to ISO 25239-3	Not recorded	Pending — qualification test plan drafted	Action planning	M. Whitlock	2026-11-04	-
CAPA-RDG-063	NCR-RDG-028	Missing leak test record — RDG-5108	Leak tester results held locally and not transferred to the ERP	Pending — root cause found, actions being planned	Action planning	P. Ionescu	2026-11-06	-
CAPA-RDG-004	NCR-RDG-204	BMW battery lid flatness out of tolerance — RDG-5104 rev C	Not recorded	Pending — fixture clamp sequence under study	Root cause	E. Vasko	2026-11-08	-
CAPA-RDG-002	NCR-RDG-202	Tunnel voids in RDG-5102 floor panel FSW seams — found at Ford teardown	Not recorded	Pending — tool shoulder wear suspected, not verified	Root cause	T. Marchetti	2026-11-11	-
CAPA-RDG-009	NCR-RDG-209	Extrusion twist out of tolerance from Cuyahoga Extrusions — four lots	Not recorded	Pending — supplier 8D requested	Root cause	S. Kaur	2026-11-15	-
CAPA-RDG-008	NCR-RDG-208	Mercedes-Benz run-at-rate failed on RDG-5108 — FSW cycle time and hole position capability	Not recorded	Pending — second run-at-rate not scheduled	Action planning	D. Morrow	2026-11-18	-
CAPA-RDG-012	NCR-RDG-212	Quality leadership split across two plants — management review not held in Q3 or Q4	Not recorded	Pending — review cadence re-established	Implementation	H. Lindgren (QD)	2026-11-25	-

5.2 Open Audit Findings

Finding	Description	Severity	Clause	Status	Assigned to	Due
FIND-RDG-001	Coolant leak escapes at 212 PPM — the leak test reaction plan was not followed	Major	IATF 16949 8.5.1.1	Open	S. Kaur	2026-11-04
FIND-RDG-002	Tunnel voids in FSW floor panels — macro-etch frequency below the control plan	Major	IATF 16949 8.5.1	Open	T. Marchetti	2026-11-11
FIND-RDG-003	Leak test results and genealogy for three weeks of July 2025 not migrated — suspect lots cannot be bounded	Major	IATF 16949 8.5.2.1	Open	J. Pemberton	2026-10-28
FIND-RDG-004	Helium reference leak and 18 other instruments past calibration	Major	IATF 16949 7.1.5.2.1	Open	F. Calloway	2026-10-14
FIND-RDG-005	Aging oven TUS failure at location 7 and SAT overdue on Furnace 1 (CQI-9 Section 3)	Major	IATF 16949 8.5.1	Open	C. Ibsen	2026-10-21
FIND-RDG-006	FSW operators producing without qualification to ISO 25239-3	Major	IATF 16949 7.2.2	Open	M. Whitlock	2026-11-04
FIND-RDG-007	FSW tool source changed without customer notification or PPAP	Minor	IATF 16949 8.5.6.1	Open	P. Ionescu	2026-10-25
FIND-RDG-008	Manufacturing process audits on FSW and leak test not performed since the ERP cutover	Minor	IATF 16949 9.2.2.3	Open	H. Lindgren (QD)	2026-11-25
FIND-RDG-009	Lid PPAP rejected — flatness not capable (Cpk 0.71)	Minor	IATF 16949 8.3.4.4	Open	E. Vasko	2026-11-08
FIND-RDG-010	Run-at-rate failed on RDG-5108 — cycle time and hole position capability	Minor	IATF 16949 8.3.4.4	Open	D. Morrow	2026-11-18
FIND-RDG-011	Cuyahoga Extrusions twist trend not escalated across 4 lots	Minor	IATF 16949 8.4.2.4	Open	S. Kaur	2026-11-15
FIND-RDG-012	Management review not held in Q3 or Q4 2025	Minor	IATF 16949 9.3.1	Open	H. Lindgren (QD)	2026-11-25

5.3 Open Risks

Risk	Description	Score	Status	Owner	Due	Mitigation
RISK-RDG-01	IATF 16949 recertification audit finds open majors — certificate not renewed	25	Identified	H. Lindgren (QD)	2026-12-09	Readiness plan: close or evidence every major finding before the audit
RISK-RDG-02	Ford escalates controlled shipping from CS1 to CS2 and places a new business hold	20	Identified	S. Kaur	2026-11-11	Daily leak escape review with the Ford STA; third-party CS1 inspection verified
RISK-RDG-07	Leak test and inspection results suspect — 19 instruments past calibration, reference leak expired	20	Identified	F. Calloway	2026-10-14	Borrow a certified reference leak; verify detectors every shift
RISK-RDG-04	BMW lid PPAP not approved before SOP — launch delay and penalties	16	Identified	E. Vasko	2026-11-15	Fixture clamp study and interim 100% laser scan

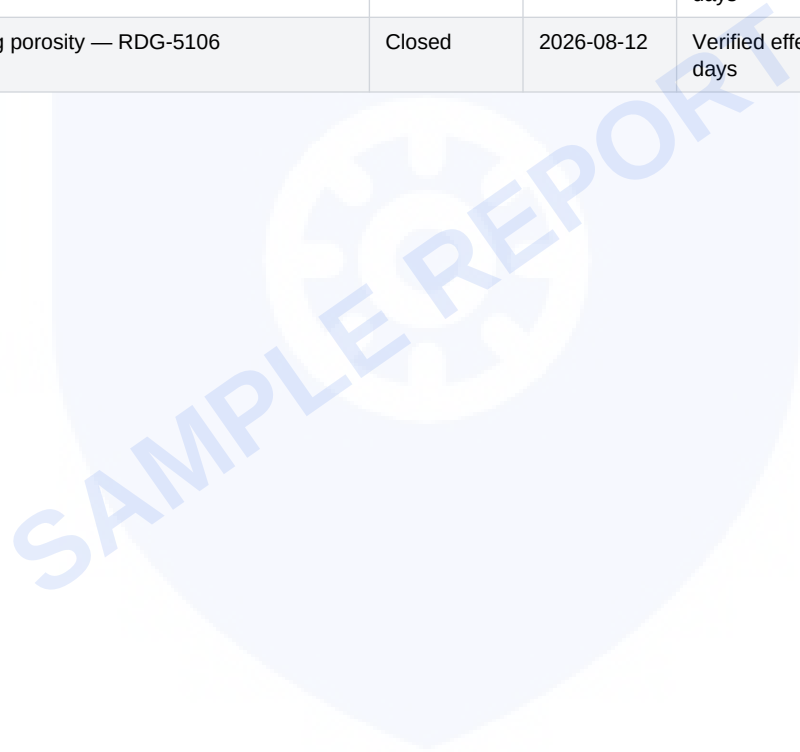
Risk	Description	Score	Status	Owner	Due	Mitigation
RISK-RDG-05	ERP cutover gap — suspect lots from July 2025 cannot be bounded	16	Identified	J. Pemberton	2026-10-28	Rebuild genealogy from the leak testers' local logs and shipping records
RISK-RDG-06	Cuyahoga Extrusions capacity and twist problems starve the tray line	16	Identified	S. Kaur	2026-12-02	Qualify a second extrusion source; weekly supplier review
RISK-RDG-03	Coolant leak in the field — pack-level warranty and recall exposure	15	Mitigating	S. Kaur	2026-11-04	Bound suspect cooling plates by braze load; 100% helium re-test
RISK-RDG-08	FSW tool wear uncontrolled — voids recur on floor panels	12	Identified	T. Marchetti	2026-11-11	Tool life counter and shoulder inspection every shift
RISK-RDG-09	Mercedes-Benz launch slips after the failed run-at-rate	12	Identified	D. Morrow	2026-11-18	Second run-at-rate with a cycle-time recovery plan
RISK-RDG-10	One qualified heat treat operator on nights	9	Identified	C. Ibsen	2026-11-25	Cross-train two operators to CQI-9 before the audit

5.4 Closed CAPAs (effectiveness checks)

CAPA	Title	Status	Closed	Effectiveness criteria
CAPA-RDG-013	Hole position out after welding — RDG-5115	Closed	2026-01-20	Verified effective — no recurrence in 45 days
CAPA-RDG-014	Adhesive bead break — RDG-5102	Closed	2025-12-25	Verified effective — no recurrence in 49 days
CAPA-RDG-015	CMT weld porosity — RDG-5107	Closed	2025-12-10	Verified effective — no recurrence in 42 days
CAPA-RDG-016	Conductivity below temper spec — RDG-5103	Closed	2025-12-31	Verified effective — no recurrence in 48 days
CAPA-RDG-017	Helium leak above spec — RDG-5110	Closed	2026-02-08	Verified effective — no recurrence in 34 days
CAPA-RDG-018	Rivet head height out — RDG-5108	Closed	2026-01-22	Verified effective — no recurrence in 45 days
CAPA-RDG-019	Missing leak test record — RDG-5103	Closed	2025-12-21	Verified effective — no recurrence in 46 days
CAPA-RDG-020	Extrusion twist over limit — RDG-5106	Closed	2026-01-18	Verified effective — no recurrence in 43 days
CAPA-RDG-021	CMT weld porosity — RDG-5110	Closed	2026-01-03	Verified effective — no recurrence in 38 days
CAPA-RDG-022	Adhesive bead break — RDG-5105	Closed	2026-01-17	Verified effective — no recurrence in 50 days
CAPA-RDG-023	Powder coat thickness low — RDG-5109	Closed	2026-02-24	Verified effective — no recurrence in 38 days
CAPA-RDG-024	CMT weld porosity — RDG-5114	Closed	2026-01-24	Verified effective — no recurrence in 32 days
CAPA-RDG-025	Flatness out of tolerance — RDG-5108	Closed	2026-01-25	Verified effective — no recurrence in 46 days

CAPA	Title	Status	Closed	Effectiveness criteria
CAPA-RDG-026	Missing leak test record — RDG-5110	Closed	2026-02-16	Verified effective — no recurrence in 35 days
CAPA-RDG-027	Casting porosity — RDG-5112	Closed	2026-02-23	Verified effective — no recurrence in 48 days
CAPA-RDG-028	Adhesive bead break — RDG-5107	Closed	2026-01-24	Verified effective — no recurrence in 42 days
CAPA-RDG-029	Helium leak above spec — RDG-5111	Closed	2026-02-10	Verified effective — no recurrence in 33 days
CAPA-RDG-030	Extrusion twist over limit — RDG-5106	Closed	2026-02-22	Verified effective — no recurrence in 33 days
CAPA-RDG-031	Extrusion twist over limit — RDG-5103	Closed	2026-01-17	Verified effective — no recurrence in 45 days
CAPA-RDG-032	Flatness out of tolerance — RDG-5115	Closed	2026-01-31	Verified effective — no recurrence in 49 days
CAPA-RDG-033	Rivet head height out — RDG-5105	Closed	2026-02-22	Verified effective — no recurrence in 48 days
CAPA-RDG-034	Casting porosity — RDG-5114	Closed	2026-04-20	Verified effective — no recurrence in 41 days
CAPA-RDG-035	Helium leak above spec — RDG-5116	Closed	2026-02-24	Verified effective — no recurrence in 41 days
CAPA-RDG-036	Rivet head height out — RDG-5111	Closed	2026-03-26	Verified effective — no recurrence in 41 days
CAPA-RDG-037	Tunnel void in FSW seam — RDG-5105	Closed	2026-03-31	Verified effective — no recurrence in 34 days
CAPA-RDG-038	Flatness out of tolerance — RDG-5112	Closed	2026-04-02	Verified effective — no recurrence in 33 days
CAPA-RDG-039	Casting porosity — RDG-5113	Closed	2026-05-18	Verified effective — no recurrence in 39 days
CAPA-RDG-040	Flatness out of tolerance — RDG-5108	Closed	2026-03-26	Verified effective — no recurrence in 31 days
CAPA-RDG-041	Hole position out after welding — RDG-5112	Closed	2026-05-07	Verified effective — no recurrence in 38 days
CAPA-RDG-042	Casting porosity — RDG-5113	Closed	2026-04-21	Verified effective — no recurrence in 30 days
CAPA-RDG-043	Conductivity below temper spec — RDG-5107	Closed	2026-06-03	Verified effective — no recurrence in 39 days
CAPA-RDG-044	Casting porosity — RDG-5112	Closed	2026-07-07	Verified effective — no recurrence in 42 days
CAPA-RDG-045	Powder coat thickness low — RDG-5109	Closed	2026-05-14	Verified effective — no recurrence in 42 days
CAPA-RDG-046	Hole position out after welding — RDG-5115	Closed	2026-06-06	Closed without an effectiveness check; none scheduled
CAPA-RDG-047	Flatness out of tolerance — RDG-5109	Closed	2026-07-06	Closed without an effectiveness check; none scheduled
CAPA-RDG-048	Adhesive bead break — RDG-5111	Closed	2026-06-13	Verified effective — no recurrence in 41 days

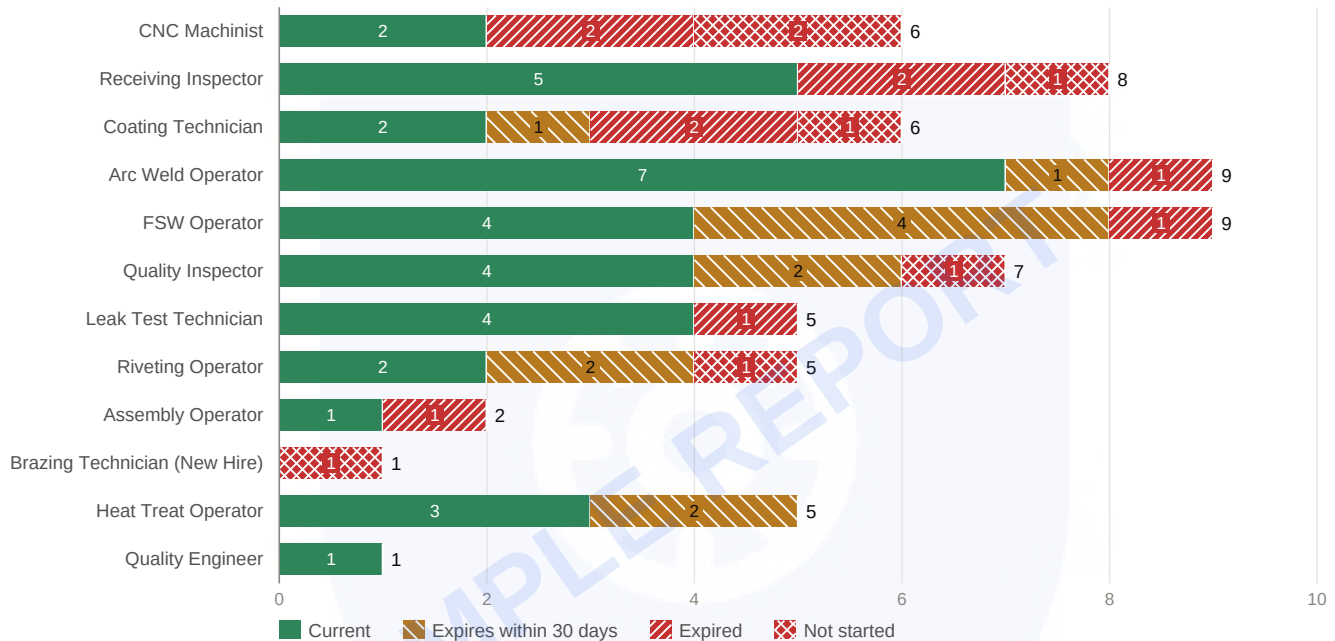
CAPA	Title	Status	Closed	Effectiveness criteria
CAPA-RDG-049	Adhesive bead break — RDG-5102	Closed	2026-06-18	Verified effective — no recurrence in 43 days
CAPA-RDG-050	Tunnel void in FSW seam — RDG-5114	Closed	2026-07-28	Verified effective — no recurrence in 49 days
CAPA-RDG-051	Extrusion twist over limit — RDG-5103	Closed	2026-07-12	Verified effective — no recurrence in 44 days
CAPA-RDG-052	Conductivity below temper spec — RDG-5101	Closed	2026-08-13	Verified effective — no recurrence in 42 days
CAPA-RDG-053	Tunnel void in FSW seam — RDG-5104	Closed	2026-08-01	Verified effective — no recurrence in 47 days
CAPA-RDG-054	Conductivity below temper spec — RDG-5112	Closed	2026-08-05	Verified effective — no recurrence in 32 days
CAPA-RDG-055	Tunnel void in FSW seam — RDG-5104	Closed	2026-07-20	Verified effective — no recurrence in 42 days
CAPA-RDG-056	Casting porosity — RDG-5106	Closed	2026-08-12	Verified effective — no recurrence in 44 days



6. Training & Competency Snapshot

Measure	Value	What it measures
Training assignment completion (%)	71.9	46 of 64 training assignments completed.
Expired qualifications	10	Qualification records whose expiry date has passed.
People with an expired qualification	10	Distinct people holding at least one expired qualification.

Qualification currency by role



Qualifications held, by the holder's role in the training matrix, on the report date. Not started: assigned and not yet begun.

CNC Machinist: 2 current, 2 expired, 2 not started; Receiving Inspector: 5 current, 2 expired, 1 not started; Coating Technician: 2 current, 1 expires within 30 days, 2 expired, 1 not started; Arc Weld Operator: 7 current, 1 expires within 30 days, 1 expired; FSW Operator: 4 current, 4 expires within 30 days, 1 expired; Quality Inspector: 4 current, 2 expires within 30 days, 1 not started; Leak Test Technician: 4 current, 1 expired; Riveting Operator: 2 current, 2 expires within 30 days, 1 not started; Assembly Operator: 1 current, 1 expired; Brazing Technician (New Hire): 1 not started; Heat Treat Operator: 3 current, 2 expires within 30 days; Quality Engineer: 1 current.

Expired qualifications

Employee	Name	Qualification	Expired
RDG-E026	Wren Jablonski	Material Cert and Temper Verification	2026-07-01
RDG-E007	Andre Baptiste	Powder Coat Process Control (CQI-12)	2026-07-08
RDG-E051	Cameron Nwosu	Powder Coat Process Control (CQI-12)	2026-07-24
RDG-E002	Priya Venkatesan	Helium Leak Test Operation and Verification	2026-07-29
RDG-E017	Harper Abernathy	GD&T per ASME Y14.5	2026-08-12
RDG-E004	Grace Achebe	AWS D8.14 Automotive Aluminum Arc Weld Quality	2026-08-25
RDG-E038	Ravi Quarles	FSW Operator Qualification per ISO 25239-3	2026-08-31

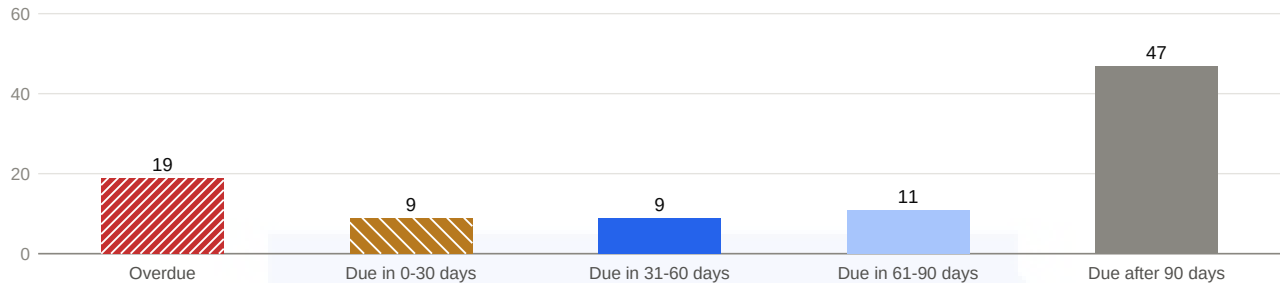
Employee	Name	Qualification	Expired
RDG-E009	Jonah Pike	Adhesive Dispense and Bead Verification	2026-09-15
RDG-E042	Lena Nwosu	GD&T per ASME Y14.5	2026-10-06
RDG-E021	Theo Gallardo	Material Cert and Temper Verification	2026-10-07



7. Calibration & MSA

Calibration Compliance: 80.0% · Past due: 19 · Failed calibration: 2

When each gauge falls due



Measuring equipment in service by when its next calibration falls due from the report date. Overdue is as your calibration register records it.

95 gauges in service: 19 overdue, 9 due in 0-30 days, 9 due in 31-60 days, 11 due in 61-90 days, 47 due after 90 days.

Failed and out-of-tolerance calibrations



From 89 calibrations recorded in the 12 months to the report date, and the impact evaluations raised for failures.

89 calibrations in 12 months: 1 failed, 2 found out of tolerance; 2 impact evaluations raised, 2 not completed.

Failed calibration and product impact (7.1.5.2.1)

Asset	Equipment	Failed	Evaluation	In doubt	Disposition
CAL-RDG-06	Measuring Microscope (FSW macro-etch)	2026-09-27	Pending evaluation	Not yet assessed	Not yet dispositioned
CAL-RDG-01	Helium Leak Detector (vacuum chamber)	2026-09-30	Pending evaluation	Not yet assessed	Not yet dispositioned

"Not yet assessed" means no measurement record links this equipment to the parts it accepted. Supply the gauge-usage records (instrument, work order, date) to size the suspect population; until then it is unbounded, not zero.

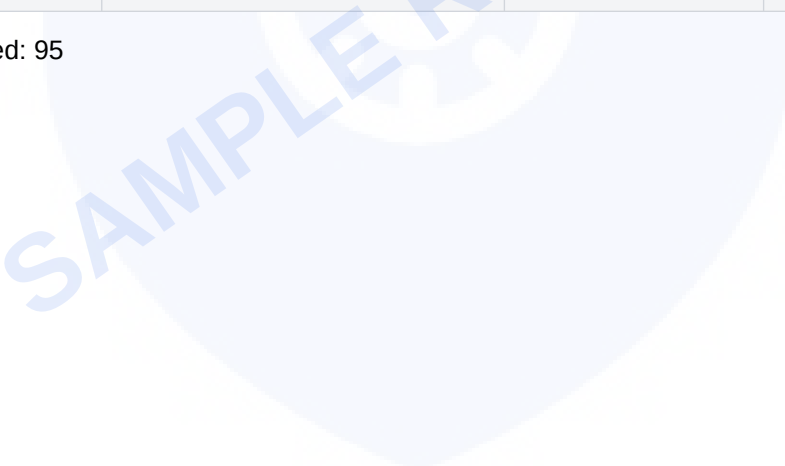
Overdue calibrations

Risk Statement: Overdue calibrations may impact product acceptance decisions made using affected measurement equipment.

Asset	Serial	Equipment	Location	Due	Days past due
CAL-RDG-014	TW-6014	Torque Wrench 5-50 Nm	Assembly Line 1	2026-05-18	143
CAL-RDG-06	MS-1407	Measuring Microscope (FSW macro-etch)	Weld Lab	2026-05-27	134
CAL-RDG-011	TW-6011	Torque Wrench 5-50 Nm	Assembly Line 1	2026-06-13	117

Asset	Serial	Equipment	Location	Due	Days past due
CAL-RDG-012	BGA-6012	Adhesive Bead Gauge	Adhesive Cell	2026-06-14	116
CAL-RDG-02	RL-3390	Helium Reference Leak 1e-5 mbar-l/s	Leak Test Cell 1	2026-06-21	109
CAL-RDG-016	DC-6016	Digital Caliper 300mm	Tray Cell	2026-07-01	99
CAL-RDG-021	DC-6021	Digital Caliper 300mm	Tray Cell	2026-07-06	94
CAL-RDG-023	FG-6023	Feeler Gauge Set	Lid Cell	2026-07-10	90
CAL-RDG-04	CMM-2240	Hexagon Global CMM	Inspection Bay 1	2026-07-19	81
CAL-RDG-013	TW-6013	Torque Wrench 5-50 Nm	Assembly Line 1	2026-07-20	80
CAL-RDG-017	WHT-6017	Webster Hardness Tester	Receiving Lab	2026-07-25	75
CAL-RDG-019	TW-6019	Torque Wrench 5-50 Nm	Assembly Line 1	2026-07-25	75
CAL-RDG-022	WHT-6022	Webster Hardness Tester	Receiving Lab	2026-07-26	74
CAL-RDG-09	HG-2305	Digital Height Gauge 0-600mm	Inspection Bay 2	2026-08-11	58
CAL-RDG-018	CTG-6018	Coating Thickness Gauge	Powder Coat Line	2026-08-11	58
CAL-RDG-05	ECM-2911	Eddy Current Conductivity Meter	Receiving Lab	2026-08-31	38
CAL-RDG-020	TC-6020	Type K Thermocouple	Solution Furnace 1	2026-09-08	30
CAL-RDG-015	DC-6015	Digital Caliper 300mm	Tray Cell	2026-09-18	20
CAL-RDG-03	LSC-2201	Laser Line Scanner (tray cell)	Inspection Bay 1	2026-10-05	3

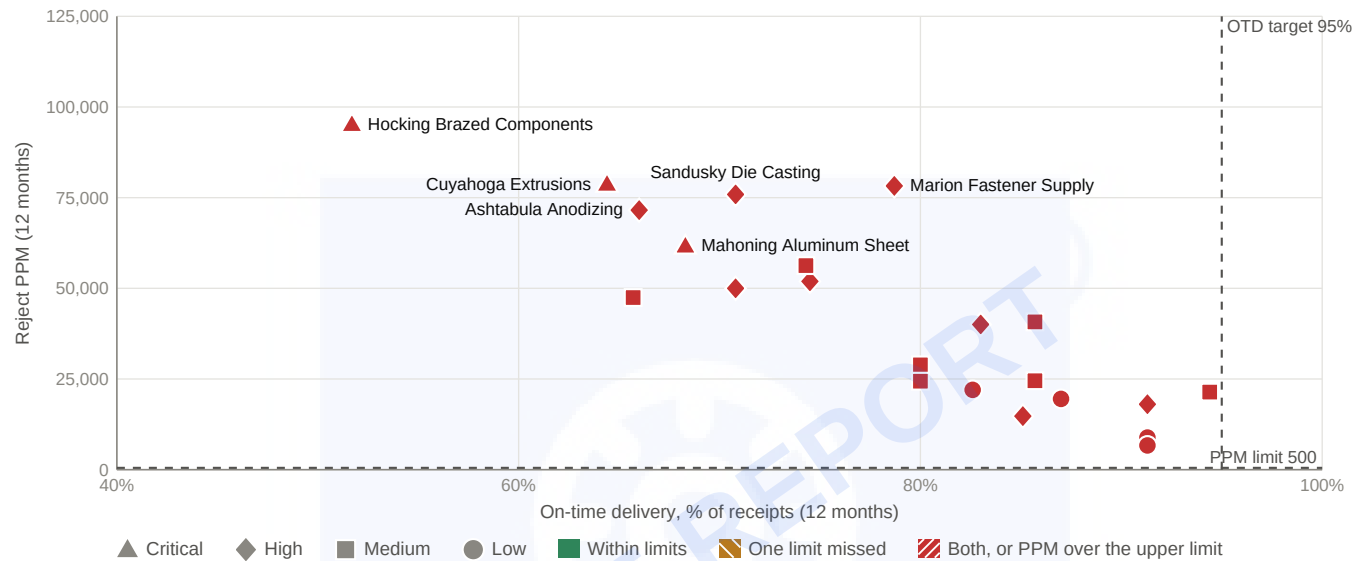
Total equipment tracked: 95



8. Supplier Governance

23 suppliers on record; 10 need attention. Each row brings together the supplier's receipts (rejected parts per million), its corrective actions (8D), its special-process assessments (8.5.1), customer-specific requirement flow-down (8.4.3.1), suspect product cases (8.7.1.3) and its risk index. Highest risk first.

Delivery against rejects



Each supplier with receipts: delivery and rejects from its receipts. Marker shape is the criticality on the supplier list; the furthest out are named.

22 suppliers with receipts, 22 missing a limit. Furthest out: Hocking Brazed Components 51.7% on time, 95,168 PPM (critical); Cuyahoga Extrusions 64.4% on time, 78,696 PPM (critical); Ashtabula Anodizing 66% on time, 71,560 PPM (high); Sandusky Die Casting 70.8% on time, 75,893 PPM (high); Mahoning Aluminum Sheet 68.3% on time, 61,640 PPM (critical); Marion Fastener Supply 78.7% on time, 78,239 PPM (high).

Supplier	Criticality	Status	OTD %	PPM	Risk index	Corrective actions (8D)	Flow-down	Special processes at risk	Suspect cases
Hocking Brazed Components	Critical	Probation	51.7	95,168	72.8 (High)	3 closed, avg 25d; 4 open, oldest 252d	0/2 (0 verified)	-	1
Mahoning Aluminum Sheet	Critical	Conditional	68.3	61,640	41.1 (Medium)	4 closed, avg 35d	none mapped	-	-
Cuyahoga Extrusions	Critical	Probation	64.4	78,696	37.5 (Medium)	2 closed, avg 33d; 2 open, oldest 223d	0/1 (0 verified)	-	-
Sandusky Die Casting	High	Conditional	70.8	75,893	33.9 (Medium)	4 closed, avg 34d	none mapped	-	-
Crawford Friction Tools	High	Conditional	70.8	50,000	33.1 (Medium)	3 closed, avg 33d; 1 open, oldest 75d	none mapped	-	-
Ashtabula Anodizing	High	Probation	66	71,560	30.5 (Medium)	2 closed, avg 22d; 2 open, oldest 278d	0/1 (0 verified)	-	-

Supplier	Criticality	Status	OTD %	PPM	Risk index	Corrective actions (8D)	Flow-down	Special processes at risk	Suspect cases
Maumee Thermal Processing	High	Conditional	74.5	51,923	19.6 (Low)	2 closed, avg 28d; 1 open, oldest 64d	0/1 (0 verified)	-	-
Marion Fastener Supply	High	Conditional	78.7	78,239	18.9 (Low)	3 closed, avg 35d	1/1 (1 verified)	-	1
Portage Precision Machining	Medium	Conditional	65.7	47,458	17.8 (Low)	2 closed, avg 34d; 1 open, oldest 33d	0/1 (0 verified)	-	-
Keystone Fastening Systems	Medium	Conditional	74.3	56,270	15.8 (Low)	3 closed, avg 35d	none mapped	-	-
Lakeview Powder Coating	High	Approved	83	40,036	7.6 (Low)	1 closed, avg 24d	none mapped	-	-
Hancock Gasket & Seal	Medium	Approved	80	28,894	6.6 (Low)	1 closed, avg 32d	none mapped	-	-
Seneca Test Laboratories	High	Approved	85.1	14,758	6.2 (Low)	1 closed, avg 14d	none mapped	-	-
Lorain Tool & Die	Medium	Approved	80	24,390	5.5 (Low)	none raised	none mapped	-	-
Tiffin Helium & Gas	Medium	Approved	85.7	24,522	5.1 (Low)	1 closed, avg 20d	none mapped	-	-
Van Wert Fixture Works	Medium	Approved	85.7	40,746	4.4 (Low)	none raised	none mapped	-	-
Blanchard River Logistics	Low	Approved	82.6	22,007	4.3 (Low)	1 closed, avg 30d	none mapped	-	-
Buckeye Adhesives	High	Approved	91.3	18,041	3.1 (Low)	none raised	none mapped	-	-
Riverside Returnable Packaging	Low	Approved	87	19,512	2.8 (Low)	none raised	none mapped	-	-
Tri-State Metrology	Medium	Approved	94.4	21,390	2.8 (Low)	1 closed, avg 37d	none mapped	-	-
Wyandot Wire & Connectors	Low	Approved	91.3	8,850	1.8 (Low)	none raised	none mapped	-	-
Defiance Label & Print	Low	Approved	91.3	6,679	1.7 (Low)	none raised	none mapped	-	-
Auglaize Extrusion Co LLC	-	Approved	-	-	Not assessed	none raised	none mapped	-	-

OTD % is the share of the last 12 months' receipts delivered on time. PPM is parts rejected at receiving inspection per million received over the last 12 months, from the receipts. The risk index is Aegis's proprietary assessment of each supplier's delivery, quality and findings, taking account of the criticality your supplier list records. Corrective actions are the findings and customer complaints raised against the supplier, with days from raised to closed.

Suppliers needing attention

Hocking Brazed Components — status probation; 1 suspect product case open: SPR-RDG-002; risk index 72.8 (High); 4 corrective action (8D)s open, the oldest 252 days; 95,168 PPM rejected at receiving (260 of 2,732 units in 12 months).

Mahoning Aluminum Sheet — status conditional; 61,640 PPM rejected at receiving (197 of 3,196 units in 12 months).

Cuyahoga Extrusions — status probation; 2 corrective action (8D)s open, the oldest 223 days; 78,696 PPM rejected at receiving (198 of 2,516 units in 12 months).

Sandusky Die Casting — status conditional; 75,893 PPM rejected at receiving (187 of 2,464 units in 12 months).

Crawford Friction Tools — status conditional; 50,000 PPM rejected at receiving (114 of 2,280 units in 12 months).

Ashtabula Anodizing — status probation; 2 corrective action (8D)s open, the oldest 278 days; 71,560 PPM rejected at receiving (156 of 2,180 units in 12 months).

Maumee Thermal Processing — status conditional; 51,923 PPM rejected at receiving (108 of 2,080 units in 12 months).

Marion Fastener Supply — status conditional; 1 suspect product case open: SPR-RDG-001; 78,239 PPM rejected at receiving (215 of 2,748 units in 12 months).

Portage Precision Machining — status conditional; 47,458 PPM rejected at receiving (56 of 1,180 units in 12 months).

Keystone Fastening Systems — status conditional; 56,270 PPM rejected at receiving (70 of 1,244 units in 12 months).

Not assessed from the records supplied: on-time-in-full (OTIF), which needs the quantity ordered on each receipt; the receipts template records the quantity received only.

9. Internal Audit Program Evidence

Internal audits: 7

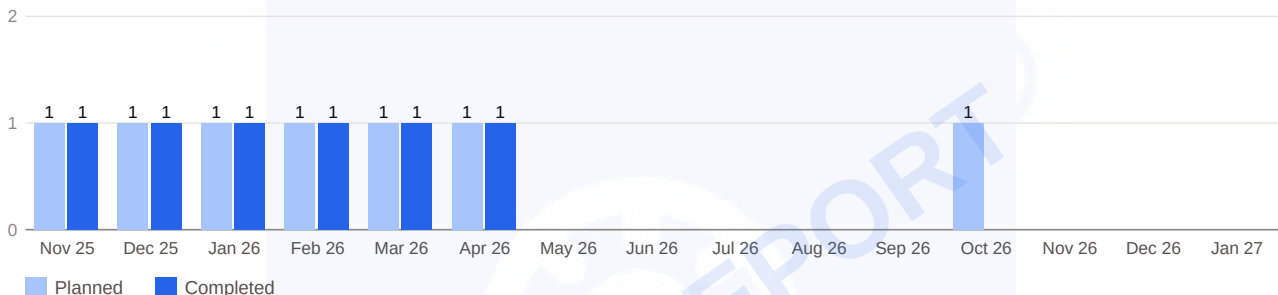
Completed: 6

In Progress: 0

Planned: 1

External, customer and certification body audits: 5 (listed in the schedule below; not part of the internal audit programme).

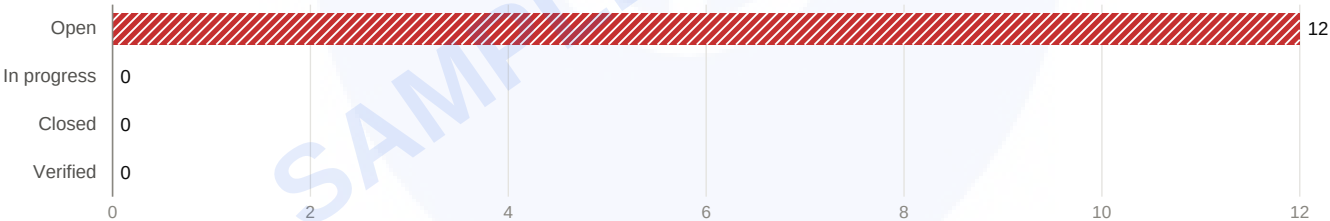
Internal audit programme: planned against completed



Internal audit programme: audits scheduled in each month and audits completed in it, twelve months back and three ahead.

7 internal audits scheduled and 6 completed from Nov 25 to Jan 27.

Findings by status



Audit findings by status, internal, external and customer.

12 audit findings: 12 open, 0 in progress, 0 closed, 0 verified.

Audit schedule

Audit	Name	Type	Status	Lead auditor	Scheduled	Findings
AUD-RDG-01	Clause 4-6 context, leadership and risk	Internal	Completed	Registrar	2025-11-26	2
AUD-RDG-02	Clause 7.1.5 measuring resources and MSA	Internal	Completed	R. Halvorsen	2025-12-25	1
AUD-RDG-03	Clause 7.2 competence and special process qualification	Internal	Completed	Registrar	2026-01-23	1
AUD-RDG-04	Clause 8.4 supplier control and monitoring	Internal	Completed	B. Okoro	2026-02-21	1
AUD-RDG-05	Manufacturing process audit — FSW floor panel cell (VDA 6.3)	Internal	Completed	R. Halvorsen	2026-03-22	1

Audit	Name	Type	Status	Lead auditor	Scheduled	Findings
AUD-RDG-06	Clause 8.7 and 10.2 nonconforming product and corrective action	Internal	Completed	Registrar	2026-04-20	0
AUD-RDG-07	Ford STA site visit	External	Completed	B. Okoro	2026-05-19	3
AUD-RDG-08	CQI-9 heat treat assessment	External	Completed	R. Halvorsen	2026-06-17	1
AUD-RDG-09	BMW supplier quality visit	External	Completed	R. Halvorsen	2026-07-16	1
AUD-RDG-10	Mercedes-Benz launch readiness review	External	Completed	B. Okoro	2026-08-14	1
AUD-RDG-11	CB surveillance (IATF certification body)	Surveillance	Completed	R. Halvorsen	2026-09-12	0
AUD-RDG-12	Clause 7.5 documented information after the ERP cutover	Internal	Planned	R. Halvorsen	2026-10-11	0

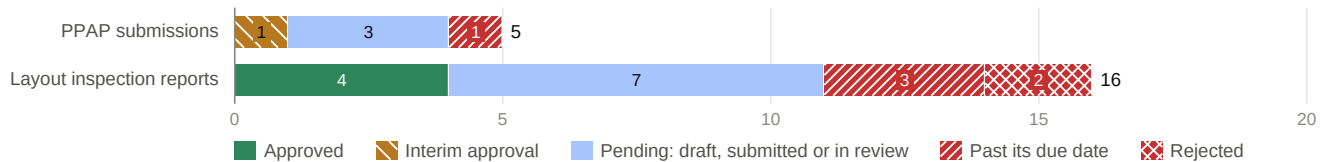
Findings Closure Rate: 0% (0/12)



10. Layout Inspection

Clause 8.6.2 — layout inspection and functional testing: every characteristic on the design record measured against its specification, the dimensional results a PPAP submits. Each report below shows its status and outcome, and whether it carries two distinct signatures, a reviewer's and an approver's.

PPAP and layout inspection status



PPAP submissions and layout inspection reports by state on the report date. Past its due date: not approved and the required or due date has passed. A PPAP is the customer's decision; this shows what the plant's registers record.

PPAP submissions: 1 interim approval, 3 pending, 1 past its due date; Layout inspection reports: 4 approved, 7 pending, 3 past its due date, 2 rejected. 5 part submission warrants not signed.

Report	Part	Type	Customer	Status	Due	Findings / disposition	Signatures
LAY-RDG-101	RDG-5101 Battery Tray Assembly	PPAP dimensional	Ford	Open	2026-10-28	Seal flange flatness 3 points over; leak test records for the build lot missing; Resubmit	Not signed: cannot be until 131 of 142 characteristics are measured; every one must be measured against its specification before the report is signed; Nonconforming results must name the nonconformance records that control them (IATF 16949 8.7)

Report	Part	Type	Customer	Status	Due	Findings / disposition	Signatures
LAY-RDG-102	RDG-5102 Battery Tray Floor Panel	Layout inspection	Ford	Rejected	2026-10-21	FSW penetration below 5.6 mm on 2 seams — tunnel void at teardown; Rejected — process revalidation required	Not signed: cannot be until A re-layout for process change must name the approved layout or PPAP it is measured against; IATF 16949 8.6.2 requires a functional verification with the layout inspection, as the control plan specifies: record pass, fail, or not required; Nonconforming results must name the nonconformance records that control them (IATF 16949 8.7)
LAY-RDG-103	RDG-5103 Side Sill Extrusion Assembly	Layout inspection	Ford	Open	2026-11-04	Extrusion twist out of tolerance from the new die at Cuyahoga; In Review	Not signed: cannot be until A re-layout for supplier change must name the approved layout or PPAP it is measured against; 44 of 58 characteristics are measured; every one must be measured against its specification before the report is signed; IATF 16949 8.6.2 requires a functional verification with the layout inspection, as the control plan specifies: record pass, fail, or not required; Nonconforming results must name the nonconformance records that control them (IATF 16949 8.7)

Report	Part	Type	Customer	Status	Due	Findings / disposition	Signatures
LAY-RDG-104	RDG-5104 Battery Lid	PPAP dimensional	BMW	Rejected	2026-10-17	Lid flatness 1.4 mm against 0.8 mm maximum at 4 points; Rejected — fixture study required	Not signed: cannot be until A re-layout for engineering change must name the approved layout or PPAP it is measured against; Nonconforming results must name the nonconformance records that control them (IATF 16949 8.7)
LAY-RDG-105	RDG-5105 Cooling Plate Assembly	Layout inspection	Ford	Open	2026-11-08	Awaiting helium leak data — reference leak expired; On Hold	Not signed: cannot be until 0 of 40 characteristics are measured; every one must be measured against its specification before the report is signed; IATF 16949 8.6.2 requires a functional verification with the layout inspection, as the control plan specifies: record pass, fail, or not required
LAY-RDG-106	RDG-5106 Cross Member Extrusion	Layout inspection	BMW	Accepted	2026-09-27	No findings; Accepted	Accepted; no signatories recorded
LAY-RDG-107	RDG-5107 Front Crash Frame	First-off	Mercedes-Benz	Accepted	2026-09-29	1 ballooning note corrected; Accepted	Accepted; no signatories recorded

Report	Part	Type	Customer	Status	Due	Findings / disposition	Signatures
LAY-RDG-108	RDG-5108 Battery Tray Assembly (MB)	PPAP dimensional	Mercedes-Benz	Open	2026-11-16	Run-at-rate failed; 5 hole positions out after FSW distortion; Resubmit	Not signed: cannot be until 102 of 150 characteristics are measured; every one must be measured against its specification before the report is signed; Nonconforming results must name the nonconformance records that control them (IATF 16949 8.7)
LAY-RDG-109	RDG-5109 Module Mounting Bracket	Layout inspection	Ford	Accepted	2026-09-23	No findings; Accepted	Accepted; no signatories recorded
LAY-RDG-110	RDG-5110 Busbar Cover	PPAP dimensional	BMW	Open	2026-11-11	Material cert and part marking records incomplete; In Review	Not signed: cannot be until 30 of 44 characteristics are measured; every one must be measured against its specification before the report is signed
LAY-RDG-111	RDG-5111 Rear Cross Member	PPAP dimensional	Mercedes-Benz	Accepted	2026-09-18	No findings; Accepted	Accepted; no signatories recorded
LAY-RDG-112	RDG-5112 Tray Frame Weldment	PPAP dimensional	Ford	Open	2026-09-11	Ballooned drawing incomplete; In Review	Not signed: cannot be until 44 of 52 characteristics are measured; every one must be measured against its specification before the report is signed

Report	Part	Type	Customer	Status	Due	Findings / disposition	Signatures
LAY-RDG-113	RDG-5113 Coolant Manifold Bracket	PPAP dimensional	BMW	Open	2026-10-30	Ballooned drawing incomplete; In Review	Not signed: cannot be until 54 of 63 characteristics are measured; every one must be measured against its specification before the report is signed; Nonconforming results must name the nonconformance records that control them (IATF 16949 8.7)
LAY-RDG-114	RDG-5114 Underbody Shield	PPAP dimensional	Mercedes-Benz	In review	2026-09-29	Ballooned drawing incomplete; In Review	Not signed: cannot be until 70 of 74 characteristics are measured; every one must be measured against its specification before the report is signed
LAY-RDG-115	RDG-5115 Vent Valve Housing	PPAP dimensional	Ford	Open	2026-10-29	Ballooned drawing incomplete; In Review	Not signed: cannot be until 25 of 30 characteristics are measured; every one must be measured against its specification before the report is signed; Nonconforming results must name the nonconformance records that control them (IATF 16949 8.7)
LAY-RDG-116	RDG-5116 Lid Seal Flange	PPAP dimensional	BMW	In review	2026-08-05	Ballooned drawing incomplete; In Review	Not signed: cannot be until 35 of 41 characteristics are measured; every one must be measured against its specification before the report is signed

Open item: 12 layout inspection reports are not yet signed by both a reviewer and an approver.

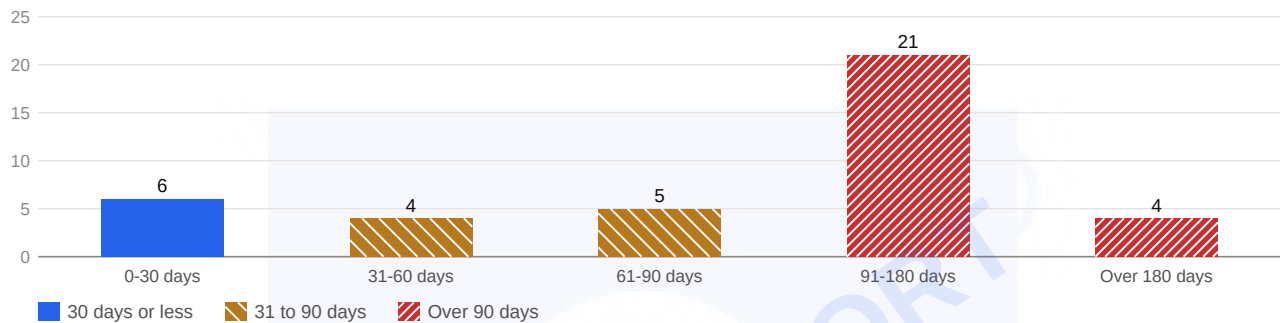
Records gap: 4 layout inspection reports are recorded as accepted with no reviewer or approver on record.



11. Nonconforming Product, Containment and Material Review

Clauses 8.7.1.1 to 8.7.1.7 — concession, suspect product, rework, repair, customer notification and disposition. Containment is the control that decides whether suspect product reached a customer, so it is reported per nonconformance rather than in summary.

Open nonconformances by age



Open nonconformances by days since detected.

40 open nonconformances by age: 6 0-30 days, 4 31-60 days, 5 61-90 days, 21 91-180 days, 4 Over 180 days.

Where the nonconformances come from



82 nonconformances. Dark bars are the vital few (together 80%); the percentage in each bar is the cumulative share, largest first.

82 nonconformances: 8 of 16 account for 82.9%: Receiving inspection (13), Adhesive dispense (10), Leak test (9), Arc weld (8), Friction stir weld (8), Fixture and clamp (8), Self-piercing rivet (7), Aging (5).

Containment status

NCR	Nonconformance	Contained	Zones outstanding	Deadline	Status
NCR-RDG-001	Casting porosity — RDG-5112	Yes	None	2025-12-25	Complete
NCR-RDG-002	Adhesive bead break — RDG-5106	Yes	None	2026-08-17	Complete
NCR-RDG-003	Powder coat thickness low — RDG-5109	Yes	None	2026-03-28	Complete
NCR-RDG-004	CMT weld porosity — RDG-5116	Yes	None	2026-07-19	Complete

NCR	Nonconformance	Contained	Zones outstanding	Deadline	Status
NCR-RDG-005	Rivet head height out — RDG-5104	Yes	None	2026-08-31	Complete
NCR-RDG-006	Conductivity below temper spec — RDG-5107	Yes	None	2026-03-10	Complete
NCR-RDG-007	CMT weld porosity — RDG-5110	Yes	None	2025-12-06	Complete
NCR-RDG-008	Adhesive bead break — RDG-5107	Yes	None	2025-12-26	Complete
NCR-RDG-009	Extrusion twist over limit — RDG-5106	Yes	None	2025-12-28	Complete
NCR-RDG-010	Conductivity below temper spec — RDG-5103	Yes	None	2025-11-05	Complete
NCR-RDG-011	Adhesive bead break — RDG-5109	Yes	None	2026-03-24	Complete
NCR-RDG-012	Helium leak above spec — RDG-5111	Yes	None	2025-12-27	Complete
NCR-RDG-013	Powder coat thickness low — RDG-5109	Yes	None	2025-12-09	Complete
NCR-RDG-014	Conductivity below temper spec — RDG-5112	Yes	None	2026-06-04	Complete
NCR-RDG-015	Tunnel void in FSW seam — RDG-5105	Yes	None	2026-02-01	Complete
NCR-RDG-016	Powder coat thickness low — RDG-5106	No	Raw material, Work in progress, Finished goods	2026-09-23	Overdue
NCR-RDG-017	Missing leak test record — RDG-5103	Yes	None	2025-11-30	Complete
NCR-RDG-018	Helium leak above spec — RDG-5114	Yes	None	2026-08-07	Complete
NCR-RDG-019	Adhesive bead break — RDG-5102	Yes	None	2025-10-26	Complete
NCR-RDG-020	Extrusion twist over limit — RDG-5114	Yes	None	2026-05-17	Complete
NCR-RDG-021	Flatness out of tolerance — RDG-5110	Yes	None	2026-09-28	Complete
NCR-RDG-022	Casting porosity — RDG-5113	Yes	None	2026-02-25	Complete
NCR-RDG-023	Rivet head height out — RDG-5108	Yes	None	2025-11-17	Complete
NCR-RDG-024	Tunnel void in FSW seam — RDG-5104	Yes	None	2026-05-31	Complete
NCR-RDG-025	CMT weld porosity — RDG-5107	Yes	None	2025-11-03	Complete
NCR-RDG-026	Adhesive bead break — RDG-5107	Yes	None	2026-06-12	Complete
NCR-RDG-027	Rivet head height out — RDG-5105	Yes	None	2026-01-11	Complete
NCR-RDG-028	Missing leak test record — RDG-5108	Yes	None	2026-09-21	Complete
NCR-RDG-029	Casting porosity — RDG-5112	Yes	None	2026-03-16	Complete
NCR-RDG-030	CMT weld porosity — RDG-5108	Yes	None	2026-05-07	Complete
NCR-RDG-031	Hole position out after welding — RDG-5115	Yes	None	2025-10-23	Complete
NCR-RDG-032	Hole position out after welding — RDG-5104	Yes	None	2026-05-18	Complete
NCR-RDG-033	Flatness out of tolerance — RDG-5112	Yes	None	2026-02-10	Complete
NCR-RDG-034	Extrusion twist over limit — RDG-5106	Yes	None	2025-12-04	Complete
NCR-RDG-035	Tunnel void in FSW seam — RDG-5104	Yes	None	2026-06-06	Complete
NCR-RDG-036	Flatness out of tolerance — RDG-5109	Yes	None	2026-04-07	Complete

NCR	Nonconformance	Contained	Zones outstanding	Deadline	Status
NCR-RDG-037	Casting porosity — RDG-5114	Yes	None	2026-01-23	Complete
NCR-RDG-038	Tunnel void in FSW seam — RDG-5114	Yes	None	2026-04-27	Complete
NCR-RDG-039	Hole position out after welding — RDG-5113	Yes	None	2026-07-01	Complete
NCR-RDG-040	Rivet head height out — RDG-5114	Yes	None	2026-07-31	Complete
NCR-RDG-041	Casting porosity — RDG-5113	Yes	None	2026-02-14	Complete
NCR-RDG-042	CMT weld porosity — RDG-5101	Yes	None	2026-09-19	Complete
NCR-RDG-043	CMT weld porosity — RDG-5114	Yes	None	2025-12-16	Complete
NCR-RDG-044	Adhesive bead break — RDG-5102	Yes	None	2026-04-22	Complete
NCR-RDG-045	Flatness out of tolerance — RDG-5108	Yes	None	2026-02-19	Complete
NCR-RDG-046	CMT weld porosity — RDG-5105	Yes	None	2026-06-08	Complete
NCR-RDG-047	Rivet head height out — RDG-5109	Yes	None	2026-07-29	Complete
NCR-RDG-048	Adhesive bead break — RDG-5105	Yes	None	2025-12-06	Complete
NCR-RDG-049	CMT weld porosity — RDG-5115	Yes	None	2026-08-25	Complete
NCR-RDG-050	Adhesive bead break — RDG-5111	Yes	None	2026-04-15	Complete
NCR-RDG-051	Flatness out of tolerance — RDG-5103	Yes	None	2026-07-21	Complete
NCR-RDG-052	Hole position out after welding — RDG-5112	Yes	None	2026-02-21	Complete
NCR-RDG-053	Casting porosity — RDG-5106	Yes	None	2026-06-16	Complete
NCR-RDG-054	Missing leak test record — RDG-5110	Yes	None	2025-12-24	Complete
NCR-RDG-055	Rivet head height out — RDG-5111	Yes	None	2026-01-28	Complete
NCR-RDG-056	Flatness out of tolerance — RDG-5108	Yes	None	2025-12-18	Complete
NCR-RDG-057	Flatness out of tolerance — RDG-5109	Yes	None	2026-08-31	Complete
NCR-RDG-058	Extrusion twist over limit — RDG-5103	Yes	None	2026-04-27	Complete
NCR-RDG-059	Conductivity below temper spec — RDG-5101	Yes	None	2026-05-29	Complete
NCR-RDG-060	Rivet head height out — RDG-5107	Yes	None	2026-10-02	Complete
NCR-RDG-061	Missing leak test record — RDG-5106	Yes	None	2026-06-17	Complete
NCR-RDG-062	Tunnel void in FSW seam — RDG-5115	Yes	None	2026-06-03	Complete
NCR-RDG-063	Hole position out after welding — RDG-5115	Yes	None	2026-04-05	Complete
NCR-RDG-064	Extrusion twist over limit — RDG-5103	Yes	None	2025-12-28	Complete
NCR-RDG-065	Helium leak above spec — RDG-5116	Yes	None	2026-01-23	Complete
NCR-RDG-066	Casting porosity — RDG-5106	No	Work in progress	2026-09-23	Overdue
NCR-RDG-067	Tunnel void in FSW seam — RDG-5105	Yes	None	2026-06-30	Complete
NCR-RDG-068	Adhesive bead break — RDG-5106	Yes	None	2026-03-13	Complete
NCR-RDG-069	Flatness out of tolerance — RDG-5115	Yes	None	2026-01-02	Complete

NCR	Nonconformance	Contained	Zones outstanding	Deadline	Status
NCR-RDG-070	Helium leak above spec — RDG-5110	Yes	None	2025-11-15	Complete
NCR-RDG-201	Coolant leak escapes on RDG-5105 cooling plates — 212 PPM at the Ford pack line	Yes	None	2026-04-17	Complete
NCR-RDG-202	Tunnel voids in RDG-5102 floor panel FSW seams — found at Ford teardown	Yes	None	2026-04-17	Complete
NCR-RDG-203	Leak test results and lot genealogy for three weeks of July 2025 lost in the ERP cutover	Yes	None	2026-04-17	Complete
NCR-RDG-204	BMW battery lid flatness out of tolerance — RDG-5104 rev C	Yes	None	2026-04-17	Complete
NCR-RDG-205	Aging oven TUS failure at location 7 and solution furnace SAT overdue	Yes	None	2026-04-17	Complete
NCR-RDG-206	Helium reference leak certificate expired — leak detectors unverified since November	Yes	None	2026-04-17	Complete
NCR-RDG-207	FSW operators producing without qualification to ISO 25239-3	Yes	None	2026-04-17	Complete
NCR-RDG-208	Mercedes-Benz run-at-rate failed on RDG-5108 — FSW cycle time and hole position capability	Yes	None	2026-04-17	Complete
NCR-RDG-209	Extrusion twist out of tolerance from Cuyahoga Extrusions — four lots	Yes	None	2026-04-17	Complete
NCR-RDG-210	Structural adhesive used past shelf life on the tray line — two drums	Yes	None	2026-04-17	Complete
NCR-RDG-211	FSW tool source changed without customer notification	Yes	None	2026-04-17	Complete
NCR-RDG-212	Quality leadership split across two plants — management review not held in Q3 or Q4	Yes	None	2026-04-17	Complete

Finding: 2 containment actions are past the 24-hour deadline. Until every zone is cleared, the population of suspect product is unbounded.

Nonconforming product dispositions (8.7.1.7)

MRB	NCR	Part	Description	Disposition	Qty	Cost	Approver	Board date	Status
MRB-RDG-301	NCR-RDG-071	RDG-5105	Cooling plate — helium leak 3.2e-4 mbar·l/s at a braze joint	Scrap	36	\$61,000	S. Kaur	2026-08-31	Closed
MRB-RDG-302	NCR-RDG-072	RDG-5102	Floor panel — tunnel void in FSW seam at macro-etch	Scrap	9	\$44,500	T. Marchetti	2026-09-05	Closed
MRB-RDG-303	NCR-RDG-073	RDG-5104	Battery lid — flatness 1.4 mm against 0.8 mm maximum	Rework	48	\$19,200	E. Vasko	2026-09-09	Closed

MRB	NCR	Part	Description	Disposition	Qty	Cost	Approver	Board date	Status
MRB-RDG-304	NCR-RDG-074	RDG-5103	Side sill extrusion — twist 0.9 mm/m against 0.5 mm/m	Return to vendor	120	\$27,600	S. Kaur	2026-09-13	Closed
MRB-RDG-305	NCR-RDG-075	RDG-5108	MB tray — hole position 0.6 mm out after FSW distortion	Rework	14	\$23,800	D. Morrow	2026-09-16	Closed
MRB-RDG-306	NCR-RDG-076	RDG-5105	Cooling plate lot shipped before its leak re-test result was recorded	Scrap	22	\$38,000	H. Lindgren (QD)	2026-09-20	Closed
MRB-RDG-307	NCR-RDG-077	RDG-5101	Tray — structural adhesive applied from a drum past its shelf life	Scrap	6	\$31,500	N. Oyelaran	2026-09-22	Pending
MRB-RDG-308	NCR-RDG-078	RDG-5106	Cross member — conductivity 38 %IACS, temper suspect (T4 not T6)	Use as is	30	\$2,400	Eng Auth (pending)	2026-09-26	Pending
MRB-RDG-309	NCR-RDG-079	RDG-5112	Tray frame weldment — CMT weld porosity at X-ray	Rework	11	\$16,700	T. Marchetti	2026-09-28	Pending
MRB-RDG-310	NCR-RDG-080	RDG-5115	Vent valve housing — casting porosity at leak test	Return to vendor	240	\$12,000	S. Kaur	2026-09-29	Pending
MRB-RDG-311	NCR-RDG-081	RDG-5110	Busbar cover — powder coat below 60 µm	Rework	75	\$6,300	L. Ferreira	2026-09-30	Pending
MRB-RDG-312	NCR-RDG-082	RDG-5109	Mounting bracket — flow drill screw torque below minimum	Rework	32	\$4,100	P. Ionescu	2026-10-01	Pending
MRB-RDG-313	NCR-RDG-003	RDG-5109	Powder coat thickness low — RDG-5109	Rework	4	\$33,000	H. Lindgren (QD)	2026-03-30	Closed
MRB-RDG-314	NCR-RDG-005	RDG-5104	Rivet head height out — RDG-5104	Rework	23	\$33,000	S. Kaur	2026-09-02	Closed
MRB-RDG-315	NCR-RDG-006	RDG-5107	Conductivity below temper spec — RDG-5107	Use as is	44	\$8,000	S. Kaur	2026-03-12	Closed
MRB-RDG-316	NCR-RDG-007	RDG-5110	CMT weld porosity — RDG-5110	Rework	12	\$20,000	H. Lindgren (QD)	2025-12-08	Closed
MRB-RDG-317	NCR-RDG-008	RDG-5107	Adhesive bead break — RDG-5107	Return to vendor	18	\$29,000	H. Lindgren (QD)	2025-12-28	Closed
MRB-RDG-318	NCR-RDG-009	RDG-5106	Extrusion twist over limit — RDG-5106	Rework	33	\$35,000	H. Lindgren (QD)	2025-12-30	Closed
MRB-RDG-319	NCR-RDG-012	RDG-5111	Helium leak above spec — RDG-5111	Scrap	15	\$24,000	S. Kaur	2025-12-29	Closed

MRB	NCR	Part	Description	Disposition	Qty	Cost	Approver	Board date	Status
MRB-RDG-320	NCR-RDG-014	RDG-5112	Conductivity below temper spec — RDG-5112	Scrap	11	\$28,000	S. Kaur	2026-06-06	Closed
MRB-RDG-321	NCR-RDG-015	RDG-5105	Tunnel void in FSW seam — RDG-5105	Rework	24	\$17,000	H. Lindgren (QD)	2026-02-03	Closed
MRB-RDG-322	NCR-RDG-017	RDG-5103	Missing leak test record — RDG-5103	Use as is	42	\$44,000	S. Kaur	2025-12-02	Closed
MRB-RDG-323	NCR-RDG-023	RDG-5108	Rivet head height out — RDG-5108	Scrap	40	\$8,000	H. Lindgren (QD)	2025-11-19	Closed
MRB-RDG-324	NCR-RDG-024	RDG-5104	Tunnel void in FSW seam — RDG-5104	Return to vendor	38	\$38,000	S. Kaur	2026-06-02	Closed
MRB-RDG-325	NCR-RDG-025	RDG-5107	CMT weld porosity — RDG-5107	Use as is	58	\$42,000	H. Lindgren (QD)	2025-11-05	Closed
MRB-RDG-326	NCR-RDG-027	RDG-5105	Rivet head height out — RDG-5105	Scrap	23	\$5,000	H. Lindgren (QD)	2026-01-13	Closed
MRB-RDG-327	NCR-RDG-028	RDG-5108	Missing leak test record — RDG-5108	Scrap	24	\$24,000	S. Kaur	2026-09-23	Closed
MRB-RDG-328	NCR-RDG-034	RDG-5106	Extrusion twist over limit — RDG-5106	Scrap	57	\$32,000	H. Lindgren (QD)	2025-12-06	Closed
MRB-RDG-329	NCR-RDG-035	RDG-5104	Tunnel void in FSW seam — RDG-5104	Return to vendor	35	\$13,000	H. Lindgren (QD)	2026-06-08	Closed
MRB-RDG-330	NCR-RDG-036	RDG-5109	Flatness out of tolerance — RDG-5109	Return to vendor	59	\$26,000	S. Kaur	2026-04-09	Closed
MRB-RDG-341	NCR-RDG-231	RDG-5102	Floor panels since the last good macro-etch: tunnel void at seam 3	Scrap	7	\$34,853	Not recorded	2026-09-18	Pending
MRB-RDG-342	NCR-RDG-232	RDG-5105	Cooling plates leak tested after a failed detector check: re-test before release	Rework	54	\$1,993	Not recorded	2026-09-18	Pending
MRB-RDG-343	NCR-RDG-233	RDG-5101	Trays bonded from an expired adhesive drum: bead removed and re-applied	Rework	9	\$1,365	Not recorded	2026-09-18	Pending

33 dispositions; recorded cost \$784,311.

4 dispositions are use-as-is or repair; either needs the customer's written authorization before the product ships (8.7.1.1, 8.7.1.5).

Open item: 20 dispositions have not met 8.7.1:

MRB	Disposition	What 8.7.1 still requires
MRB-RDG-303	Rework	Record the rework confirmation to the original specification, per the control plan (IATF 16949 8.7.1.4)
MRB-RDG-305	Rework	Record the rework confirmation to the original specification, per the control plan (IATF 16949 8.7.1.4)
MRB-RDG-306	Scrap	Nonconforming product has shipped: notify the customer promptly and follow with the detailed documentation (IATF 16949 8.7.1.6)
MRB-RDG-307	Scrap	The board's quality approval is not recorded; Verify the scrapped product was rendered unusable before disposal (IATF 16949 8.7.1.7)
MRB-RDG-308	Use as is	The board's quality approval is not recorded; Use as is needs the customer's authorization by concession before the product is used or processed further (IATF 16949 8.7.1.1)
MRB-RDG-309	Rework	The board's quality approval is not recorded; Record the rework confirmation to the original specification, per the control plan (IATF 16949 8.7.1.4)
MRB-RDG-310	Return to vendor	The board's quality approval is not recorded
MRB-RDG-311	Rework	The board's quality approval is not recorded; Record the rework confirmation to the original specification, per the control plan (IATF 16949 8.7.1.4)
MRB-RDG-312	Rework	The board's quality approval is not recorded; Record the rework confirmation to the original specification, per the control plan (IATF 16949 8.7.1.4)
MRB-RDG-313	Rework	Record the rework confirmation to the original specification, per the control plan (IATF 16949 8.7.1.4)
MRB-RDG-314	Rework	Record the rework confirmation to the original specification, per the control plan (IATF 16949 8.7.1.4)
MRB-RDG-315	Use as is	Record the expiry date or the quantity the concession authorizes (IATF 16949 8.7.1.1)
MRB-RDG-316	Rework	Record the rework confirmation to the original specification, per the control plan (IATF 16949 8.7.1.4)
MRB-RDG-318	Rework	Record the rework confirmation to the original specification, per the control plan (IATF 16949 8.7.1.4)
MRB-RDG-321	Rework	Record the rework confirmation to the original specification, per the control plan (IATF 16949 8.7.1.4)
MRB-RDG-322	Use as is	Record the expiry date or the quantity the concession authorizes (IATF 16949 8.7.1.1)
MRB-RDG-325	Use as is	Record the expiry date or the quantity the concession authorizes (IATF 16949 8.7.1.1)
MRB-RDG-341	Scrap	The board's quality approval is not recorded; Verify the scrapped product was rendered unusable before disposal (IATF 16949 8.7.1.7)
MRB-RDG-342	Rework	The board's quality approval is not recorded; Record the rework confirmation to the original specification, per the control plan (IATF 16949 8.7.1.4)
MRB-RDG-343	Rework	The board's quality approval is not recorded; Record the rework confirmation to the original specification, per the control plan (IATF 16949 8.7.1.4)

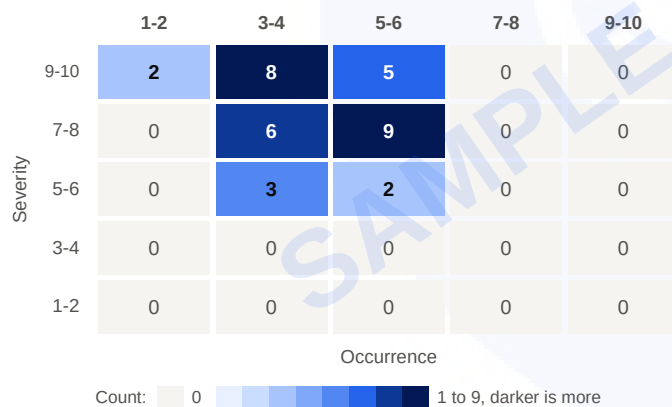
12. Advanced Product Quality Planning and PPAP

Clauses 8.3.5.2, 8.5.1.1 and 8.3.4.4 — the process FMEA, the control plan raised from it, and customer product approval. Failure modes are ranked by Action Priority (AIAG-VDA) rather than risk priority number, because multiplying severity, occurrence and detection lets a safety-critical failure share a score with a cosmetic one.

Failure mode registers

FMEA	Part / Process	Type	Rev	Modes	High Priority	Status
PFMEA-RDG-5101	RDG-5101 Battery Tray Assembly joining, heat treat and leak test	PFMEA	C	7	6	Draft
PFMEA-RDG-5102	RDG-5102 Battery Tray Floor Panel joining, heat treat and leak test	PFMEA	C	7	6	Draft
PFMEA-RDG-5103	RDG-5103 Side Sill Extrusion Assembly joining, heat treat and leak test	PFMEA	C	7	6	Draft
PFMEA-RDG-5104	RDG-5104 Battery Lid joining, heat treat and leak test	PFMEA	C	7	6	Draft
PFMEA-RDG-5105	RDG-5105 Cooling Plate Assembly joining, heat treat and leak test	PFMEA	C	7	6	Draft

Process failure modes by severity and occurrence



35 process failure modes by severity and occurrence band, from the PFMEA registers. Severity 9-10 is the top row.

35 failure modes; 30 with severity 7 or above; 0 with severity and occurrence both 7 or above.

Control plans

Plan	Part	Stage	Rev	Characteristics	Status
CP-RDG-5101	RDG-5101 Battery Tray Assembly	Production	B	5	Draft
CP-RDG-5103	RDG-5103 Side Sill Extrusion Assembly	Production	B	5	Draft
CP-RDG-5105	RDG-5105 Cooling Plate Assembly	Production	B	5	Draft

PFMEA to control plan (8.5.1.1)

Finding: the failure modes below carry a High action priority and no control plan raised from their FMEA names them. A register and a control plan that disagree are two documents, not one control.

FMEA	Part / Process	High-priority modes	Not in a control plan
PFMEA-RDG-5101	RDG-5101 Battery Tray Assembly joining, heat treat and leak test	6	2
PFMEA-RDG-5102	RDG-5102 Battery Tray Floor Panel joining, heat treat and leak test	6	6
PFMEA-RDG-5103	RDG-5103 Side Sill Extrusion Assembly joining, heat treat and leak test	6	2
PFMEA-RDG-5104	RDG-5104 Battery Lid joining, heat treat and leak test	6	6
PFMEA-RDG-5105	RDG-5105 Cooling Plate Assembly joining, heat treat and leak test	6	2

Production part approval (8.3.4.4)

PPAP	Part	Customer	Level	Elements	Warrant	Status	Required
PPAP-RDG-5101	RDG-5101 Battery Tray Assembly	Ford	3	None recorded	Not Started	Draft	2026-11-19
PPAP-RDG-5102	RDG-5102 Battery Tray Floor Panel	Ford	3	None recorded	Not Started	Interim	2026-09-22
PPAP-RDG-5103	RDG-5103 Side Sill Extrusion Assembly	Ford	3	None recorded	In Progress	Interim	2026-11-17
PPAP-RDG-5104	RDG-5104 Battery Lid	BMW	3	None recorded	Not Started	Submitted	2026-11-22
PPAP-RDG-5105	RDG-5105 Cooling Plate Assembly	Ford	3	None recorded	In Progress	Draft	2026-11-20

Open item: 5 submissions have no signed Part Submission Warrant. Until the warrant is signed nobody has attested that the required elements are complete.

13. Process Capability and Measurement Systems

Clauses 9.1.1.1 and 7.1.5.1.1. Capability indices are derived from the recorded measurements, and control limits are computed from the process rather than the drawing tolerance. Cpk uses the within-subgroup spread and Ppk the overall spread, so a gap between them means the process is capable but not being held.

Capability studies

Study	Part	Characteristic	Spec	Cpk	Ppk	Chart	Gauge
SPC-5102-PEN	RDG-5102	FSW penetration depth	5.6 - 6.0 mm	5.44	1.00	X-bar R	MS-1407
SPC-5104-FLANGE	RDG-5104	Lid flange height	12.35 - 12.65 mm	2.18	2.13	X-bar R	HG-2305
SPC-5101-DEPTH	RDG-5101	Tray pocket depth	119.5 - 120.5 mm	1.68	1.13	X-bar R	LSC-2201
SPC-5110-COAT	RDG-5110	Powder coat thickness	60.0 - 90.0 μ m	1.89	1.93	X-bar R	CTG-2710
SPC-5106-COND	RDG-5106	Conductivity	40.0 - 46.0 %IACS	0.89	0.86	X-bar R	ECM-2911

1 characteristic shows Cpk materially above Ppk: the process is capable within a subgroup and drifting between them, which is a control problem rather than a capability problem.

Measurement system analysis

Study	Gauge	Gauge ID	Type	%GR&R	Repeat	Reprod	ndc	Expires
GRR-HG2305	Digital Height Gauge	HG-2305	GR&R	5.1%	4.6%	2.2%	27	2027-04-26
GRR-LSC2201	Laser Line Scanner	LSC-2201	GR&R	14.4%	14.4%	0.0%	9	2027-04-26
GRR-ECM2911	Eddy Current Conductivity Meter	ECM-2911	GR&R	55.3%	44.7%	32.5%	2	2027-04-26

Finding: the capability claims below rest on a measurement system that cannot carry them. A Cpk measured with a gauge consuming a third of the total variation is a statement about the gauge.

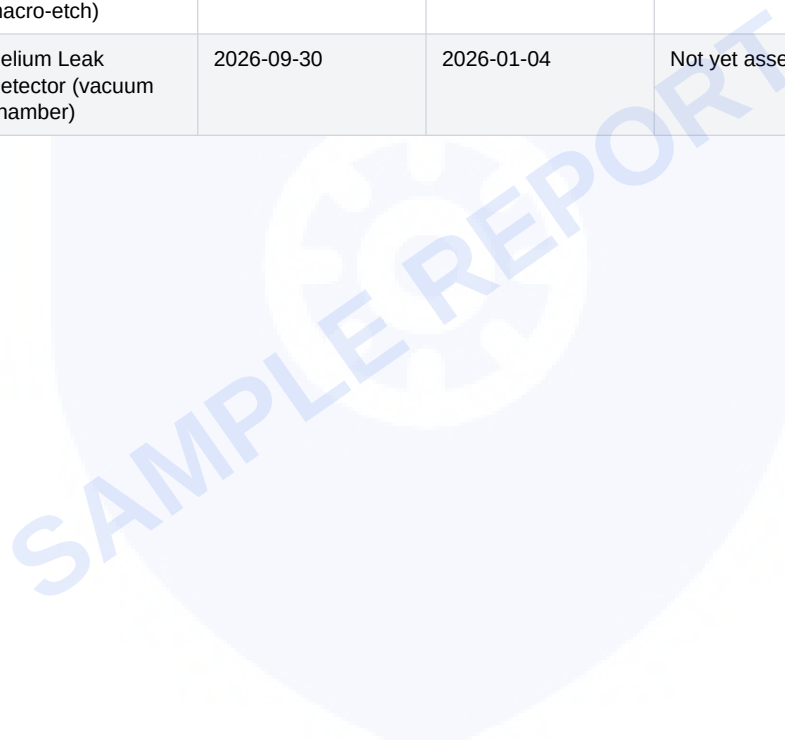
Characteristic	Part	Cpk	Gauge	Reason
FSW penetration depth	RDG-5102	5.44	MS-1407	gauge MS-1407 has no Gage R&R study
Powder coat thickness	RDG-5110	1.89	CTG-2710	gauge CTG-2710 has no Gage R&R study

Characteristic	Part	Cpk	Gauge	Reason
Conductivity	RDG-5106	0.89	ECM-2911	gauge ECM-2911 consumes 55.3% of total variation; this study measures the gauge

Measurements in doubt (7.1.5.2.1)

Finding: 2 calibration failures have no completed impact evaluation. Every measurement taken with the affected equipment since its last passing calibration remains in doubt, and product accepted on those measurements has not been dispositioned.

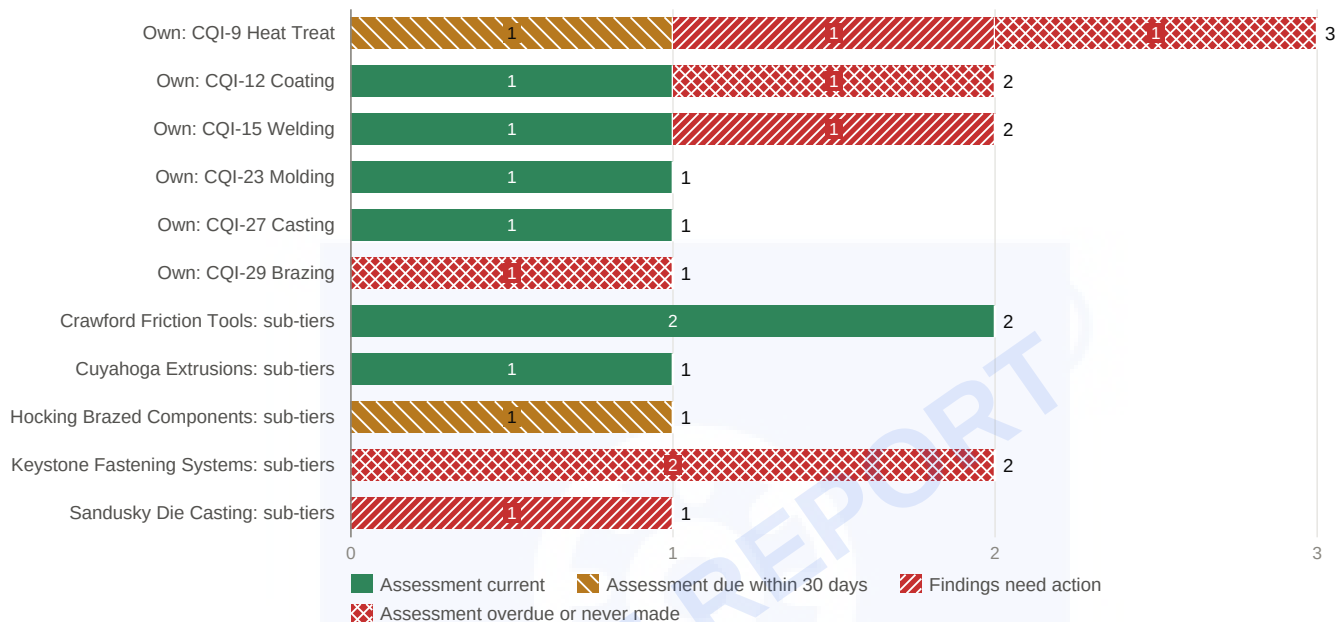
Asset	Equipment	Failed	Last good	Measurements in doubt	Evaluation
CAL-RDG-06	Measuring Microscope (FSW macro-etch)	2026-09-27	2025-11-25	Not yet assessed	Pending evaluation
CAL-RDG-01	Helium Leak Detector (vacuum chamber)	2026-09-30	2026-01-04	Not yet assessed	Pending evaluation



14. Special Processes, Flow-down and Traceability

Special processes (8.5.1, 4.3.2)

CQI special process assessments



The organisation's own special processes by CQI standard, and the special processes each supplier's sub-tiers perform, by assessment state. A CQI assessment is the customer's requirement and the organisation's own check, not a certificate.

Own: CQI-9 Heat Treat: 1 assessment due within 30 days, 1 findings need action, 1 assessment overdue or never made; Own: CQI-12 Coating: 1 assessment current, 1 assessment overdue or never made; Own: CQI-15 Welding: 1 assessment current, 1 findings need action; Own: CQI-23 Molding: 1 assessment current; Own: CQI-27 Casting: 1 assessment current; Own: CQI-29 Brazing: 1 assessment overdue or never made; Crawford Friction Tools: sub-tiers: 2 assessment current; Cuyahoga Extrusions: sub-tiers: 1 assessment current; Hocking Brazed Components: sub-tiers: 1 assessment due within 30 days; Keystone Fastening Systems: sub-tiers: 2 assessment overdue or never made; Sandusky Die Casting: sub-tiers: 1 findings need action. 8 without a current assessment or with findings to close: 5 own, 3 sub-tier.

Process	Specification	CQI standard	Assessed by	Status	Assessment due	Assessment	At risk because
SP-RDG-01 Solution Heat Treat — Furnace 1 (T6)	6061/6082 solution 530-545°C; CQI-9 process table	CQI-9	C. Ibsen	Active	2026-10-27	Due soon	Not at risk
SP-RDG-02 Artificial Aging — Aging Oven 2	Aging 175°C 8 h; conductivity 40-46 %IACS	CQI-9	C. Ibsen	Active	2026-10-27	Due soon	1 open Needs Immediate Action finding
SP-RDG-03 Friction Stir Welding — Floor Panel Cell	Penetration 5.6-6.0 mm; macro-etch each shift	CQI-15	T. Marchetti	Active	2027-02-02	Current	1 open Needs Immediate Action finding
SP-RDG-04 CMT Arc Welding — Tray Frame Cell	AWS D8.14; cut-check each shift	CQI-15	T. Marchetti	Active	2027-04-26	Current	Not at risk

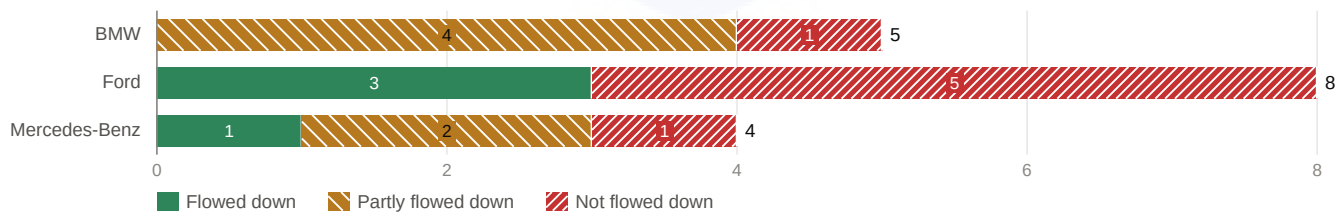
Process	Specification	CQI standard	Assessed by	Status	Assessment due	Assessment	At risk because
SP-RDG-05 Powder Coat — Busbar Covers and Shields	Film 60-90 µm; cure per powder supplier window	CQI-12	L. Ferreira	Active	2026-09-14	Overdue	CQI-12 assessment was due 2026-09-14; 1 open Needs Immediate Action finding
SP-RDG-06 E-Coat — Tray Assemblies	Film 18-28 µm; cure per paint supplier window	CQI-12	L. Ferreira	Active	2027-03-22	Current	Not at risk
SP-RDG-07 Vacuum Brazing — Cooling Plates (Hocking Brazed Components)	Braze joints to the customer leak spec ($\leq 1\text{e-}5$ mbar-l/s helium)	CQI-29	S. Kaur	Active	2026-07-19	Overdue	CQI-29 assessment was due 2026-07-19; 2 open Needs Immediate Action findings
SP-RDG-08 HPDC Aluminium Casting (Sandusky Die Casting)	AlSi10MnMg; porosity per customer X-ray standard	CQI-27	S. Kaur	Active	2027-06-14	Current	Not at risk
SP-RDG-09 Outsourced Aging (Maumee Thermal Processing)	Aging per drawing; conductivity check each load	CQI-9	S. Kaur	Active	2026-07-06	Overdue	CQI-9 assessment was due 2026-07-06
SP-RDG-10 Injection Molded Seal Carriers (Hancock Gasket & Seal)	PA66-GF30; dimensional per drawing	CQI-23	S. Kaur	Active	2027-03-08	Current	Not at risk

Open item: the parts that need 5 special processes are at risk: no current CQI assessment, or findings from one still open.

Customer-specific requirements (4.3.2, 8.4.3.1)

17 customer-specific requirements in scope; 4 recorded as flowed to their supplier or in place in their process.

Customer-specific requirements flowed down



Customer-specific requirements by whether they reached the supplier or the process they apply to, by customer, from the flow-down register. Partly flowed down: the register records part of it in place.

17 customer-specific requirements: 4 flowed down, 6 partly, 7 not flowed down; 13 are not fully in place.

Finding: the requirements below are not recorded as flowed to the supplier, or in place in the process, they apply to. A requirement the organisation accepted and did not pass on is a requirement nobody is meeting.

Requirement	Customer	Requirement text	Flowed to	Gap
CSR-RDG-01	Ford Customer-Specific Requirements for IATF 16949	Leak specification and special characteristics flowed to the brazing supplier	Supplier: Hocking Brazed Components	The PO still quotes the superseded rev B leak specification
CSR-RDG-02	Ford Customer-Specific Requirements for IATF 16949	Special process sources hold a current AIAG CQI assessment	Supplier: Hocking Brazed Components	The brazing supplier's CQI-29 assessment expired in December
CSR-RDG-03	Ford Customer-Specific Requirements for IATF 16949	Ford approval before a process or source change	Process: Change Control	FSW tool source changed without an SREA
CSR-RDG-04	BMW Group CSR for IATF 16949	Part approval for every engineering change	Process: PPAP	Lid rev C submitted without a capability study
CSR-RDG-05	BMW Group CSR for IATF 16949	Measurement systems analysis for every gauge in the control plan	Process: Inspection	No Gage R&R on file for the laser scanner or the conductivity meter
CSR-RDG-06	Mercedes-Benz CSR for IATF 16949	Traceability to the batch for safety-relevant parts	Process: Traceability	Genealogy for July 2025 was not migrated to the new ERP
CSR-RDG-07	Mercedes-Benz CSR for IATF 16949	Run-at-rate before part submission	Process: Launch	Run-at-rate failed; the second run is not scheduled
CSR-RDG-09	BMW Group CSR for IATF 16949	Coating process controls per CQI-12	Process: Powder Coat	Cure oven profile not run since the cutover
CSR-RDG-10	Ford Customer-Specific Requirements for IATF 16949	Qualification of special process operators	Process: FSW	FSW operators are not qualified to ISO 25239-3
CSR-RDG-11	Ford Customer-Specific Requirements for IATF 16949	Special characteristics flowed to sub-tier suppliers	Supplier: Cuyahoga Extrusions	Special characteristics not identified on POs to Cuyahoga Extrusions
CSR-RDG-12	BMW Group CSR for IATF 16949	Right of access for BMW and its supplier quality engineers	Supplier: Portage Precision Machining	Recorded as Partial in the register: not all of it is in place
CSR-RDG-15	BMW Group CSR for IATF 16949	Heat treat sources hold a current CQI-9 assessment	Supplier: Maumee Thermal Processing	Recorded as Partial in the register: not all of it is in place
CSR-RDG-16	Mercedes-Benz CSR for IATF 16949	Anodizing sources hold a current CQI-11 assessment	Supplier: Ashtabula Anodizing	Recorded as Partial in the register: not all of it is in place

Traceability and suspect product (8.5.2.1, 8.7.1.3)

34 material certificates on record for this scope: 5 flagged, 29 verified.

Finding: 5 lots do not have a valid material certificate. Material without traceable certification is suspect product until proven otherwise (8.7.1.3).

Record	Supplier	Lot / heat	What was found
TRC-RDG-01	Hocking Brazed Components	RDG-L25-071 / B-HBC-5512	Missing — leak test data lost in the cutover
TRC-RDG-02	Hocking Brazed Components	RDG-L25-074 / B-HBC-5519	Missing — braze furnace chart not attached

Record	Supplier	Lot / heat	What was found
TRC-RDG-03	Cuyahoga Extrusions	RDG-L26-006 / H-CX-4410	Complete
TRC-RDG-04	Cuyahoga Extrusions	RDG-L26-009 / H-CX-4416	Missing — conductivity not reported
TRC-RDG-08	Sandusky Die Casting	RDG-L26-010 / Shot 8811	Pending — X-ray report

Finding: 2 suspect product cases are recorded against this scope.

Case	Part	Supplier	Stage	What was found
SPR-RDG-001	M6 flow drill screw, Zn-Ni	Marion Fastener Supply	Quarantine	Head marking does not match the manufacturer; cert issued by an unlisted distributor
SPR-RDG-002	RDG-5105	Hocking Brazed Components	Reporting	Leak test results for braze loads 5512 and 5519 were lost in the ERP cutover; plates shipped



15. Ownership, Actions and Management Review

Clauses 5.3, 9.1.1, 9.3 and 10.2.3. An auditor asking who is accountable for a clause is asking for the RACI; asking what is being done about a finding is asking for the action register.

Assigned ownership (5.3)

1 RACI matrix defines accountability for the processes in scope.

Open actions (10.2.3)

Action	Owner	Due	Priority	Status
[ACT-RDG-001] Stand up the recovery room — 6:30 AM review of yesterday's leak and FSW results	Recovery Lead	2026-10-05	High	In Progress
[ACT-RDG-004] Hold every cooling plate tested on HLD-3301 since its last good verification	S. Kaur	2026-10-05	High	In Progress
[ACT-RDG-003] Borrow a certified helium reference leak and re-verify every detector	F. Calloway	2026-10-06	High	In Progress
[ACT-RDG-005] Send Ford the containment and bounding plan for the leak escapes	S. Kaur	2026-10-06	Critical	Not Started
[ACT-RDG-002] Merge the legacy eQMS and the spreadsheet registers into one CAPA list and rank all 19 open	H. Lindgren (QD)	2026-10-07	High	Not Started
[ACT-RDG-012] Load the plant's registers into Aegis AutoAssure® and publish the baseline	Recovery Lead	2026-10-07	Medium	Not Started
[ACT-RDG-015] Quarantine adhesive drums by batch and add expiry checks at the dispense station	N. Oyelaran	2026-10-07	Medium	In Progress
[ACT-RDG-013] Submit the FSW tool source change to Ford for approval	P. Ionescu	2026-10-12	Critical	Not Started
[ACT-RDG-008] Build the recertification gap analysis — the ten findings most likely to be majors	H. Lindgren (QD)	2026-10-14	High	Not Started
[ACT-RDG-010] Re-survey Aging Oven 2 and run the SATs on Furnace 1	C. Ibsen	2026-10-14	High	Not Started
[ACT-RDG-014] Schedule the second Mercedes-Benz run-at-rate	D. Morrow	2026-10-19	Medium	Not Started
[ACT-RDG-007] Qualify the FSW operators to ISO 25239-3; pull unqualified operators off the floor panel cell	M. Whitlock	2026-10-21	High	Not Started
[ACT-RDG-009] Rebuild July 2025 genealogy from the leak testers' local logs and shipping records	J. Pemberton	2026-10-21	High	Not Started

Quality objectives (9.1.1)

1 scorecard tracks quality objectives against target.

Management review (9.3)

No management reviews on record for this scope.

Management review records in the document register:

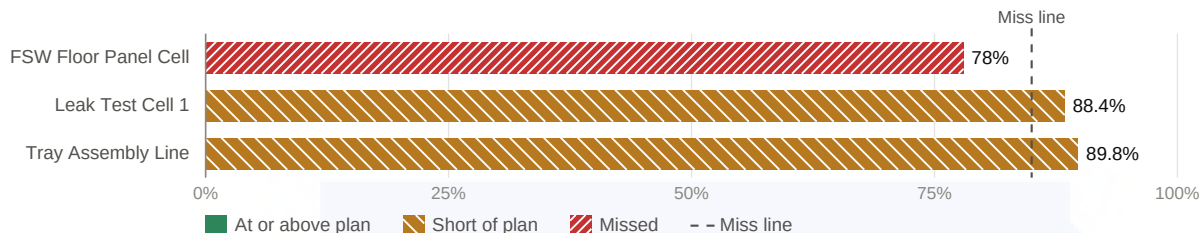
Document	Title	Type	Status
DOC-RDG-036	RPT-225 Management Review Minutes	Record	Approved



16. Operations: the Floor, Material Review and the Supply Base

The floor, hour by hour

30 hour-by-hour boards: 2187 made against a plan of 2560 for the hours recorded (85.4% attainment); 130 hours missed and 59 escalations.



Actual as a share of plan for the hours recorded, every shift. Dashed: the miss line.

Attainment to plan: FSW Floor Panel Cell 78%, Leak Test Cell 1 88.4%, Tray Assembly Line 89.8%.

Cell	09-14	09-15	09-16	09-17	09-18	Week	No reason written	Missed twice running
FSW Floor Panel Cell	77%	79%	79%	76%	79%	78%	12	15
Leak Test Cell 1	88%	91%	89%	91%	84%	88.4%	7	5
Tray Assembly Line	88%	88%	90%	91%	94%	89.8%	8	12

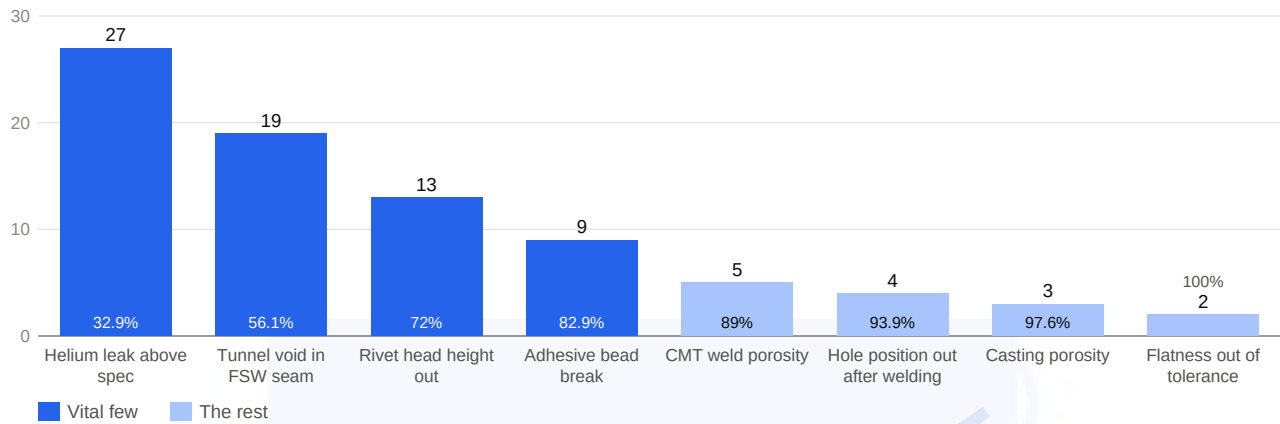
Every escalation, by hour, is on the Floor board page and in the Audit Defense Pack's 13_Operations workbook; 59 escalations this period.

Shift handover: issues still open

Cell	Logged	Shift	Type	Issue	Assigned to	Days open
FSW Floor Panel Cell	2026-09-15	1st	quality	First RDG-5102 macro-etch of the shift shows a tunnel void at seam 3; panels since the last good etch quarantined, NCR-RDG-231 raised and MRB requested	Quality engineering (T. Marchetti)	23
Leak Test Cell 1	2026-09-16	2nd	equipment	HLD-3301 failed its start-of-shift check against the reference leak; cooling plates tested since 15:00 held for re-test	Metrology (F. Calloway)	22
Tray Assembly Line	2026-09-17	1st	material	Cuyahoga side sills short 40 pieces on today's delivery; the Mercedes-Benz tray build moved to Friday	Supplier quality (S. Kaur)	21
Tray Assembly Line	2026-09-18	2nd	quality	Adhesive drum 2 opened past its shelf-life date; bead laid 16:40 to 17:20 to be removed and re-applied	Production (N. Oyelaran)	20

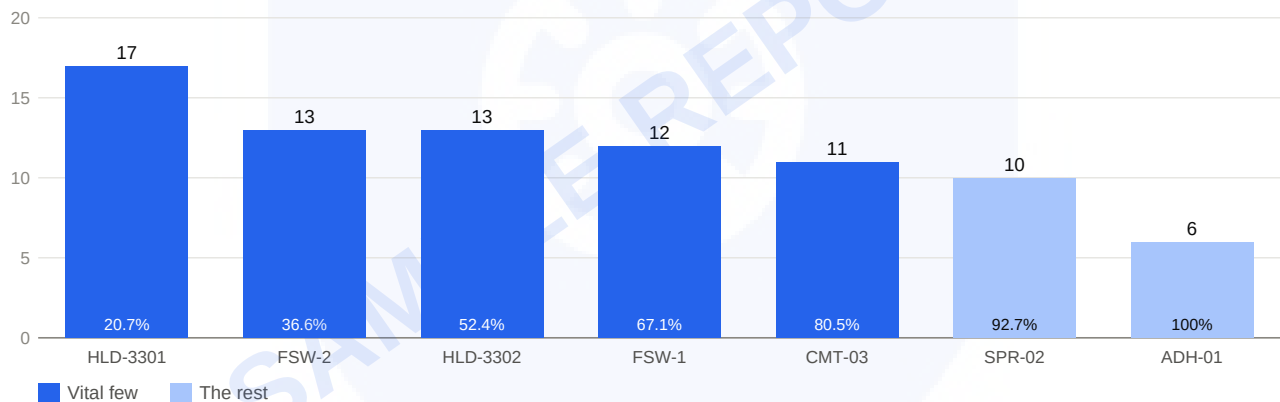
Where the scrap comes from

82 parts scrapped over 52 entries. 4 of 8 reasons make up 82.9%: Helium leak above spec (27), Tunnel void in FSW seam (19), Rivet head height out (13), Adhesive bead break (9).



82 parts scrapped, by reason. Dark bars are the vital few (together 80%); the percentage in each bar is the cumulative share, largest first.

82 parts scrapped, by reason: 4 of 8 account for 82.9%: Helium leak above spec (27), Tunnel void in FSW seam (19), Rivet head height out (13), Adhesive bead break (9).



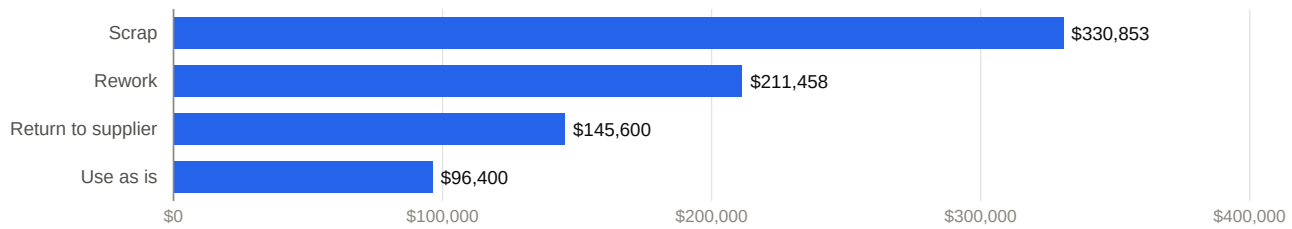
82 parts scrapped, by machine. Dark bars are the vital few (together 80%); the percentage in each bar is the cumulative share, largest first.

82 parts scrapped, by machine: 5 of 7 account for 80.5%: HLD-3301 (17), FSW-2 (13), HLD-3302 (13), FSW-1 (12), CMT-03 (11).

Reason	Parts	Share	Cumulative	By machine
Helium leak above spec	27	32.9%	32.9%	HLD-3301 16, HLD-3302 11
Tunnel void in FSW seam	19	23.2%	56.1%	FSW-1 11, FSW-2 8
Rivet head height out	13	15.9%	72%	SPR-02 6, CMT-03 4, ADH-01 3
Adhesive bead break	9	11%	82.9%	SPR-02 4, ADH-01 3, CMT-03 2
CMT weld porosity	5	6.1%	89%	CMT-03 5
Hole position out after welding	4	4.9%	93.9%	FSW-2 4
Casting porosity	3	3.7%	97.6%	HLD-3302 2, HLD-3301 1
Flatness out of tolerance	2	2.4%	100%	FSW-1 1, FSW-2 1

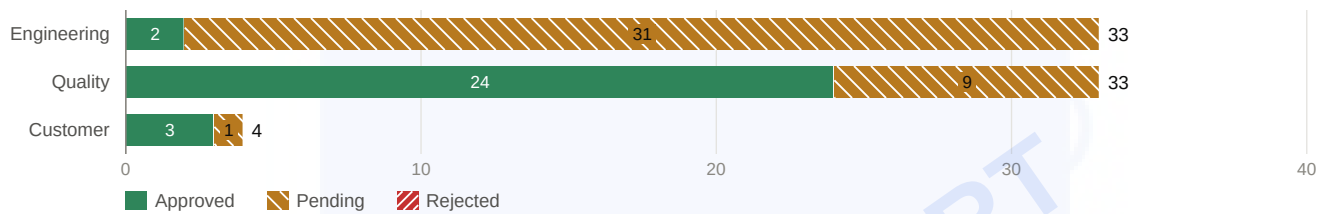
MRB: cost and approvals

33 dispositions with \$784,311 of cost impact recorded; 9 awaiting a board decision.



Cost impact recorded on the dispositions, by disposition.

MRB cost impact \$784,311: Scrap \$330,853, Rework \$211,458, Return to supplier \$145,600, Use as is \$96,400.



Dispositions by reviewer and decision. Customer counts only dispositions that need it.

MRB approvals: Engineering 2 approved, 31 pending, 0 rejected; Quality 24 approved, 9 pending, 0 rejected; Customer 3 approved, 1 pending, 0 rejected.

Disposition	Part	Type	Qty	Cost impact	Eng	Quality	Customer
NCR-RDG-077	RDG-5101	Scrap	6	\$31,500	pending	pending	n/a
NCR-RDG-078	RDG-5106	Use as is	30	\$2,400	pending	pending	pending
NCR-RDG-079	RDG-5112	Rework	11	\$16,700	pending	pending	n/a
NCR-RDG-080	RDG-5115	Return to supplier	240	\$12,000	pending	pending	n/a
NCR-RDG-081	RDG-5110	Rework	75	\$6,300	pending	pending	n/a
NCR-RDG-082	RDG-5109	Rework	32	\$4,100	pending	pending	n/a
NCR-RDG-231	RDG-5102	Scrap	7	\$34,853	pending	pending	n/a
NCR-RDG-232	RDG-5105	Rework	54	\$1,993	approved	pending	n/a
NCR-RDG-233	RDG-5101	Rework	9	\$1,365	approved	pending	n/a

Supplier onboarding



Supplier intakes by stage, from link sent to decision.

3 supplier intakes: 1 invited, 0 in progress, 1 submitted, 1 approved, 0 rejected.

Supplier	Stage	Red flags found	Decided by
Auglaize Extrusion Co	approved	None	H. Lindgren (QD)
Scioto Vacuum Brazing	submitted	Special process 'brazing' offered without a CQI-29 assessment for it; 1 root-cause analyses (8D) not closed: 8D-SVB-14; Stated scrap rate 1.5% but the imported data shows 9.8%	
Muskingum Coatings	invited	None	

Sub-tier special processes (CQI assessments)

The CQI assessment state of every sub-tier special process is charted in Section 14, beside the organisation's own.

Supplier	Sub-tier	Process	CQI	Status	Assessment	Due
Keystone Fastening Systems	Mercer Fastener Finishing	plating	CQI-11	Overdue	MFF-CQI11-305	2026-09-03
Keystone Fastening Systems	Mercer Fastener Finishing	heat treating	CQI-9	Not assessed	—	—
Sandusky Die Casting	Fremont Impregnation & Heat Treat	heat treating	CQI-9	Needs Immediate Action open	FIH-CQI9-2210	2027-07-10
Hocking Brazed Components	Athens Conversion Coatings	coating	CQI-12	Due within 30 days	ACC-CQI12-077	2026-10-26

17. Data Completeness

Complete. Every register supplied for this report was read in full; no section is blank because records could not be read.

Every figure in this pack is drawn from the registers and records your organisation supplied, as they stood when the pack was generated. A section with nothing to report says so in words.

18. Appendices

A. Controlled Document List

Doc ID	Title	Type	Revision	Status	Owner
DOC-RDG-001	QMS-001 Quality Management System Manual	Procedure	Rev 3	Approved	H. Lindgren (QD)
DOC-RDG-002	FSW-110 Friction Stir Welding Set-Up and Tool Life	Work instruction	Rev 2	Under review	T. Marchetti
DOC-RDG-003	LT-210 Helium Leak Test Operation and Daily Verification	Work instruction	Rev 4	Under review	N. Oyelaran
DOC-RDG-004	HT-310 Solution Heat Treat and Aging per CQI-9	Procedure	Rev 3	Obsolete	C. Ibsen
DOC-RDG-005	ADH-410 Structural Adhesive Handling and Shelf Life	Work instruction	Rev 1	Approved	N. Oyelaran
DOC-RDG-006	CAL-510 Calibration and Measurement Systems Control	Procedure	Rev 5	Under review	F. Calloway
DOC-RDG-007	NCP-610 Nonconforming Product and MRB	Procedure	Rev 2	Approved	H. Lindgren (QD)
DOC-RDG-008	TRN-710 Operator Qualification and Records	Procedure	Rev 1	Obsolete	M. Whitlock
DOC-RDG-009	TRC-810 Lot Genealogy and Traceability	Procedure	Rev 2	Under review	J. Pemberton
DOC-RDG-010	PPAP-910 Production Part Approval and Run-at-Rate	Procedure	Rev 3	Approved	D. Morrow
DOC-RDG-011	FRM-120 Nonconformance Report Form	Form	Rev 4	Approved	H. Lindgren (QD)
DOC-RDG-012	RPT-220 Weekly Quality and Launch Report	Record	Rev 1	Under review	Recovery Lead
DOC-RDG-013	QP-4.1 Context and Interested Parties	Procedure	Rev 6	Approved	C. Ibsen
DOC-RDG-014	QP-5.3 Roles and Authorities	Procedure	Rev 6	Approved	J. Pemberton
DOC-RDG-015	QP-6.1 Risk and Contingency Planning	Procedure	Rev 3	Under review	D. Morrow
DOC-RDG-016	QP-7.1.5 Monitoring and Measuring Resources	Procedure	Rev 5	Under review	S. Kaur
DOC-RDG-017	QP-7.2 Competence	Procedure	Rev 3	Under review	C. Ibsen
DOC-RDG-018	QP-7.5 Control of Documented Information	Procedure	Rev 2	Approved	J. Pemberton
DOC-RDG-019	QP-8.4 Control of External Providers	Procedure	Rev 1	Approved	E. Vasko
DOC-RDG-020	QP-8.5.2 Identification and Traceability	Procedure	Rev 5	Approved	F. Calloway
DOC-RDG-021	QP-8.5.6 Control of Changes	Procedure	Rev 5	Approved	T. Marchetti
DOC-RDG-022	QP-8.7 Control of Nonconforming Output	Procedure	Rev 5	Approved	N. Oyelaran
DOC-RDG-023	QP-8.7.1.3 Control of Suspect Product	Procedure	Rev 1	Approved	D. Morrow
DOC-RDG-024	QP-9.2 Internal Audit and Layered Process Audits	Procedure	Rev 3	Approved	N. Oyelaran
DOC-RDG-025	QP-9.3 Management Review	Procedure	Rev 6	Approved	T. Marchetti
DOC-RDG-026	QP-10.2 Nonconformity and Corrective Action	Procedure	Rev 2	Under review	J. Pemberton
DOC-RDG-027	QP-4.3.2 Customer-Specific Requirements Matrix	Procedure	Rev 4	Approved	T. Marchetti

Doc ID	Title	Type	Revision	Status	Owner
DOC-RDG-028	QP-8.3.4.4 Production Part Approval (PPAP)	Procedure	Rev 5	Under review	D. Morrow
DOC-RDG-029	WI-FSW-115 FSW Macro-Etch Sampling	Work instruction	Rev 3	Approved	P. Ionescu
DOC-RDG-030	WI-PC-330 Powder Coat Oven Profiling	Work instruction	Rev 4	Approved	T. Marchetti
DOC-RDG-031	WI-SPR-140 Rivet Joint Cross-Section	Work instruction	Rev 5	Approved	T. Marchetti
DOC-RDG-032	WI-HT-315 Aging Oven Load Thermocouples	Work instruction	Rev 2	Approved	J. Pemberton
DOC-RDG-033	FRM-205 Supplier Scorecard	Form	Rev 6	Approved	D. Morrow
DOC-RDG-034	FRM-310 Calibration Recall Notice	Form	Rev 5	Approved	F. Calloway
DOC-RDG-035	FRM-415 Layout Inspection Report	Form	Rev 4	Under review	F. Calloway
DOC-RDG-036	RPT-225 Management Review Minutes	Record	Rev 2	Approved	J. Pemberton

B. Acronyms & Definitions

Acronym	Definition
8D	Eight Disciplines problem solving
APQP	Advanced Product Quality Planning
CAPA	Corrective and Preventive Action
CQI	AIAG Continuous Quality Improvement special process assessment
CSR	Customer-Specific Requirement
FMEA	Failure Mode and Effects Analysis (PFMEA: process FMEA)
GR&R	Gauge Repeatability and Reproducibility
IATF 16949	Quality management system requirements for automotive production and relevant service parts organizations
IATF	International Automotive Task Force
ISO	International Organization for Standardization
KPI	Key Performance Indicator
MRB	Material Review Board
MSA	Measurement System Analysis
NCR	Nonconformance Report
NDT	Non-Destructive Testing
OEM	Original Equipment Manufacturer (the vehicle manufacturer)
OJT	On-the-Job Training
OTD	On-Time Delivery
PPAP	Production Part Approval Process
PPM	Parts Per Million (defective)
PSW	Part Submission Warrant
QBR	Quarterly Business Review

Acronym	Definition
QMS	Quality Management System
RACI	Responsible, Accountable, Consulted, Informed
RCA	Root Cause Analysis
SHA-256	Secure Hash Algorithm (256-bit): the file fingerprint in the manifest
SOP	Standard Operating Procedure
SPC	Statistical Process Control

C. Pack Generation

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D. Your Audit Checklist

69 questions an IATF 16949 auditor asks about the 21 clauses where your own records show a gap, critical first. Walk each clause with its process owner before the audit: tick a question only when the evidence is in hand, and note where it is kept. The full set of 251 questions is in the ZIP (11_Self_Audit/IATF16949_Self_Audit_Questions.xlsx), with these marked.

Clauses 8.7.1.3 and 8.7.1.7: Control of suspect product

What your records showed

What was found	Records	Which
Suspect product case open	2	SPR-RDG-002, SPR-RDG-001
Containment past its deadline	2	NCR-RDG-016, NCR-RDG-066
Nonconforming product disposition incomplete	4	RDG-5105 (MRB-RDG-306), RDG-5115 (MRB-RDG-310), RDG-5101 (MRB-RDG-307), RDG-5102 (MRB-RDG-341)

	Question	Priority	Evidence location / initials
☐	IATF-8.7-001 Does the organization ensure nonconforming outputs are identified, controlled, and prevented from unintended use or delivery?	Critical	
☐	IATF-8.7-003 Is scrap material identified, controlled, and prevented from being used in production?	High	
☐	IATF-8.7-004 Does the organization track and trend nonconformance data to identify systemic issues and drive improvement?	High	
☐	IATF-8.7-002 Does the organization maintain records of nonconforming product disposition and corrective actions taken?	Medium	

	Have ready		
☐	Nonconforming material procedure	☐	Nonconformance trend data

Have ready			
☐	Quarantine area controls	☐	Pareto analysis
☐	Red tag/hold identification system	☐	Systemic issue identification
☐	Disposition records	☐	Improvement actions from trend analysis
☐	Scrap identification procedure	☐	Nonconformity log
☐	Scrap area controls	☐	Corrective action linkage
☐	Scrap tracking records	☐	NC trend analysis
☐	Physical barriers to prevent reuse		
Red flags an auditor looks for			
No quarantine system		No NC trend analysis	
Nonconforming product mixed with good product		Same types of NCs recurring without action	
Scrap mixed with production material		No NC documentation	
No scrap tracking		Repeat NCs without corrective action	

Clause 8.5.1: Control of production and service provision

What your records showed

What was found	Records	Which
Special process with an open Needs Immediate Action finding	2	Friction Stir Welding — Floor Panel Cell, Artificial Aging — Aging Oven 2
Customer requirement not flowed down	1	CSR-RDG-09 (BMW) in Powder Coat
FIND-RDG-002 Tunnel voids in FSW floor panels — macro-etch frequency below the control plan	-	
FIND-RDG-005 Aging oven TUS failure at location 7 and SAT overdue on Furnace 1 (CQI-9 Section 3)	-	

	Question	Priority	Evidence location / initials
☐	IATF-8.5.1.1-001 Does the organization develop and maintain control plans at system, subsystem, component, and/or material level?	Critical	
☐	IATF-8.5.1.1-003 Does the control plan include the reaction plan for each characteristic when an out-of-control condition is detected?	Critical	
☐	IATF-CQI-001 Does the control plan include all required elements per IATF Annex A including reaction plans, measurement frequency, and statistical evaluation methods?	Critical	
☐	IATF-8.5-008B Are standardized work instructions available at all workstations and do they include setup verification, first piece inspection, and reaction plans?	High	
☐	IATF-8.5-009B Are production job setups verified using documented methods including first-off/last-off comparison and statistical validation where applicable?	High	

	Question	Priority	Evidence location / initials
☐	IATF-8.5-010B Does the organization have a total productive maintenance (TPM) system including planned maintenance, predictive maintenance, and maintenance objectives?	High	
☐	IATF-8.5-011B Are production tooling, gauges, and test equipment managed with identification, storage, periodic verification, and change management?	High	
☐	IATF-8.5.1.1-002 Are control plans reviewed and updated when product, manufacturing process, measurement, logistics, supply sources, production volume changes, or FMEA updates occur?	High	
☐	IATF-8.5.1.2-001 Are standardized work instructions and visual standards available at the workstation for all employees having responsibility for the operation of processes?	High	
☐	IATF-8.5.1.3-001 Is verification of job set-ups performed whenever a set-up is executed including initial material run, material changeover, and job changeover?	High	
☐	IATF-8.5.1.5-001 Does the organization implement a total productive maintenance (TPM) system including maintenance objectives, documented maintenance plans, and packaging of maintenance parts?	High	
☐	IATF-8.5.1.6-001 Does the organization manage production tooling including tool design, fabrication, repair, and replacement?	High	
☐	IATF-8.5.1.2-002 Are work instructions verified by a qualified person to ensure they accurately reflect the production process?	Medium	
☐	IATF-8.5.1.4-001 Does the organization verify production after a planned or unplanned shutdown to confirm product meets specifications upon restart?	Medium	
☐	IATF-8.5.1.5-002 Are OEE (Overall Equipment Effectiveness) targets established for key production equipment and tracked regularly?	Medium	
☐	IATF-8.5.1.6-002 Are customer-owned tooling and equipment identified, verified, protected, and safeguarded?	Medium	
☐	IATF-8.5.1.7-001 Does the organization ensure production scheduling meets customer orders/demand including a scheduling information system?	Medium	
☐	IATF-CQI-007 Does the organization have documented standard work for all production operations including cycle times, work sequence, and quality checks at each station?	Medium	

	Have ready		
☐	Control plans (prototype, pre-launch, production)	☐	Operator training records on work instructions
☐	PFMEA cross-reference	☐	Instruction revision control
☐	Process flow diagram cross-reference	☐	Set-up verification procedure
☐	Reaction plans for nonconformance	☐	Last piece retention records
☐	Control plan reaction plans	☐	Set-up approval documentation
☐	Operator awareness of reaction plans	☐	Preventive maintenance schedules
☐	Evidence of reaction plan execution	☐	OEE tracking records
☐	Containment action records	☐	Maintenance spare parts inventory
☐	Control plan per IATF Annex A format	☐	Tooling identification/markings
☐	Reaction plans for each control	☐	Customer-owned tooling records
☐	Measurement frequency justification	☐	Tooling maintenance/repair records

Have ready			
☐	Statistical method identification	☐	Tooling life tracking
☐	Standardized work instructions at workstations	☐	Work instruction approval records
☐	Setup verification checklists	☐	Instruction validation records
☐	First piece inspection records	☐	Process engineer sign-off
☐	Reaction plans for out-of-spec conditions	☐	Operator feedback on instructions
☐	Setup verification procedure	☐	Shutdown/start-up verification procedure
☐	First-off inspection records	☐	Start-up inspection records
☐	Last-off part retention	☐	Re-qualification records after extended shutdown
☐	Statistical setup validation where applicable	☐	OEE calculation records
☐	TPM program documentation	☐	OEE targets
☐	Planned maintenance schedules	☐	OEE trend data
☐	Predictive maintenance records	☐	Improvement actions for low OEE
☐	Maintenance KPIs (OEE, MTBF, MTTR)	☐	Customer tooling register
☐	Spare parts management	☐	Tooling identification markings
☐	Tooling register	☐	Tooling condition records
☐	Tool identification system	☐	Customer notification of tooling issues
☐	Tool storage conditions	☐	Production scheduling system
☐	Tool life management records	☐	Customer order tracking
☐	Tool change management records	☐	Capacity planning records
☐	Control plan revision history	☐	Schedule adherence metrics
☐	Change linkage records (FMEA to CP)	☐	Standard work documents at each station
☐	Engineering change notices triggering CP updates	☐	Cycle time documentation
☐	Work instructions at workstations	☐	Work sequence documentation
☐	Visual standards/aids	☐	Quality check integration in standard work

Red flags an auditor looks for	
No control plans	No set-up verification
Control plans severely outdated	Production running without first piece approval
Reaction plans not defined	No preventive maintenance program
No reaction plans	Chronic equipment breakdowns affecting delivery
Operators unaware of what to do for nonconformance	Customer tooling unidentified
Control plans incomplete per Annex A	Instructions not approved
No work instructions at workstations	Instructions don't match actual process
Operators not following documented instructions	Production start without verification
No setup verification	No shutdown procedure
Production starts without setup approval	No OEE tracking

Red flags an auditor looks for	
No planned maintenance program	OEE consistently below 60% without action
Frequent unplanned downtime	Damaged customer tooling not reported
No tooling management system	No production scheduling system
Worn tools continuing in production	Chronic schedule adherence issues
Control plans with no revision for 2+ years	No standard work
Recent process changes not in control plan	Operators doing work differently than documented
Operators following different methods than documented	

Clause 4.3.2: Customer-specific requirements

What your records showed

What was found	Records	Which
Special process with an open Needs Immediate Action finding	2	Powder Coat — Busbar Covers and Shields, Vacuum Brazing — Cooling Plates (Hocking Brazed Components)
Special process without a current CQI assessment	1	Outsourced Aging (Maumee Thermal Processing)
Customer requirement not flowed down	1	CSR-RDG-12 (BMW) to Portage Precision Machining

	Question	Priority	Evidence location / initials
☐	IATF-CQI-012 Has the organization identified and documented all customer-specific requirements (CSRs) and integrated them into the quality management system?	Critical	
☐	IATF-4.2-002 Are customer-specific requirements (CSRs) identified, understood, and integrated into the QMS?	High	

	Have ready		
☐	CSR register per customer	☐	CSR register per OEM customer
☐	CSR integration into QMS processes	☐	Evidence of CSR review meetings
☐	CSR communication to relevant functions	☐	Process updates linked to CSR changes
☐	CSR review and update records		

Red flags an auditor looks for	
No CSR register	CSR register not maintained
CSRs unknown to operational staff	Operations unaware of key CSR requirements

Clause 8.4.2.1: Type and extent of control — supplemental

What your records showed

What was found	Records	Which
Sub-tier special process with an open Needs Immediate Action finding	1	Fremont Impregnation & Heat Treat (heat treating for Sandusky Die Casting)
Sub-tier special process without a current CQI assessment	2	Mercer Fastener Finishing (heat treating for Keystone Fastening Systems), Mercer Fastener Finishing (plating for Keystone Fastening Systems)

	Question	Priority	Evidence location / initials
☐	IATF-8.4.2.1-001 Does the organization ensure externally provided processes remain within the control of its QMS through second-party audits?	High	

	Have ready		
☐	Supplier audit schedule (risk-based)	☐	Audit finding corrective actions
☐	Supplier audit reports	☐	Supplier audit competency records

Red flags an auditor looks for	
No supplier audit program	Critical suppliers never audited

Clause 8.3.4.4: Product approval process

What your records showed

What was found	Records	Which
Part submission warrant unsigned	5	RDG-5102 (PPAP-RDG-5102), RDG-5105 (PPAP-RDG-5105), RDG-5103 (PPAP-RDG-5103), RDG-5101 (PPAP-RDG-5101), RDG-5104 (PPAP-RDG-5104)
Customer requirement not flowed down	2	CSR-RDG-04 (BMW) in PPAP, CSR-RDG-07 (Mercedes-Benz) in Launch
FIND-RDG-009 Lid PPAP rejected — flatness not capable (Cpk 0.71)	-	
FIND-RDG-010 Run-at-rate failed on RDG-5108 — cycle time and hole position capability	-	

	Question	Priority	Evidence location / initials
☐	IATF-8.3.4.4-001 Does the organization use a product approval process (PPAP) conforming to customer requirements?	Critical	
☐	IATF-8.3-008B Is the product approval process (PPAP/PPA) applied to externally provided products, materials, and services?	High	

	Have ready		
☐	PPAP procedure	☐	Supplier PPAP requirements

Have ready			
☐	PPAP submission records	☐	Supplier PPAP status tracker
☐	Customer PPAP approval records (PSW)	☐	Supplier PPAP submissions
☐	PPAP timing relative to production start	☐	Customer-specific PPAP requirements
Red flags an auditor looks for			
Shipping production parts without PPAP approval		No supplier PPAP requirements	
No PPAP process		Production materials used without PPAP approval	

Clause 8.5.1.1: Control plan

What your records showed

What was found	Records	Which
High-priority failure mode with no control	5	RDG-5102 (PFMEA-RDG-5102), RDG-5104 (PFMEA-RDG-5104), RDG-5105 (PFMEA-RDG-5105), RDG-5101 (PFMEA-RDG-5101), RDG-5103 (PFMEA-RDG-5103)
FIND-RDG-001 Coolant leak escapes at 212 PPM — the leak test reaction plan was not followed	-	

	Question	Priority	Evidence location / initials
☐	IATF-8.5.1.1-001 Does the organization develop and maintain control plans at system, subsystem, component, and/or material level?	Critical	
☐	IATF-8.5.1.1-003 Does the control plan include the reaction plan for each characteristic when an out-of-control condition is detected?	Critical	
☐	IATF-CQI-001 Does the control plan include all required elements per IATF Annex A including reaction plans, measurement frequency, and statistical evaluation methods?	Critical	
☐	IATF-8.5.1.1-002 Are control plans reviewed and updated when product, manufacturing process, measurement, logistics, supply sources, production volume changes, or FMEA updates occur?	High	

Have ready			
☐	Control plans (prototype, pre-launch, production)	☐	Control plan per IATF Annex A format
☐	PFMEA cross-reference	☐	Reaction plans for each control
☐	Process flow diagram cross-reference	☐	Measurement frequency justification
☐	Reaction plans for nonconformance	☐	Statistical method identification
☐	Control plan reaction plans	☐	Control plan revision history
☐	Operator awareness of reaction plans	☐	Change linkage records (FMEA to CP)
☐	Evidence of reaction plan execution	☐	Engineering change notices triggering CP updates
☐	Containment action records		

Red flags an auditor looks for	
No control plans	Operators unaware of what to do for nonconformance
Control plans severely outdated	Control plans incomplete per Annex A
Reaction plans not defined	Control plans with no revision for 2+ years
No reaction plans	Recent process changes not in control plan

Clause 7.1.5.2.1: Calibration/verification records

What your records showed

What was found	Records	Which
Measurements in doubt after a calibration failure	2	Helium Leak Detector (vacuum chamber) (CAL-RDG-01), Measuring Microscope (FSW macro-etch) (CAL-RDG-06)
FIND-RDG-004 Helium reference leak and 18 other instruments past calibration	-	

	Question	Priority	Evidence location / initials
☐	IATF-7.1-008B Are calibration records maintained and do they include measurement uncertainty analysis for critical measurements?	High	
☐	IATF-7.1.5.2.1-001 Are calibration/verification records maintained for all monitoring and measuring devices including employee-owned and customer-owned equipment?	High	

	Have ready		
☐	Calibration records	☐	Calibration database/register
☐	Measurement uncertainty analysis	☐	Calibration certificates
☐	Calibration certificates traceable to national standards	☐	Calibration labels/stickers on equipment
☐	Out-of-calibration investigation records	☐	Out-of-calibration action procedure

Red flags an auditor looks for	
No calibration records	Overdue calibrations in use
Measurement uncertainty not considered for critical measurements	No calibration program for employee tools

Clause 8.5.2.1: Identification and traceability — supplemental

What your records showed

What was found	Records	Which
Customer requirement not flowed down	1	CSR-RDG-06 (Mercedes-Benz) in Traceability
FIND-RDG-003 Leak test results and genealogy for three weeks of July 2025 not migrated — suspect lots cannot be bounded	-	

	Question	Priority	Evidence location / initials
☐	IATF-8.5.2.1-001 Does the organization use identification and traceability systems to enable detection of product that is suspect or nonconforming?	High	
☐	IATF-8.5.2.1-002 Can the organization demonstrate full traceability from raw material through finished product for a randomly selected production lot?	High	

Have ready	
☐ Traceability procedure	☐ Lot/batch traceability records
☐ Lot/batch identification records	☐ Material certificates linked to lots
☐ Containment action examples	☐ In-process identification
☐ Material traceability from receipt to shipment	☐ Shipment records with lot linkage

Red flags an auditor looks for	
No traceability system	Cannot complete traceability
Cannot trace product to material lots	Traceability takes days not hours

Clause 7.2.2: Competence — on-the-job training

What your records showed

What was found	Records	Which
FIND-RDG-006 FSW operators producing without qualification to ISO 25239-3	-	

	Question	Priority	Evidence location / initials
☐	IATF-7.2-004B Does the organization have an on-the-job training process including adequate supervision until competence is demonstrated for new or modified jobs affecting quality?	High	

Have ready	
☐ OJT procedure	☐ Supervision log during training
☐ OJT records per employee	☐ Competence verification records post-OJT

Red flags an auditor looks for	
Untrained operators on quality-critical processes	No OJT records

Clause 8.6.2: Layout inspection and functional testing

What your records showed

What was found	Records	Which
Layout inspection report not signed	12	RDG-5105 (LAY-RDG-105), RDG-5104 (LAY-RDG-104), RDG-5101 (LAY-RDG-101), RDG-5114 (LAY-RDG-114), RDG-5102 (LAY-RDG-102), RDG-5112 (LAY-RDG-112), RDG-5113 (LAY-RDG-113), RDG-5108 (LAY-RDG-108), RDG-5110 (LAY-RDG-110), RDG-5115 (LAY-RDG-115), RDG-5116 (LAY-RDG-116), RDG-5103 (LAY-RDG-103)
Accepted layout inspection report with no signatures on record	4	RDG-5109 (LAY-RDG-109), RDG-5107 (LAY-RDG-107), RDG-5106 (LAY-RDG-106), RDG-5111 (LAY-RDG-111)

	Question	Priority	Evidence location / initials
☐	IATF-8.6-002B Does the organization perform layout inspection and functional testing per customer requirements at frequencies defined in the control plan?	High	
☐	IATF-8.6.3-001 Does the organization perform layout inspection and functional verification per customer engineering standards at frequencies specified in the control plan?	High	

	Have ready		
☐	Layout inspection schedule	☐	Layout inspection reports
☐	Layout inspection results	☐	Customer frequency requirements
☐	Functional test records	☐	Full dimensional reports
☐	Customer-specified frequencies		

Red flags an auditor looks for	
No layout inspections performed	No layout inspection program
Critical dimensions missed in layout	Years between layout inspections

Clause 7.1.5.1.1: Measurement systems analysis

What your records showed

What was found	Records	Which
Customer requirement not flowed down	1	CSR-RDG-05 (BMW) in Inspection
Capability claim resting on an unqualified gauge	3	Conductivity on RDG-5106 measured with ECM-2911, FSW penetration depth on RDG-5102 measured with MS-1407, Powder coat thickness on RDG-5110 measured with CTG-2710

	Question	Priority	Evidence location / initials
☐	IATF-7.1.5.1.1-001 Has the organization conducted measurement system analysis (MSA) studies for all monitoring and measuring equipment referenced in the control plan?	Critical	
☐	IATF-CQI-003 Are Gage R&R studies conducted for all measurement systems referenced in the control plan, and are acceptance criteria per AIAG MSA manual applied?	Critical	
☐	IATF-7.1.5.1.1-002 Are MSA studies repeated when there are changes to the measurement system including gage modifications, repairs, or operator changes?	High	
☐	IATF-9.1.1.3-001 Has the organization verified the effectiveness of all purchased inspection, test, and measuring equipment per customer requirements?	High	

Have ready	
☐ MSA study records (GR&R, bias, linearity, stability)	☐ MSA re-study trigger criteria
☐ Control plan with gage identification	☐ Re-study records after changes
☐ MSA acceptance criteria	☐ Gage modification/repair logs
☐ Action plans for failed MSA studies	☐ Equipment qualification records
☐ GR&R study records per control plan	☐ MSA studies for new equipment
☐ Acceptance criteria documentation	☐ Customer acceptance of measurement methods
☐ Action plans for marginal measurement systems	☐ Correlation studies between sites
☐ Attribute MSA studies	

Red flags an auditor looks for	
No MSA program	MSA never repeated
Control plan gages without MSA studies	Gages modified without re-study
Measurement decisions based on unvalidated gages	Equipment in use without qualification
No GR&R studies	No MSA on measurement equipment
Measurement systems with >30% GR&R still in use	

Clause 8.7.1.1: Customer authorization for concession

What your records showed

What was found	Records	Which
Nonconforming product disposition incomplete	4	RDG-5107 (MRB-RDG-325), RDG-5107 (MRB-RDG-315), RDG-5103 (MRB-RDG-322), RDG-5106 (MRB-RDG-308)

Question	Priority	Evidence location / initials
☐ IATF-8.7.1.1-001 Is customer authorization obtained prior to further processing or shipment of product that does not conform to requirements?	Critical	

Have ready	
☐ Customer concession approvals	☐ Shipment hold/release records
☐ Concession request records	

Red flags an auditor looks for	
Shipping without customer concession	No concession process

Clause 10.2.3: Problem solving

What your records showed

What was found	Records	Which
Corrective action past due	4	CAPA-RDG-057, CAPA-RDG-058, CAPA-RDG-059, CAPA-RDG-060

Question	Priority	Evidence location / initials
☐ IATF-10.2-005B Does the organization apply a disciplined problem-solving methodology (8D, A3, DMAIC) with root cause analysis verified by testing the root cause theory?	Critical	
☐ IATF-10.2.3-001 Does the organization use a systematic problem-solving methodology (e.g., 8D, A3) for root cause analysis?	Critical	
☐ IATF-10.2.3-002 Does the organization have trained problem-solving facilitators who can lead 8D, A3, or equivalent structured problem-solving?	Medium	

Have ready	
☐ Problem-solving records using structured methodology	☐ Root cause analysis records
☐ Root cause verification evidence	☐ Verification of root cause elimination
☐ Containment action effectiveness	☐ Problem-solving training records
☐ Recurrence prevention verification	☐ Problem-solving facilitator list
☐ Problem-solving procedure	☐ Facilitator training records

Have ready			
☐	8D/A3 reports	☐	Assignment records for major issues
Red flags an auditor looks for			
No structured problem-solving		No verification of corrective action effectiveness	
Root causes never verified		No trained facilitators	
No formal problem-solving methodology		Problem-solving done without facilitation	
Root cause always 'operator error'			

Clause 8.4.2.3: Supplier quality management system development

What your records showed

What was found	Records	Which
Customer requirement not flowed down	3	CSR-RDG-02 (Ford) to Hocking Brazed Components, CSR-RDG-16 (Mercedes-Benz) to Ashtabula Anodizing, CSR-RDG-15 (BMW) to Maumee Thermal Processing

	Question	Priority	Evidence location / initials
☐	IATF-8.4.2.4.1-001 Does the organization require regulatory compliance verification for embedded software in automotive electronic products?	High	
☐	IATF-8.4-007B Does the organization have a supplier quality management system development process for suppliers not yet third-party certified to IATF 16949?	Medium	
☐	IATF-8.4.1.2-001 Is the organization monitoring external provider QMS development with the goal of IATF 16949 compliance?	Medium	
☐	IATF-8.4.2.3.1-001 Does the organization require external providers of automotive products/services to develop, implement, and improve their QMS?	Medium	

Have ready			
☐	Software development process assessments	☐	Supplier QMS certification status list
☐	ASPICE/CMMI compliance records	☐	Supplier development roadmaps
☐	Software validation records	☐	QMS audit results
☐	Cybersecurity assessment	☐	Certification timeline tracking
☐	Supplier development procedure	☐	Supplier QMS development plans
☐	Supplier certification roadmap	☐	Self-assessment questionnaires
☐	Second-party audit schedule for uncertified suppliers	☐	Supplier improvement project records
☐	Development milestones and progress	☐	IATF certification roadmaps

Red flags an auditor looks for			
No software development process assessment		No supplier QMS tracking	
Safety-critical software without validation		Key suppliers without QMS certification	
No supplier development program		Suppliers with no QMS improvement plan	

Red flags an auditor looks for	
Suppliers with no certification path	

Clause 8.4.3.1: Information for external providers — supplemental

What your records showed

What was found	Records	Which
Customer requirement not flowed down	2	CSR-RDG-01 (Ford) to Hocking Brazed Components, CSR-RDG-11 (Ford) to Cuyahoga Extrusions

	Question	Priority	Evidence location / initials
☐	IATF-8.4.2.3-001 Does the organization require external providers to cascade applicable requirements including customer-specific requirements to sub-tiers?	High	
☐	IATF-CQI-013 Do purchase orders and agreements with suppliers include requirements for cascading applicable requirements to sub-tier suppliers including customer-specific requirements?	High	

	Have ready		
☐	Supplier quality agreements with flowdown clauses	☐	Supplier agreements with cascade clauses
☐	Sub-tier audit evidence	☐	Sub-tier supplier management requirements
☐	Customer requirement flowdown verification	☐	Evidence of flowdown verification
☐	Purchase order templates with flowdown requirements		

Red flags an auditor looks for	
No sub-tier management	No flowdown requirements
Quality agreements without flowdown requirements	No visibility into sub-tier supply chain

Clause 8.5.6.1: Control of changes — supplemental

What your records showed

What was found	Records	Which
Customer requirement not flowed down	1	CSR-RDG-03 (Ford) in Change Control
FIND-RDG-007 FSW tool source changed without customer notification or PPAP	-	

	Question	Priority	Evidence location / initials
☐	IATF-8.5-014B Does the organization use a systematic approach for temporary change management including alternative controls during deviation from standard process?	Critical	

	Question	Priority	Evidence location / initials
☐	IATF-8.5.6.1-001 Does the organization have a process to control and react to changes impacting product realization including process changes, software changes, and supplier changes?	Critical	
☐	IATF-8.5.6.1-002 Is there a process for verifying that all changes, including those made by suppliers, are validated before implementation?	Critical	
☐	IATF-8.5.6.1.1-001 Does the organization manage temporary changes to process controls including backup or alternative methods?	High	

Have ready	
☐ Temporary change procedure	☐ Re-PPAP records for changes
☐ Temporary change register	☐ Change validation procedure
☐ Alternative control documentation	☐ Supplier change notification records
☐ Time limits and review dates	☐ Validation study results
☐ Return-to-standard verification	☐ Customer approval for changes
☐ Change management procedure	☐ Temporary change/deviation procedure
☐ Change request/approval records	☐ Temporary change log
☐ Impact assessment for changes	☐ Additional inspection during temporary changes
☐ Customer notification records	☐ Time limit and re-validation records

Red flags an auditor looks for	
No temporary change management	Changes implemented without validation
Temporary changes running indefinitely	Supplier changes not tracked
Changes without approval	Permanent temporary changes
No customer notification process for changes	No tracking of temporary deviations

Clause 7.2.1: Competence — supplemental

What your records showed

What was found	Records	Which
Customer requirement not flowed down	1	CSR-RDG-10 (Ford) in FSW

	Question	Priority	Evidence location / initials
☐	IATF-7.2.1-001 Has the organization determined training needs including awareness of relevant automotive customer requirements for all personnel?	High	
☐	IATF-7.2-003B Has the organization conducted a training needs analysis including awareness of relevant IATF 16949 requirements for personnel affecting product quality?	Medium	

Have ready			
☐	Training needs analysis	☐	Skills matrix
☐	Training plan/matrix	☐	Automotive core tools training records
☐	Customer requirement awareness training records	☐	IATF 16949 awareness training records
☐	On-the-job training records		
Red flags an auditor looks for			
No training needs analysis		Quality-affecting personnel untrained on core tools	
Customer requirements not part of employee training			

Clause 8.7.1.4: Control of reworked product

What your records showed

What was found	Records	Which
Nonconforming product disposition incomplete	12	RDG-5110 (MRB-RDG-316), RDG-5108 (MRB-RDG-305), RDG-5109 (MRB-RDG-313), RDG-5104 (MRB-RDG-303), RDG-5106 (MRB-RDG-318), RDG-5105 (MRB-RDG-321), RDG-5104 (MRB-RDG-314), RDG-5105 (MRB-RDG-342), RDG-5112 (MRB-RDG-309), RDG-5110 (MRB-RDG-311), RDG-5101 (MRB-RDG-343), RDG-5109 (MRB-RDG-312)

	Question	Priority	Evidence location / initials
☐	IATF-8.7.1.2-001 Is reworked product re-verified using the original control plan or a separate rework control plan?	High	
☐	IATF-CQI-008 Does the organization have a defined process for management of reworked product including re-inspection, re-test, and traceability requirements?	High	
☐	IATF-8.7.1.4-001 Does the organization have a documented process to confirm that previously nonconforming product is properly dispositioned before reuse in manufacturing?	Medium	

Have ready			
☐	Rework instructions	☐	Re-inspection records
☐	Rework control plan	☐	Rework traceability records
☐	Re-inspection records after rework	☐	Re-introduction procedure
☐	Rework tracking records	☐	MRB records
☐	Rework procedure	☐	Sign-off records before reuse
☐	Rework instructions per product	☐	Traceability of reworked product

Red flags an auditor looks for			
Rework without instructions		Reworked product returned without sign-off	
No re-inspection after rework		No traceability of reworked items	
Uncontrolled rework			

Clause 8.4.2.4: Supplier monitoring

What your records showed

What was found	Records	Which
FIND-RDG-011 Cuyahoga Extrusions twist trend not escalated across 4 lots	-	

	Question	Priority	Evidence location / initials
☐	IATF-8.4-008B Does the organization monitor supplier performance using defined KPIs including PPM, delivery, premium freight, and customer disruptions?	High	
☐	IATF-8.4.1.3-001 Are customer-designated special status sources (e.g., Q1, quality-hold) managed with appropriate controls?	High	
☐	IATF-CQI-005 Does the organization conduct periodic supplier audits including product audits, process audits, and QMS audits based on supplier risk and performance?	High	
☐	IATF-8.4.2.4-001 Does the organization use second-party audits to evaluate external providers?	Medium	
☐	IATF-8.4.2.5-001 Does the organization use statistical data and other methods to evaluate external provider performance?	Medium	

Have ready	
☐ Supplier scorecards	☐ Process audit records
☐ PPM tracking	☐ QMS audit records
☐ Delivery performance tracking	☐ Audit finding closure tracking
☐ Premium freight tracking	☐ Supplier audit program
☐ Customer disruption tracking	☐ VDA 6.3 audit reports
☐ Customer quality status notifications	☐ Audit schedule with risk-based frequency
☐ Enhanced incoming inspection for affected suppliers	☐ Auditor qualifications
☐ Controlled shipping records	☐ Supplier PPM tracking
☐ Supplier corrective action status	☐ Delivery performance data
☐ Supplier audit schedule	☐ Cost of quality per supplier
☐ Supplier risk classification	☐ Statistical analysis of incoming inspection data
☐ Product audit records	

Red flags an auditor looks for	
No supplier performance monitoring	Audit findings not closed
Chronically poor suppliers without escalation	No second-party audit program
Unaware of supplier quality status	Critical suppliers never audited
No enhanced controls for problem suppliers	No supplier performance data

Red flags an auditor looks for	
No supplier audits	No statistical evaluation of quality

Clause 9.2.2.3: Manufacturing process audit

What your records showed

What was found	Records	Which
FIND-RDG-008 Manufacturing process audits on FSW and leak test not performed since the ERP cutover	-	

	Question	Priority	Evidence location / initials
☐	IATF-9.2.2.2-001 Does the audit program include manufacturing process audits to determine process effectiveness including risk-based scheduling?	High	

	Have ready		
☐	Manufacturing process audit schedule	☐	Risk-based frequency justification
☐	Process audit reports (VDA 6.3 or equivalent)	☐	Process audit findings and actions

Red flags an auditor looks for	
No manufacturing process audits	Only system audits conducted

Clause 9.3.1: Management review: general

What your records showed

What was found	Records	Which
FIND-RDG-012 Management review not held in Q3 or Q4 2025	-	

	Question	Priority	Evidence location / initials
☐	IATF-9.3-001 Does top management review the QMS at planned intervals to ensure its continuing suitability, adequacy, effectiveness, and alignment with strategic direction?	High	
☐	IATF-9.3-002 Does management review include assessment of the effectiveness of actions taken to address risks and opportunities?	Medium	

	Have ready		
☐	Management review minutes	☐	Management review agenda with risk items
☐	Management review agenda (IATF inputs)	☐	Risk treatment effectiveness data
☐	Action items from management review	☐	Updated risk register
☐	Action completion tracking		

Red flags an auditor looks for	
No management review	Risk not reviewed by management
Management review is a paperwork exercise	Risk register stale

